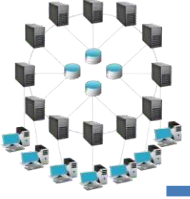


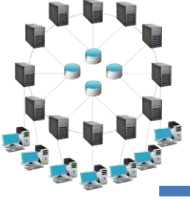
Introduction to Distributed Systems

Assoc.Prof.James Rwigema



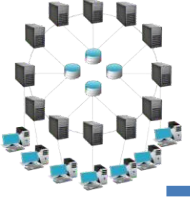
Contents

1. Introduction
2. Why Distributed Computing
3. Advantages and Disadvantages of DS
4. Goals of DS
5. DS challenges
6. DS Architectural Model
7. Types of DS



1. Introduction-1

- Distributed Systems have been in existence since the start of the universe.
- In the real world, there is a need of increase of the information usages and their accessibility regardless where there located.
- The proliferation of the internet and the emerging global village, the notion of distributed computing systems as a useful and widely deployed tool is becoming a reality.
- The scenario is **you know you are using one when the crash of a computer you have can't prevent you from doing the work.**



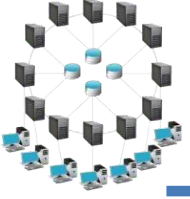
Introduction-2

- A **distributed system** is a collection of independent entities that cooperate to solve a problem that cannot be individually solved.

- Distributed system can be characterized as a collection of mostly autonomous processors communicating over a communication network.

- **Distributed computing** refers to the mean by which a single computer program runs in more than one computer at the same time.

- In distributed computing a program is split up into parts than run simultaneously on multiple computers communicating over a network.



Introduction-3

- **Distributed Computing**

  - A system in which hardware or software components located at **networked computers** communicate and coordinate their actions only by passing messages.

  - A collection of **independent computers** that appear to its users as a single coherent system.

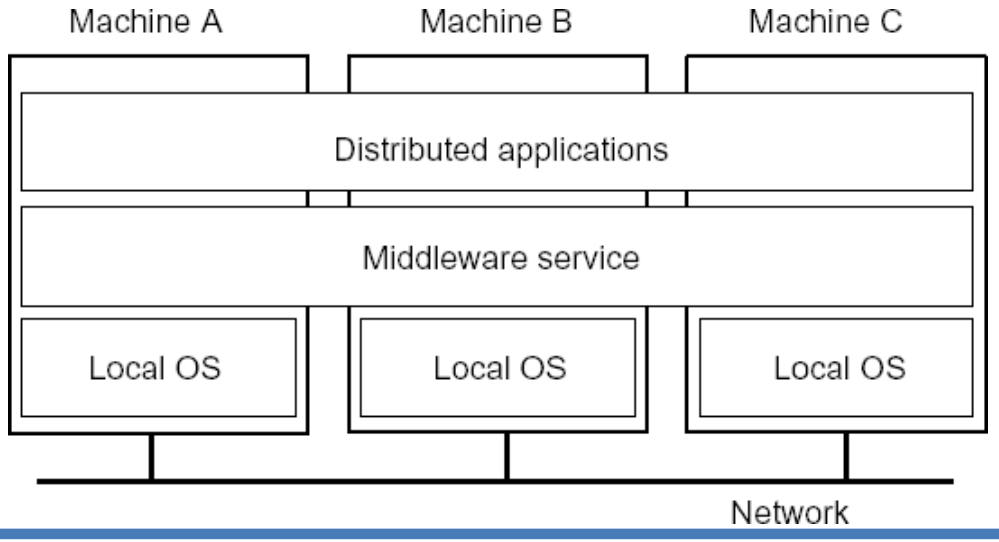
  - A parallel computer is implicitly a distributed system.

- Examples:
  - Network of workstations
  - Distributed manufacturing system (e.g., automated assembly line)
  - Network of branch office computers.

Introduction-3

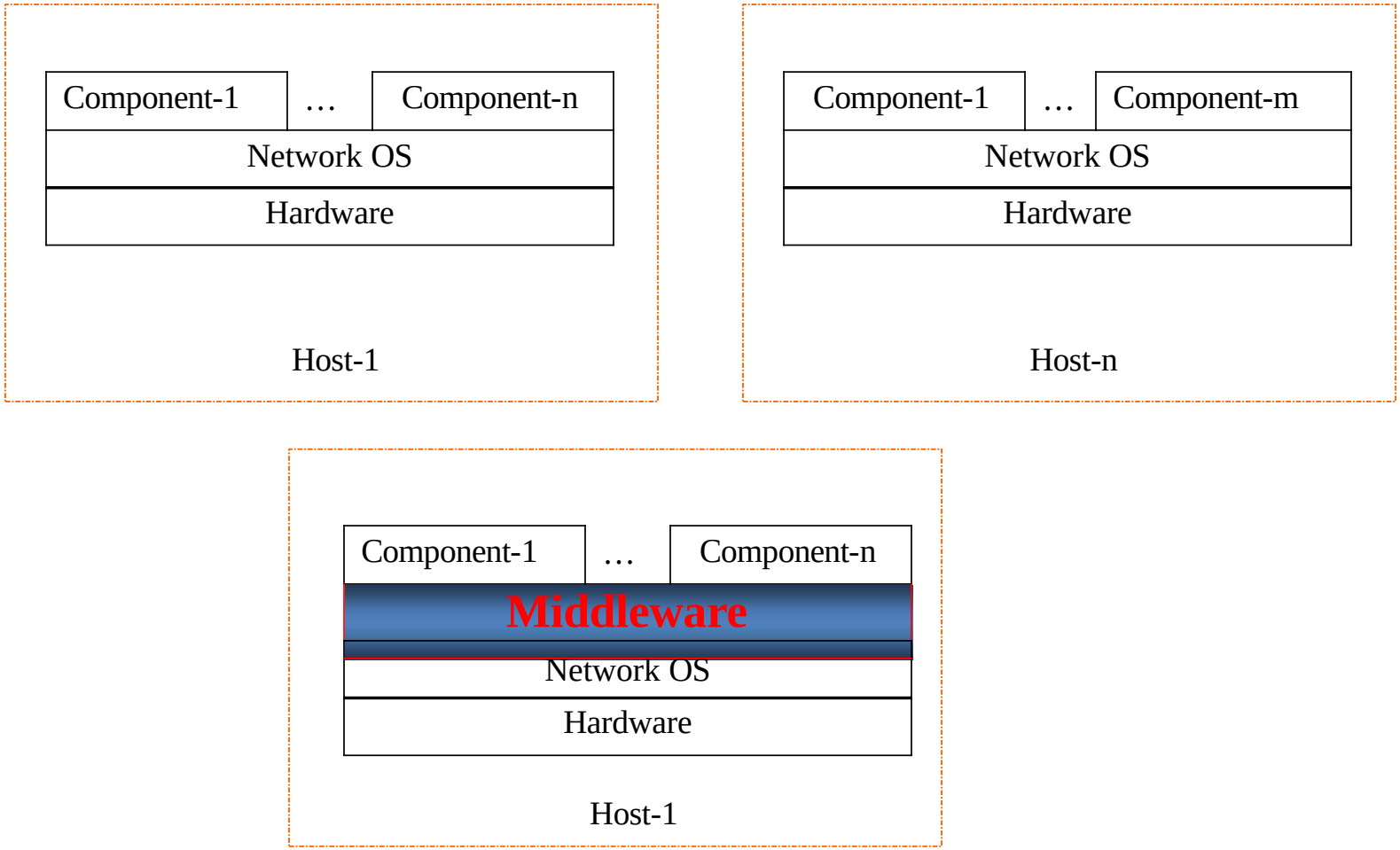
- *Distributed Computing with hardware and software systems deals containing more than one **processing element** or **storage element**, **concurrent processes**, or multiple programs running under a loosely or tightly controlled regime.*

- *Two aspects:*
  1. *Hardware*: - autonomous computers
  2. *Software*: - users think they are dealing with a single system

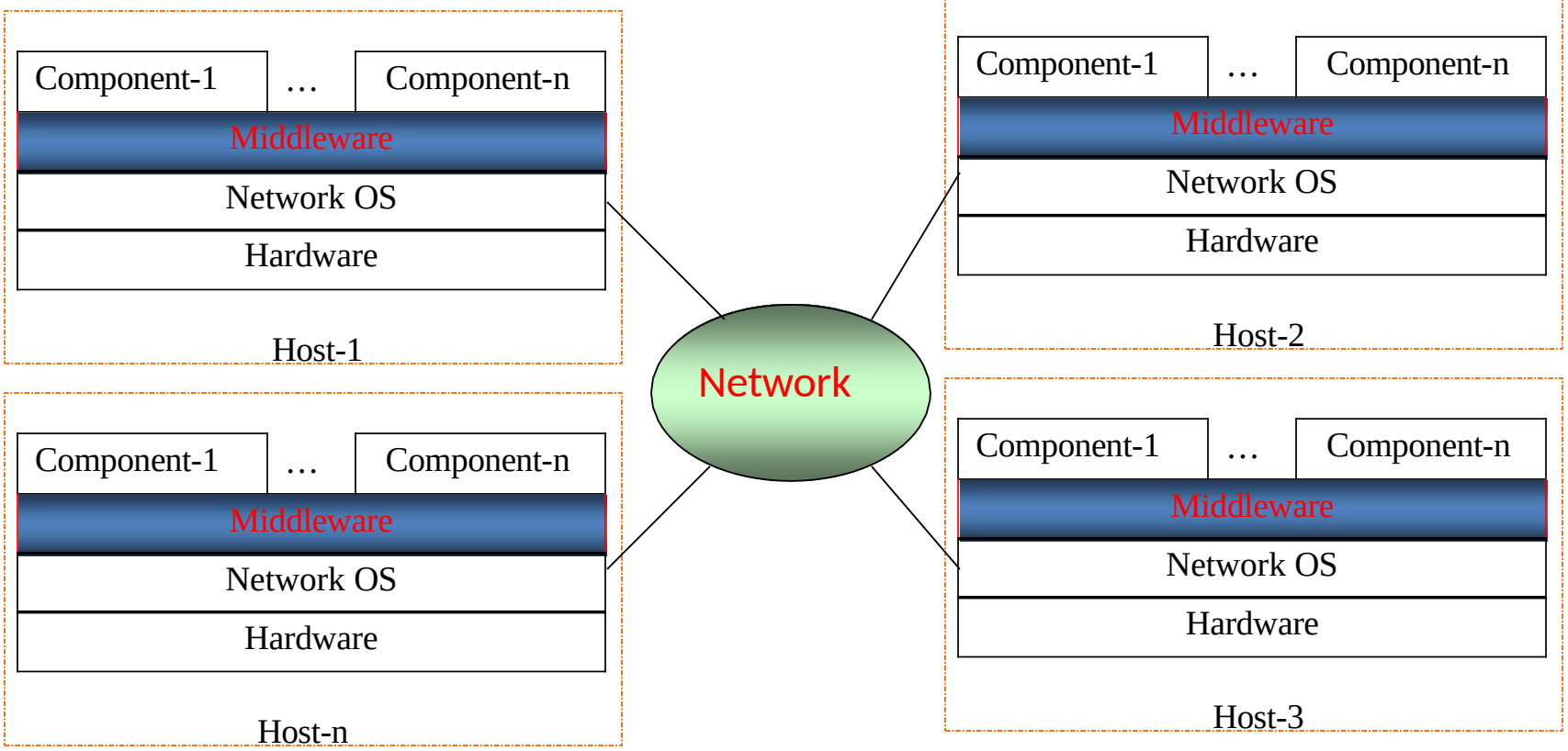


Introduction-4

- Ingredients of a Distributed System

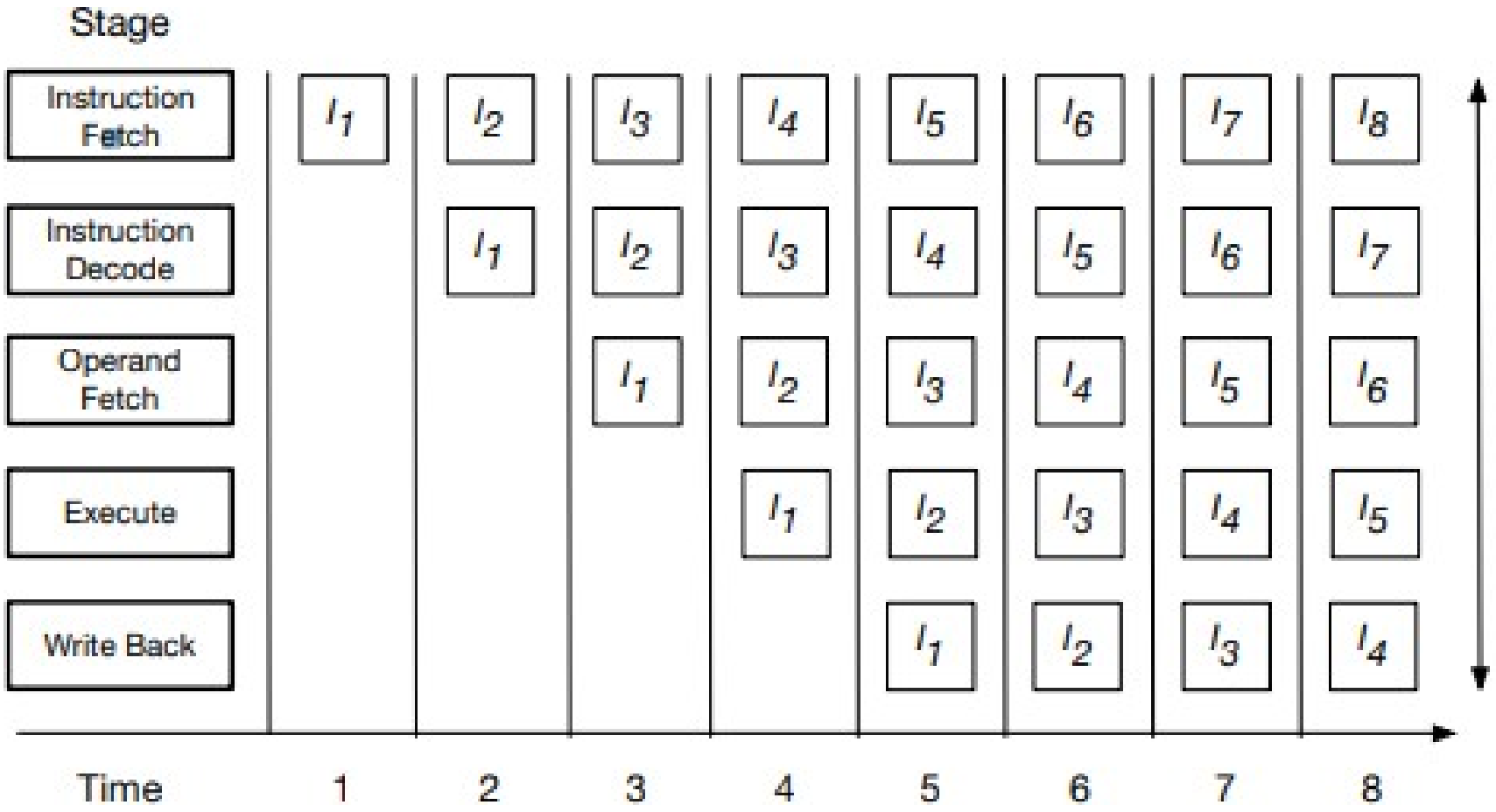


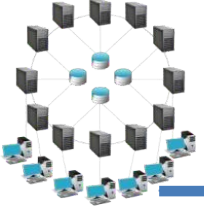
Introduction-5



Why Parallel and distributed Computing-2 ?

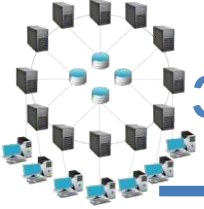
– *Instruction level parallelism (ILP) (90's)*





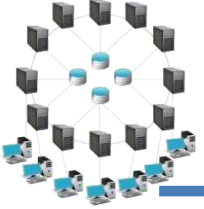
- In computation, parallel systems with multi-core processors are going to over perform mainstream.

  - Lots of hardware, software, and human issues in mainstream parallel/distributed computing
    - ✓ **How to make use a large number of processors simultaneously**
    - ✓ **How to write parallel programs easily and efficiently?**
    - ✓ **What kind of software supports is needed?**
    - ✓ **Analyzing and optimizing parallel programs is still a hard problem.**
    - ✓ **How to debug concurrent program?**



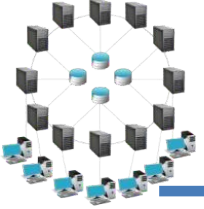
3. Advantages of Distributed System over Centralized systems

1. **Economics**: a collection of microprocessors offer a better price/performance than mainframes.
2. **Speed**: a distributed system may have more total computing power than a mainframe. Ex. 10,000 CPU chips, each running at 50 MIPS. Enhanced performance through load distributing.
3. **Inherent distribution**: Some applications are inherently distributed. Ex. a supermarket chain.
4. **Reliability**: If one machine crashes, the system as a whole can still survive. Higher availability and improved reliability.
5. **Incremental growth**: Computing power can be added in small increments. Modular expandability.
6. **Another deriving force**: the existence of large number of personal computers, the need for people to collaborate and share information.



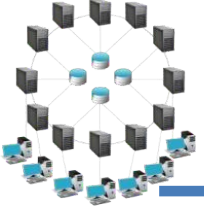
Advantages of Distributed Systems over Independent PCs

- **Data sharing:** allow many users to access to a common distributed database.
- **Resource Sharing**: expensive peripherals like color printers.
- **Communication**: enhance human-to-human communication, E.g.: email, chat.
- **Flexibility**: spread the workload over the available machines.



Disadvantages of Distributed Systems

1. **Software**: difficult to develop software for distributed systems
2. **Network**: Saturation, lossy transmissions
3. **Security**: Easy access also applies to secret data



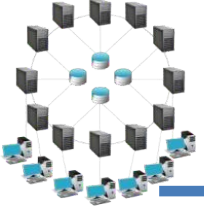
4. Distributed System Goals

- *Allow users to access and share resources:*

1. ***Transparency***
   - To hide fact that its **processes** and **resources** are physically distributed across multiple computers.

2. ***Openness***
   - To offer services according to standard rules that describe the syntax and semantics of those services.
   - E.g.: Specify interfaces using an interface definition language (IDL)

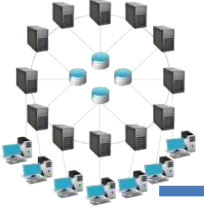
3. ***Scalability***
   - Possibility of making the system expends in any dimension



Distributed Systems Goals in details

1. Transparency characteristics are shown bellow

Transparency	Description
Access	Hides differences in data representation and invocation mechanisms
Location	Hides where an object resides
Migration	Hides from an object the ability of a system to change that object's location
Relocation	Hides from a client the ability of a system to change the location of an object to which the client is bound
Replication	Hides the fact that an object or its state may be replicated and that replicas reside at different locations
Concurrency	Hides the coordination of activities between objects to achieve consistency at a higher level
Failure	Hides failure and possible recovery of objects
Persistence	Hides the fact that an object may be (partly) passivated by the system



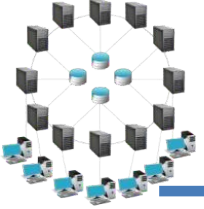
1. Transparency (cont'd)

- Aiming at full transparency may be too much:

- Users may be located in different continents; distribution is apparent
and not something you want to hide.

- Completely hiding failures of networks and nodes is (theoretically
and practically) impossible.
  - You cannot distinguish a slow computer from a failing one.
  - You can never be sure that a server actually performed an operation before a crash.

- Full transparency will cost performance, exposing distribution of the
system



2. Openness

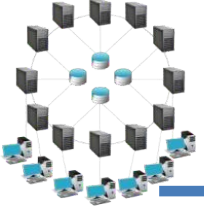
a. Open distributed system: Be able to interact with services from
other open systems, irrespective of the underlying environment:

- Systems should conform to well-defined **interfaces**
- Systems should support **portability** of applications
- Systems should easily **interoperate**

b. The challenge will be how to achieve openness:

❖ At least make the distributed system independent from
heterogeneity of the underlying environment:

- Hardware
- Platforms
- Languages



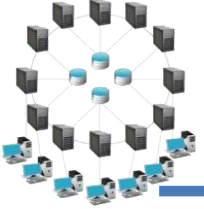
2. Openness (cont'd)

c. **Implementing openness:** Requires support for different **policies** specified by applications and users:

- ✓ What level of consistency do we require for client cached data?
- ✓ Which operations do we allow downloaded code to perform?
- ✓ Which QoS requirements do we adjust in the face of varying bandwidth?
- ✓ What level of secrecy do we require for communication?

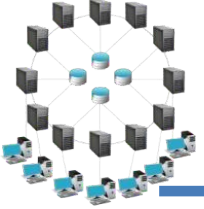
➢ Ideally, a distributed system provides only **mechanisms**:

- ✓ Allow (dynamic) setting of caching policies, preferably per cacheable item.
- ✓ Support different levels of trust for mobile code
- ✓ Provide adjustable QoS parameters per data stream
- ✓ Offer different encryption algorithms



3. Scalability

- Many developers of modern distributed systems easily use the adjective "scalable" without making clear *why* their system actually scales.
- At least three components must be cared about:
  - Number of users and/or processes **(size scalability)**
  - Maximum distance between nodes **(geographical scalability)**
  - Number of administrative domains **(administrative scalability)**
- Most systems account only, to a certain extent, for size scalability. The (non)solution: ***powerful servers***.
- Today, the challenge lies in geographical and administrative scalability.



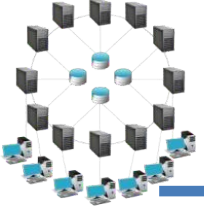
3. Scalability: Techniques of Scaling

a. **Hiding Communication Latency**
   - Move computations to clients

b. **Distribution**: Partition data and computations across multiple machines:
   - Move computations to clients (Java applets)
   - Decentralized naming services (DNS)
   - Decentralized information systems (WWW)

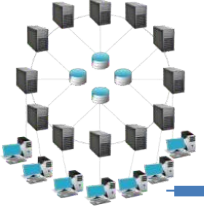
c. **Replication**: Make copies of data available at different machines:
   - Replicated file servers (mainly for fault tolerance)
   - Replicated databases
   - Mirrored Web sites
   - Large-scale distributed shared memory systems

d. **Caching**: Allow client processes to access local copies:
   - Web caches (browser/Web proxy)
   - File caching (at server and client)



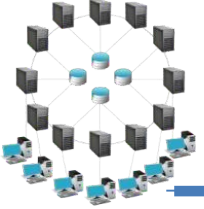
3. Scalability: Scaling problem

- Applying scaling techniques is easy, except for one thing:
  - *Having multiple copies (cached or replicated), leads to **inconsistencies**: modifying one copy makes that copy different from the rest.*
- *Always keeping copies consistent and in a general way requires **global synchronization** on each modification.*
  - Global synchronization makes large-scale solutions practically impossible.
  - If we can tolerate inconsistencies, we may reduce the need for global synchronization.
    - Tolerating inconsistencies is application dependent.



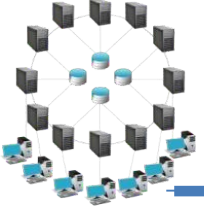
5. Distributed Systems Challenges-1

1. **Heterogeneity**: They must be constructed from a variety of different networks, OS, Computer HW, and programming languages.
2. **Openness**: DS should be extensible, the integration of components written by different programmer is a challenge.
3. **Security**: Encryption is used to provide protection to shared resources and keeping sensitive information secret when transmitted. Denial of service (DoS) attacks are still a problem.
4. **Scalability**: A DS is scalable if the cost of adding a user is a constant amount in terms of the resources that must be added.
5. **Failure handling**: any process, computer or network may fail independently of the others. Each component need to be aware of the possible ways. DS need to be designed to deal with each of these failures.



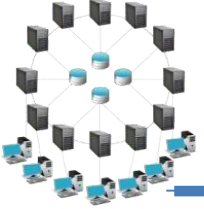
Distributed Systems Challenges-2

6. **Concurrency**: Presence of multiple users is a source of concurrent requests to the resources. Each resource must be designed to be safe in a concurrent environment.
7. **Transparency**: The aim is to make certain aspects of distribution invisible to the application programmer so that they need only be concerned with the design of their particular application.
8. **Quality of Service (QoS):** It is not sufficient to provide access to services in distributed systems. It is also important to provide guarantees regarding the qualities associated with such service access. **(like performance, security, and reliability).**



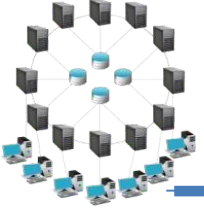
False assumptions when developing DS

1. **The network is reliable**
2. **The network is secured**
3. **The network is homogeneous**
4. **The topology does not change**
5. **Latency is zero**
6. **Bandwidth is infinite**
7. **Transport cost is zero**
8. **There is one administrator**



6. Distributed Systems: Hardware concepts

- **Multiprocessors**
- **Multicomputer**
- **Networks of computers**



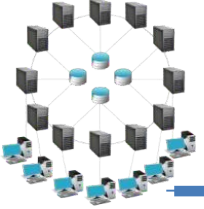
Distributed Systems: Hardware concepts

1. Multiprocessors

- Multiprocessing is the use of two or more CPUs within a single computer system.
- Multiprocessor is a computer system having two or more processing units (multiple processors) each sharing main memory and peripherals, in order to simultaneously process programs.

➢ **Processor symmetry:**

- ✓ In a multiprocessing system, all CPUs may be equally participating , or some may be reserved for special purposes.
- ✓ Systems that treat all CPUs equally are called Symmetric multiprocessing (SMP) systems
- ✓ Asymmetric Multiprocessing (ASMP).
- ✓ Non- Uniform Memory Access (NUMA)
- ✓ Clustered Multiprocessing



Distributed Systems: Hardware concepts

- **Multiprocessors Communication Architecture**

– **Message Passing**

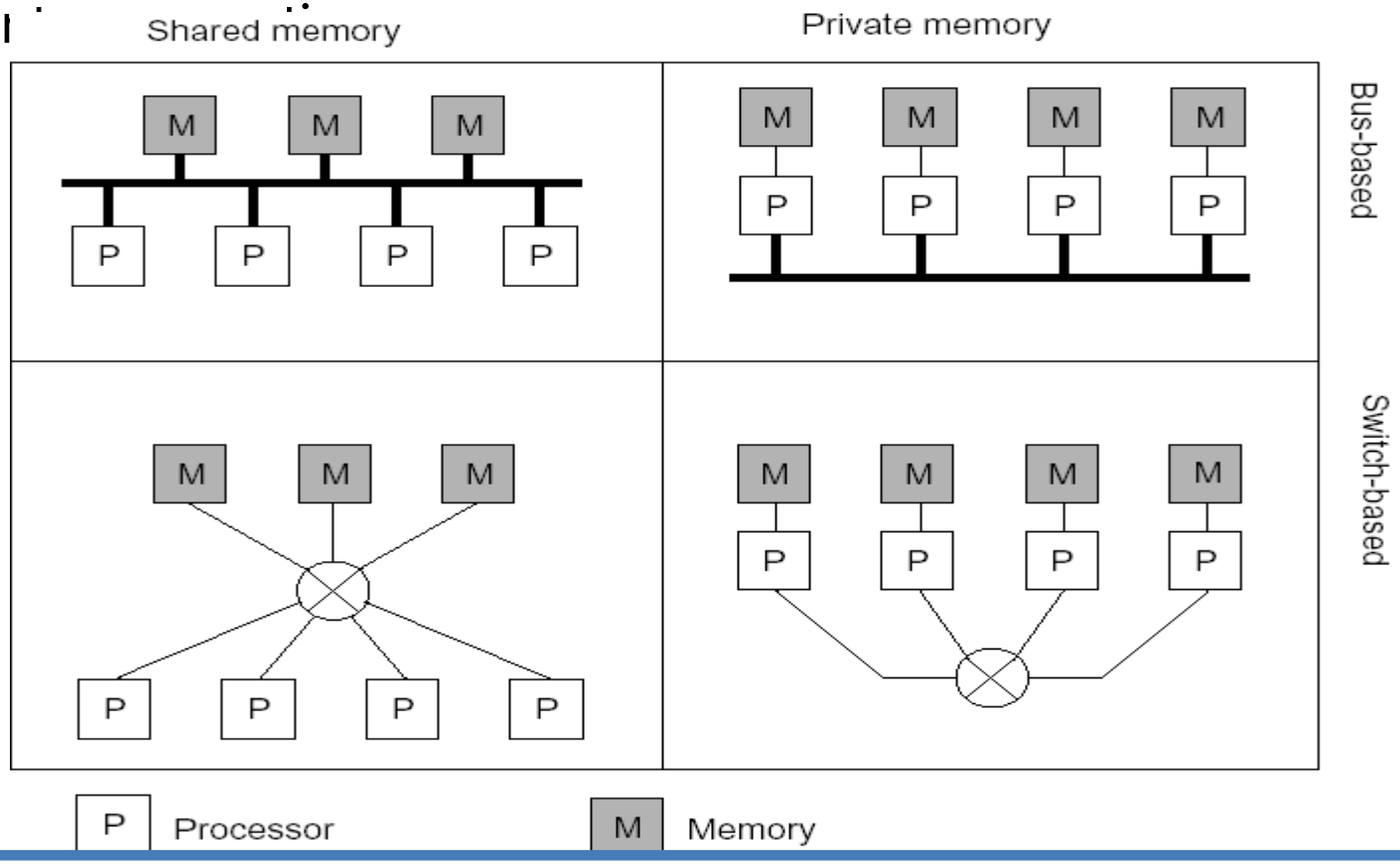
- Separate address space for each processor
- Processors communicate via message passing
- Processors provide local message queue memories
- Focus attention on costly non-local operations

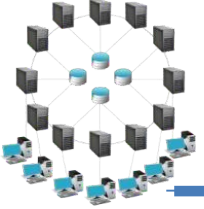
– **Shared Memory**

- Processors communicate with shared address space
- Processors communicate by memory read/write
- Easy on small-scale machines
- Lower latency

Distributed Systems: Hardware concepts

- **Distinguished features:**
  - Private vs shared memory
  - Bus vs switched



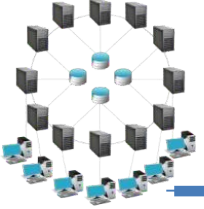


3. Networks of Computers

- **High degree of node heterogeneity:**
  - ✓ High-performance parallel systems (multiprocessors as well as multicomputer)
  - ✓ High-end PCs and workstations (servers)
  - ✓ Simple network computers (offer users only network access)
  - ✓ Mobile computers (palmtops, laptops)

- **High degree of network heterogeneity:**
  - ✓ Local-area gigabit networks
  - ✓ Wireless connections
  - ✓ Long-haul, high-latency POTS connections
  - ✓ Wide-area switched megabit connections

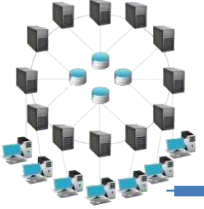
- **Observation:** Ideally, a distributed system hides these differences



Distributed Systems: Software concepts

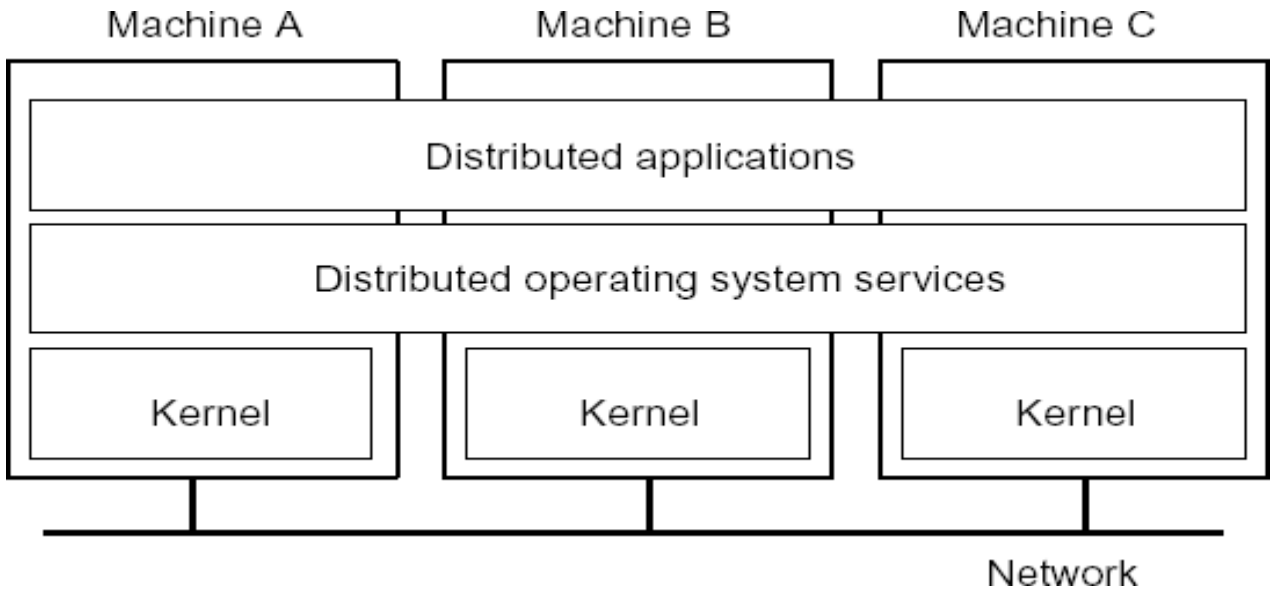
- **Distributed Operating System**
- **Network Operating System**
- **Middleware**

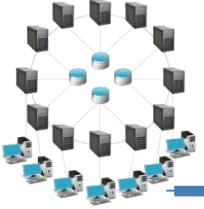
System	Description	Main goal
DOS	Tightly-coupled OS for multiprocessors and homogeneous multicomputers	Hide and manage hardware resources
NOS	Loosely-coupled OS for heterogeneous multicomputers (LAN and WAN)	Offer local services to remote clients
Middle-ware	Additional layer atop of NOS implementing general-purpose services	Provide distribution transparency



Distributed Systems: Software concepts

- **Distributed Operating System**
  - **Some characteristics**
    - OS on each computer knows about the other computers
    - Services are generally (transparently) distributed across computers





Distributed Systems: Software concepts

- **Multicomputer Operating System**
  - **Harder than traditional (multiprocessor) OS:** Because memory is not shared, emphasis shifts to processor communication by message passing:

    - Often no simple global communication:
      - Only bus-based multi-computers provide hardware broadcasting
      - Efficient broadcasting may require network interface programming techniques
    - No simple system-wide synchronization mechanisms
    - Virtual (distributed) shared memory requires OS to maintain global memory map in software
    - Inherent distributed resource management: no central point where allocation decisions can be made

- **Practice:** Only very few truly multicomputer operating systems exist (example: **Amoeba**)

Distributed Systems: Software concepts

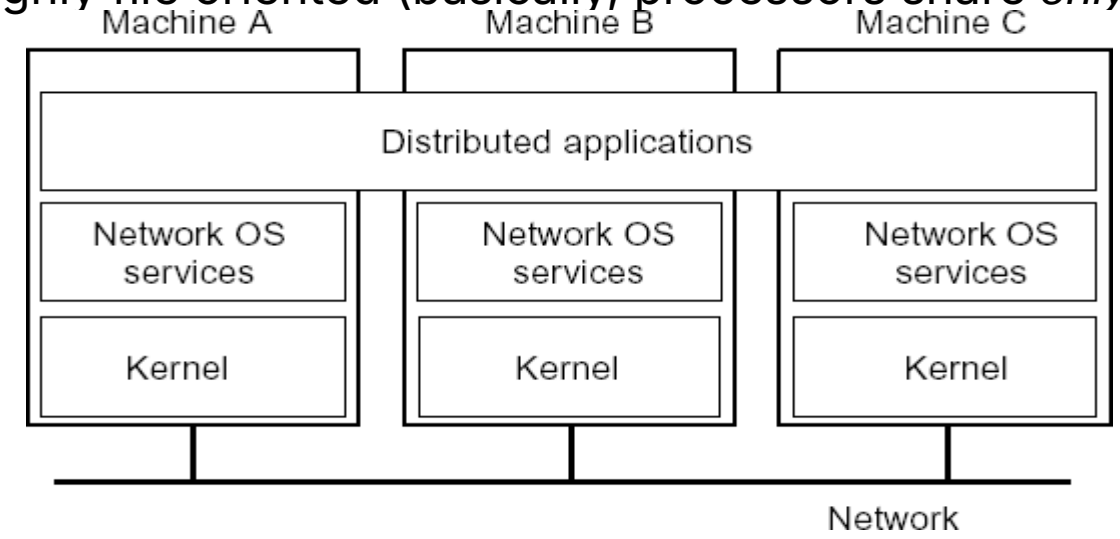
- **Network Operating Systems**
  - **Some characteristics:**

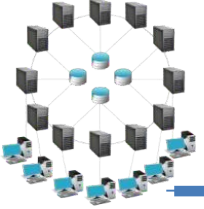
❖ Each computer has its own operating system with networking facilities

❖ Computers work independently (i.e., they may even have different operating systems)

❖ Services are tied to individual nodes (ftp, telnet,WWW)

❖ Highly file oriented (basically, processors share *only* files)



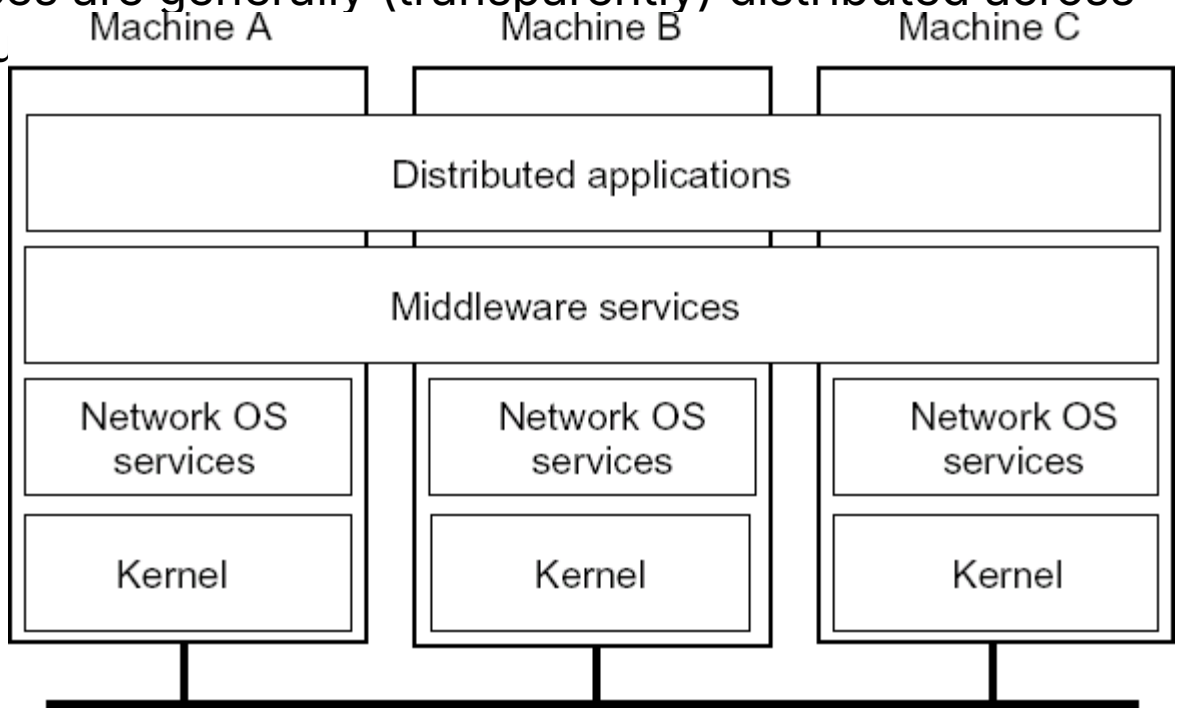


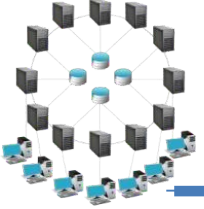
Distributed Systems: Software concepts

- **Middleware**
  - Is the software that connects software components or enterprise applications.
  - **Middleware** is the software layer that lies between the operating system and the applications on each side of a distributed computer network.

Distributed Systems: Software concepts

- **Middleware**
  - **Some characteristics:**
    - OS on each computer need not know about the other computers
    - OS on different computers need not generally be the same
    - Services are generally (transparently) distributed across compu

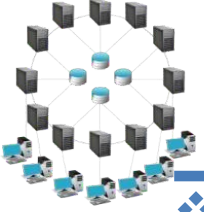




Distributed Systems: Software concepts

❖ Needs for Middleware

- **Motivation:** Too many networked applications were difficult to integrate:
  - ✓ Departments are running different NOSs
  - ✓ Integration and interoperability only at level of primitive NOS Services
- Need for federated information systems:
  - ✓ Combining different databases providing a single view to but applications
  - ✓ Setting up enterprise-wide Internet services, making use of existing information systems
  - ✓ Allow transactions across different databases
  - ✓ Allow extensibility for future services (e.g., mobility, tele-working)
- **Constraint**: use the existing operating systems, and treat them as the underlying environment

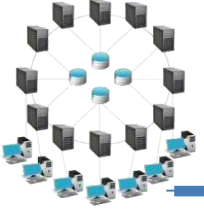


Distributed Systems: Software concepts

❖ Middleware Services

1. ***Communication services*:** Abandon primitive socket-based message passing in favor of:
   - ✓ Procedure calls across networks
   - ✓ Remote-object method invocation
   - ✓ Message-queuing systems
   - ✓ Advanced communication streams
   - ✓ Event notification service

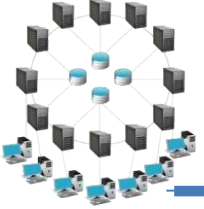
2. ***Information system services*:** Services that help manage data in a distributed system:
   - ✓ Large-scale, system-wide naming services
   - ✓ Advanced directory services (search engines)
   - ✓ Location services for tracking mobile objects
   - ✓ Persistent storage facilities
   - ✓ Data caching and replication



Distributed Systems: Software concepts

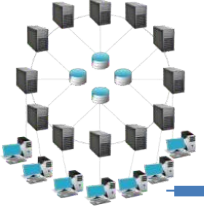
3. ***Control services*:** Services giving applications control over when,
where, and how they access data:
   - ✓ Distributed transaction processing
   - ✓ Code migration

4. ***Security services*:** Services for secure processing and communication:
   - ✓ Authentication and authorization services
   - ✓ Simple encryption services
   - ✓ Auditing service



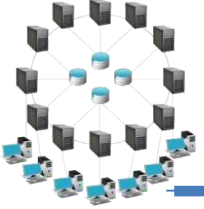
Distributed Systems: Software concepts

- **Comparison of NOS, DOS, and Middleware**
  1. Degree of transparency.
  2. Same operating system on each node?
  3. Number of copies of the operating system.
  4. Basis for communication.
  5. How are resources managed?
  6. Is the system easy to scale?
  7. How open is the system?



7. DS Architecture Models

- The architecture of a system is its structure in terms of separately specified components and their interrelationships.
- The overall goal is to ensure that the structure will meet present and likely future demands on it.
- The architectural design of a building has similar aspects.

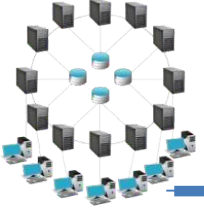


Architecture Models-2

- Architectural Elements

➢ To understand the fundamental building block of DS, it is necessary
to consider four key questions:

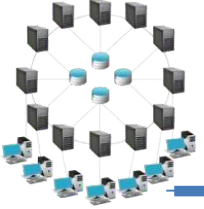
1. **What are the entities that are communicating in the DS?**
2. **How do they communicate, or , what paradigm is used?**
3. **What roles and responsibilities do they have in the overall architecture?**
4. **How are they mapped on to the physical distributed infrastructure (placement)?**



Architectural Elements

1. What are the entities that are communicating in the DS?

- The entities that communicate in a DS are processes. Reading to inter-processes communication paradigms.
- In some primitive environments, such as sensor networks, the entities that communicate are nodes.
- In most distributed Systems environments, processes are supplemented by threads, it is threads that are the endpoints of communication.
- Problem-oriented abstractions have been proposed:
  - OBJECTS: Proposed to encourage the use of OO approaches in DS.
  - COMPONENTS: resemble objects in that they offer problem-oriented abstractions for building DS and are also accessed through interface.



Architectural Elements

2. How do they communicate, or , what communication paradigm is used?

- Communication paradigm

  a. **Inter-process communication**: Communication between processes in DS, including ***message-passing*** primitives, direct access to the API offered by Internet protocol (Socket Programming)

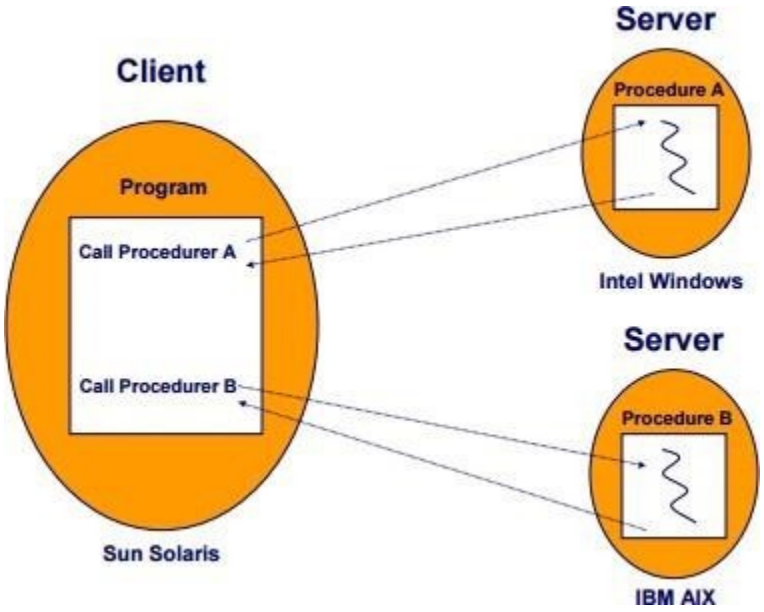
  b. **Remote invocation**: Based on a two-way exchange between communicating entities in a DS and resulting in the calling of a remote operation, procedure or method.

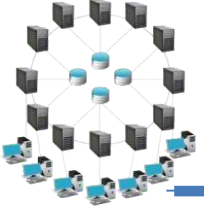
  - ***Request-reply protocols***: a pattern imposed on an underlying message-passing service to support client-server computing.

Architectural Elements

- Communication paradigm

  - ***Remote procedure calls (RPC)***: the procedures in processes on remote computers can be called as if they are procedures in the local address space.



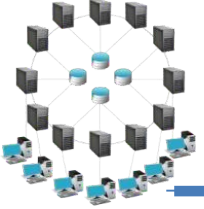


Architectural Elements

- Communication paradigm

  - ***Remote method invocation (RMI)***: resembles RPC but
    in a world of distributed objects.

=> This communication represents a two-way relationship between SENDER and RECEIVER, with senders explicitly directing messages/invocations to the associated receivers.



Architectural Elements

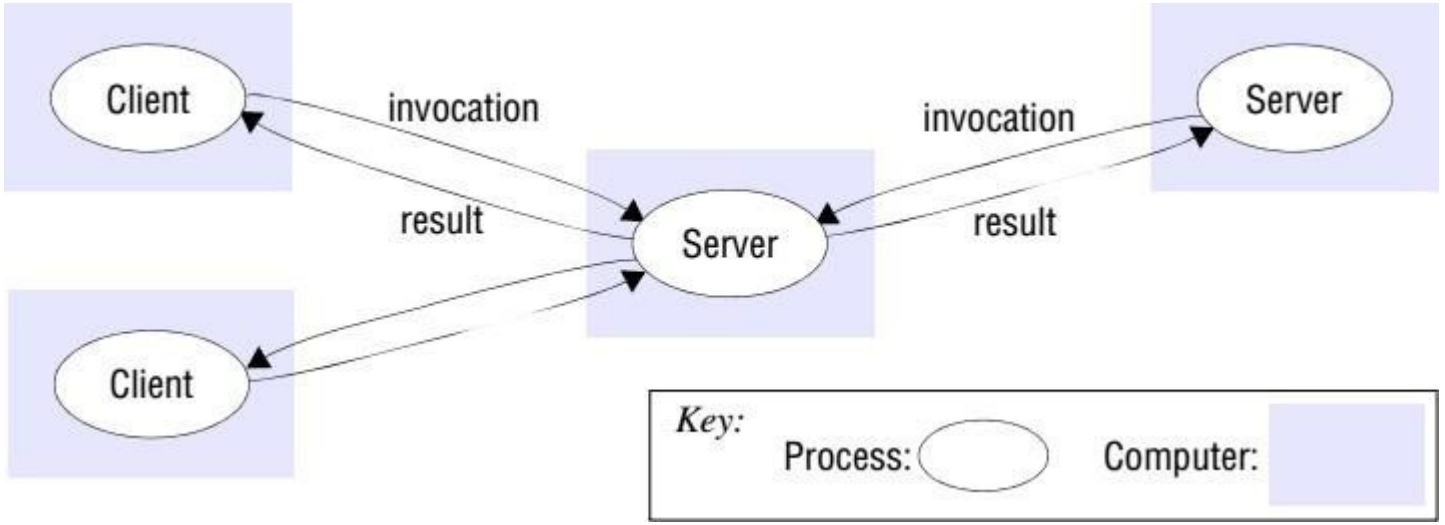
c. Indirect communication

- ✓ Senders do not need to know who they are sending to (**space uncoupling**)
- ✓ Senders and receivers do not need to exist at the same time (**time coupling**)

Architectural Elements

3. Roles and responsibilities

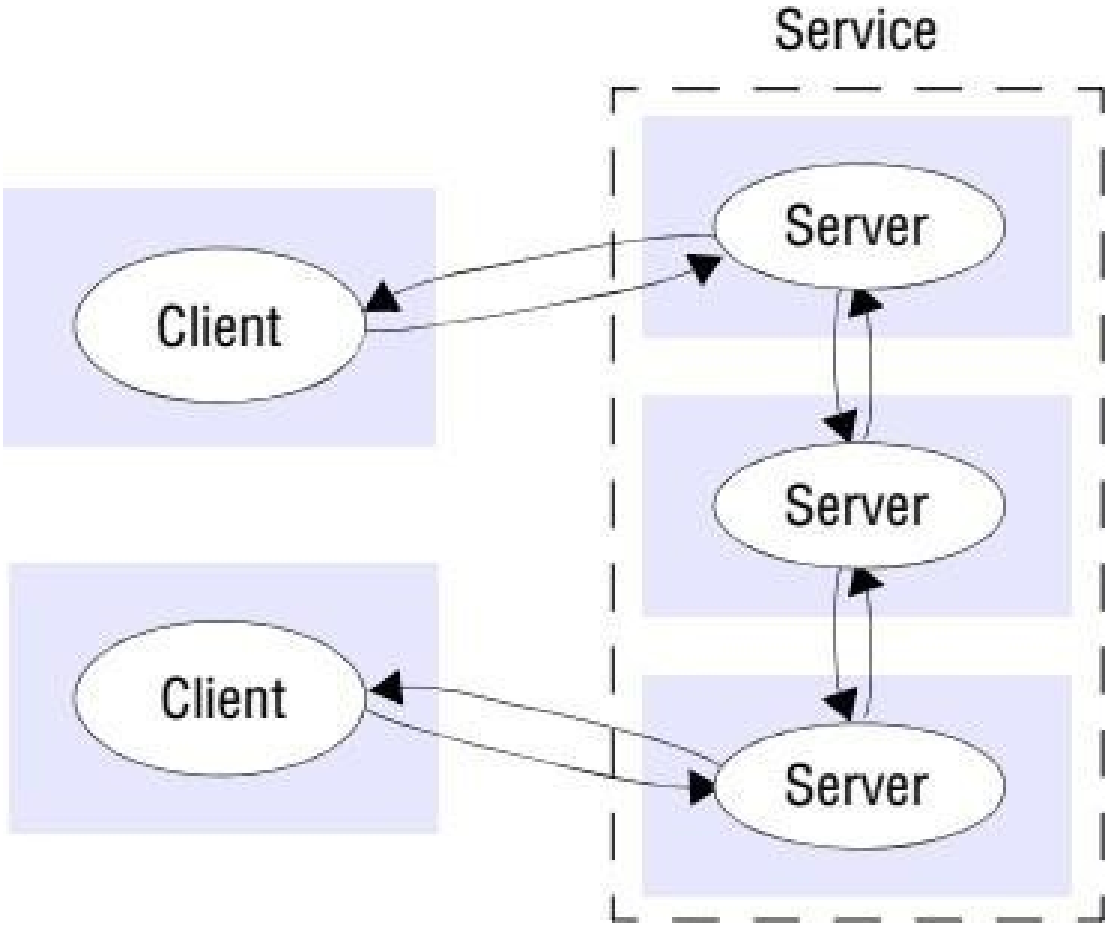
✓ **Client – Server**



- Client processes interact with individual server processes in order to access the shared resources that they manage.
- Servers may in turn be clients of other servers.
- Eg,: WEB SERVER

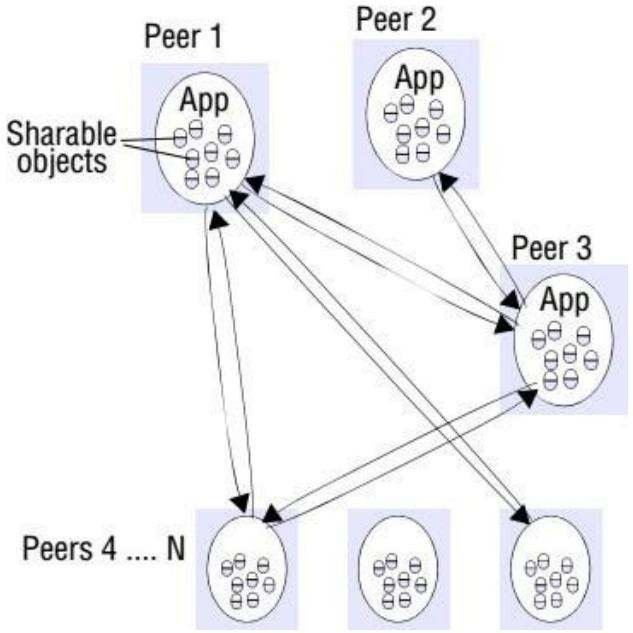
Architectural Elements

- A service provided by multiple servers



Architectural Elements

Peer-to-peer



- All the processes involved in a task or activity play similar roles.
- Interacting cooperatively as peers without any distinction between client and server processes or the computers on which they run.

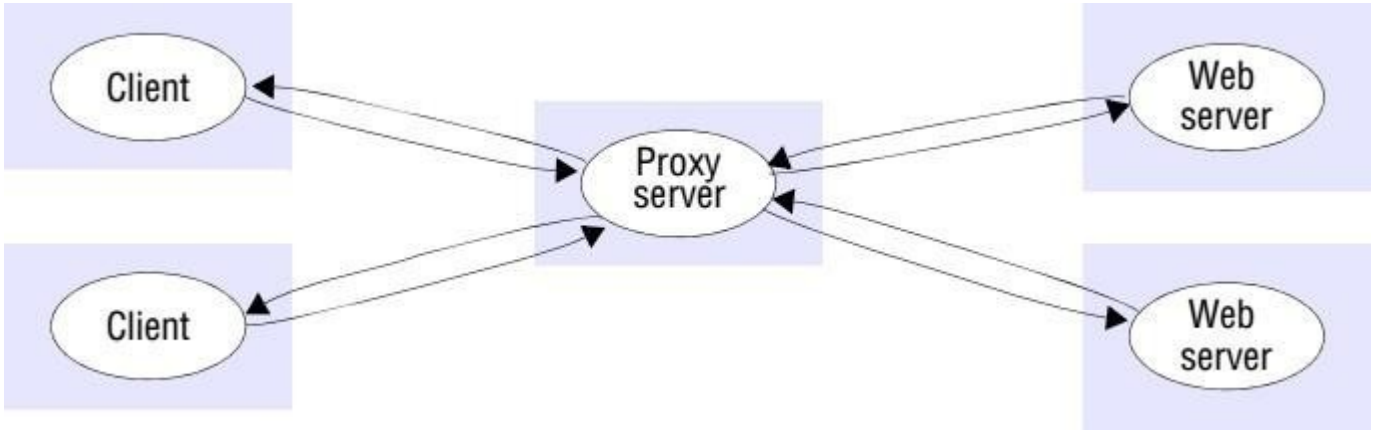
Architectural Elements

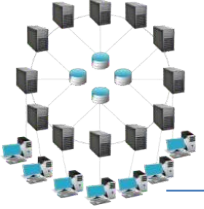
4. Placement

- The final issues to be considered is how entities such as objects or services map on to the underlying physical distributed infrastructure and there are few universal guidelines placement strategies
- Placement must be determined with strong application knowledge

1. Mapping of Services to multiple servers

✓ Services may be implemented as several server processes in separate host computers interacting as necessary to provide a service to client processes





Architectural Elements

2. Caching

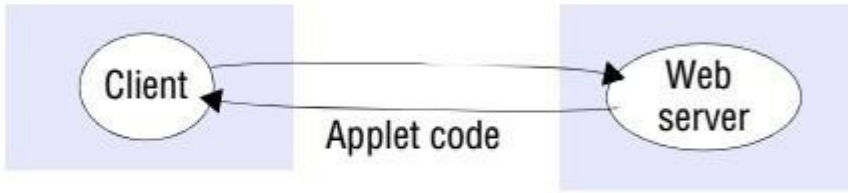
- ✓ A cache is a store of recently used data objects that is closer to a particular set of clients than the objects themselves.
- ✓ When a new object is received from a server it is added to the local cache store, replacing some existing objects if necessary.
- ✓ When an object is needed by a client process, the caching service first checks the cache and supplies the object from there if an up-to-date copy is available.
- ✓ If not, an up-to-date copy is fetched.
- ✓ Caches may be co-located with each client or they may be located in a proxy server that can be shared by several clients.

Architectural Elements

- Caching example

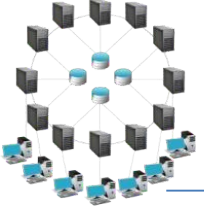
Web applets

a) client request results in the downloading of applet code



b) client interacts with the applet





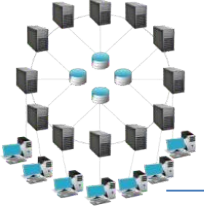
Architectural Elements

3. Mobile code

- ✓ Accessing services means running code that can invoke their operations.
- ✓ Some services are likely to be so standardized that we can access them with an existing and well-known application – the Web is the most common example of this, but even there, ***some web sites use functionality not found in standard browsers and require the downloading of additional code.***

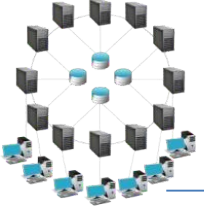
4. Mobile agents

- ✓ A mobile agent is a running program (including both code and data) that travels from one computer to another in a network carrying out a task on someone's behalf, such as collecting information, and eventually returning with the results.



Types of Distributed Systems-1

- The nodes in the distributed computing can be arranged in the form of:
  - Client/Server systems
  - Peer to peer Systems



Client-Server Model

- **Servers:** Generally provide services related to a *shared resource*:

✓ Servers for file systems, databases, implementation repositories, etc.

✓ Servers for shared, linked documents (Web, VoD)

✓ Servers for shared applications

✓ Servers for shared distributed objects

- **Clients:** Allow remote service access:

✓ Programming interface transforming client's local service calls to request/reply messages

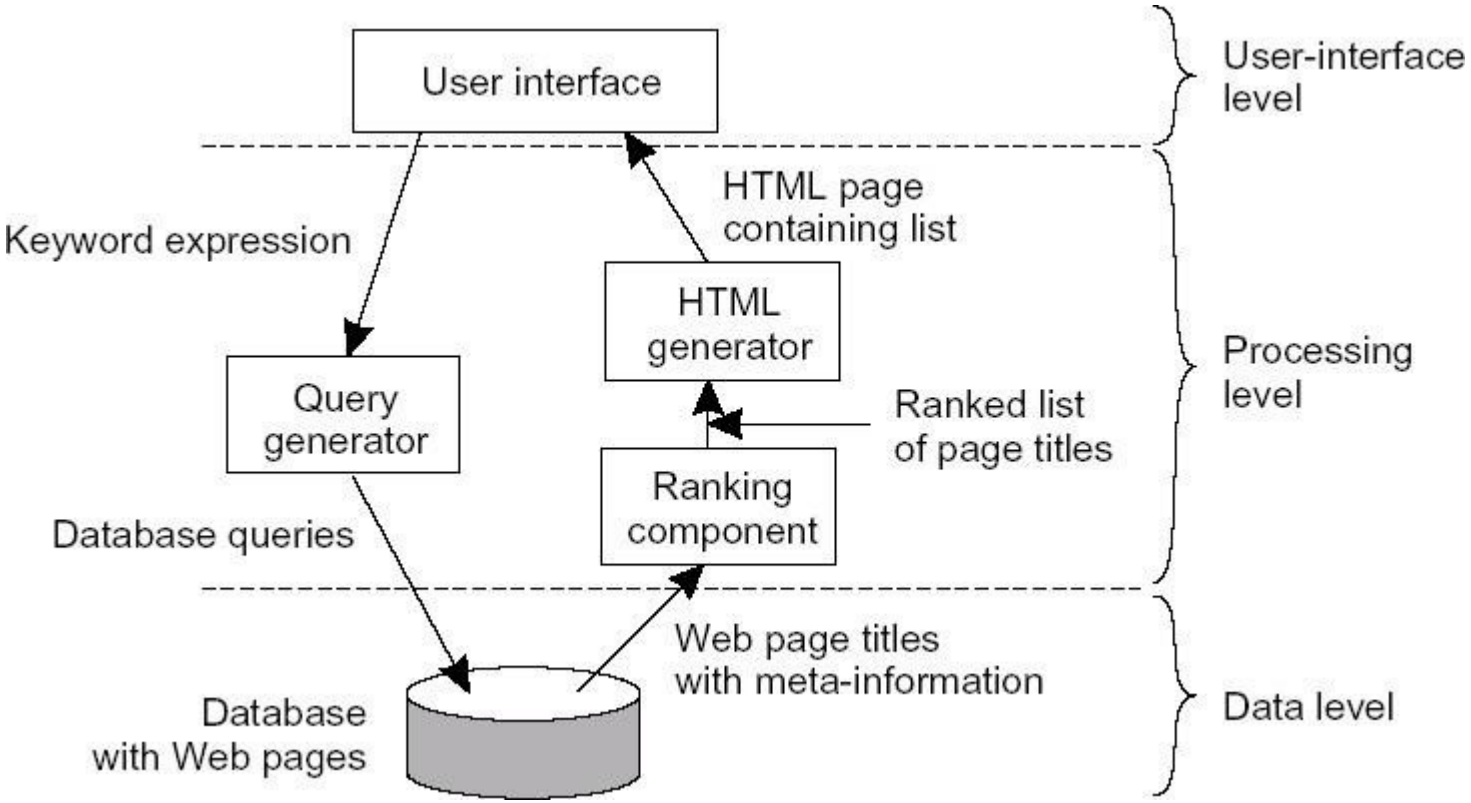
✓ Devices with (relatively simple) digital components (barcode readers, teller machines, hand-held phones)

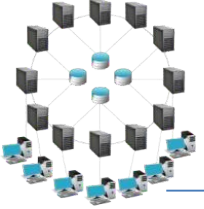
✓ Computers providing independent user interfaces for specific services

✓ Computers providing an integrated user interface for related services (compound documents)

Client-Server Model

Application Layering



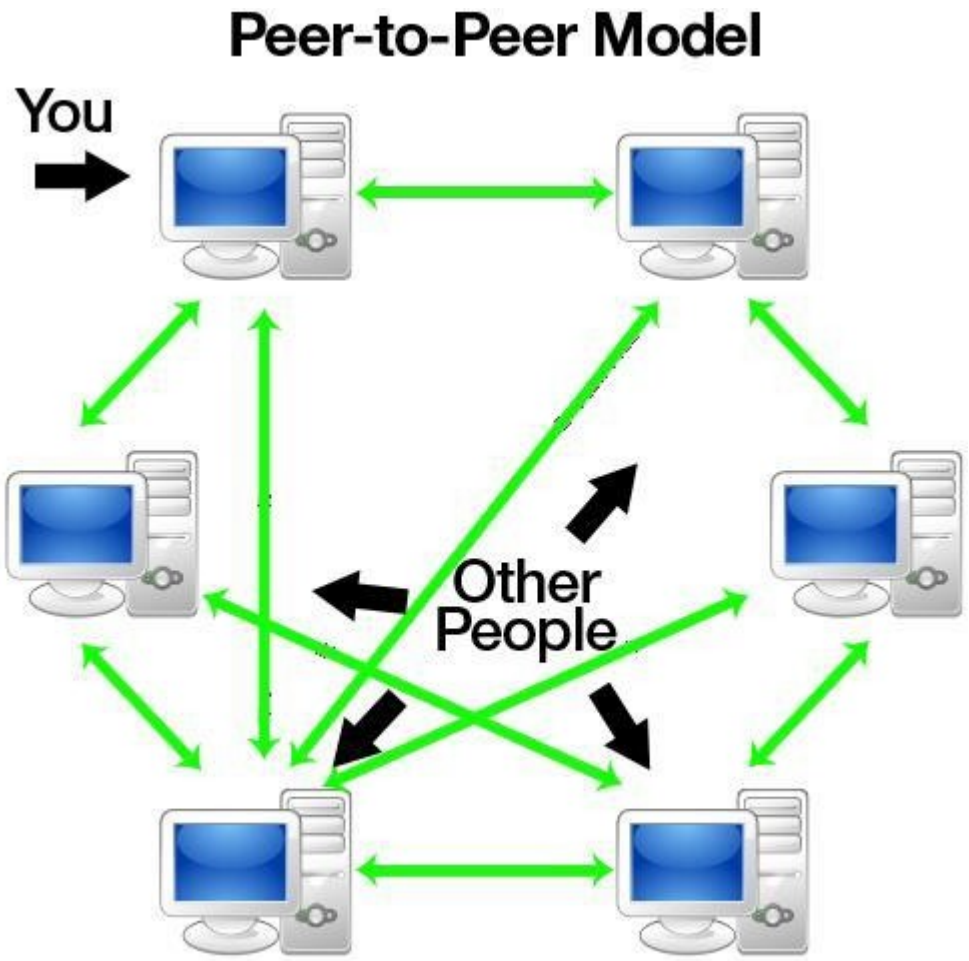


P2P Model

- **Peer-to-peer** (**P2P**) computing or networking is a distributed application architecture that partitions tasks or workloads between peers.
- Peers are equally privileged, equipotent participants in the application. They are said to form a peer-to-peer network of nodes.
- Peers make a portion of their resources, such as processing power, disk storage or network bandwidth, directly available to other network participants, without the need for central coordination by servers or stable hosts

P2P Model

Peer to Peer DS

- 

End



CHAPTER 1 : INTRODUCTIONS TO SYSTEM DESTRIBUTION

SYSTEMS INTEGRATION

Assoc.Prof. JAMES RWIGEMA

Functional Silos

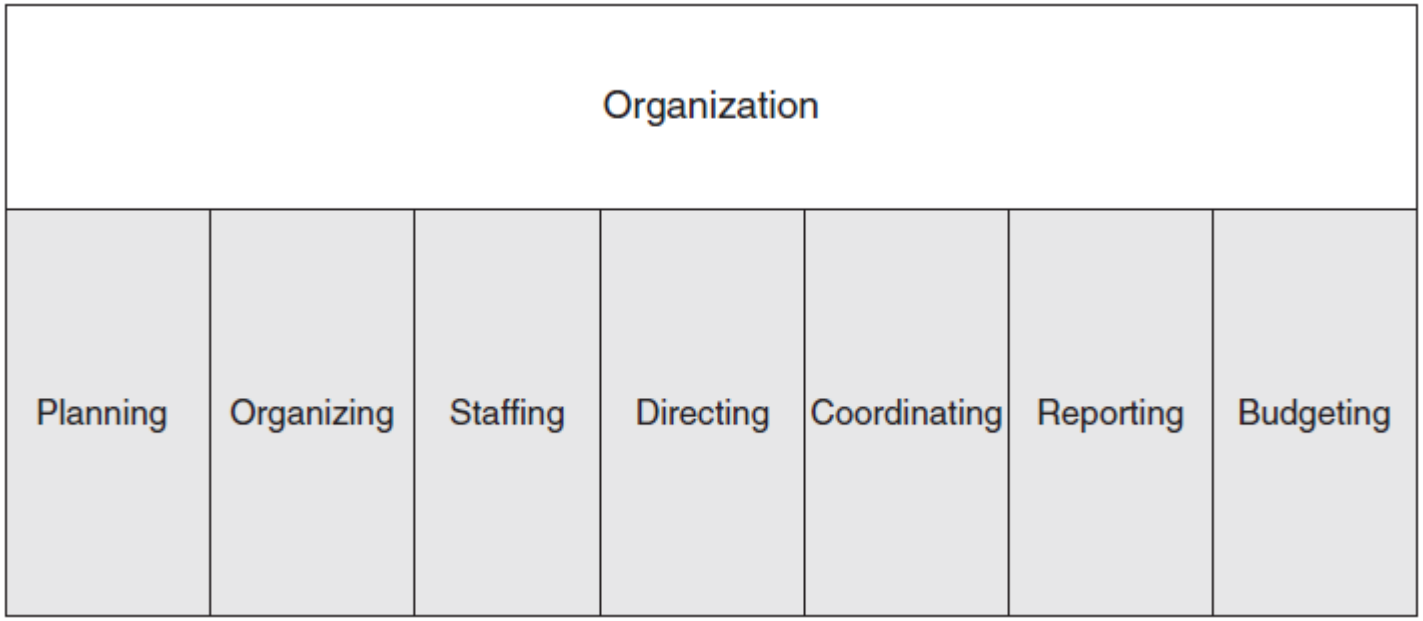
- Silos are basically compartmentalized operating units isolated from their environment.

Horizontal Silos

- The POSDCORB (Planning, Organizing, Staffing, Directing, Coordinating, Reporting and Budgeting) categorization by Luther Gulick led to a set of formal organization functions such as control, management, supervision, and administration starting in late 1930s.

- Classification of organizations into departments like Accounting and Human Resources, reflects the breaking of complex tasks into smaller manageable tasks that could be assigned to a group of people who could then be held responsible.

Figure 2-1 Functional Model of Organization (POSDCORB)



Functional Silos (Cont'd)

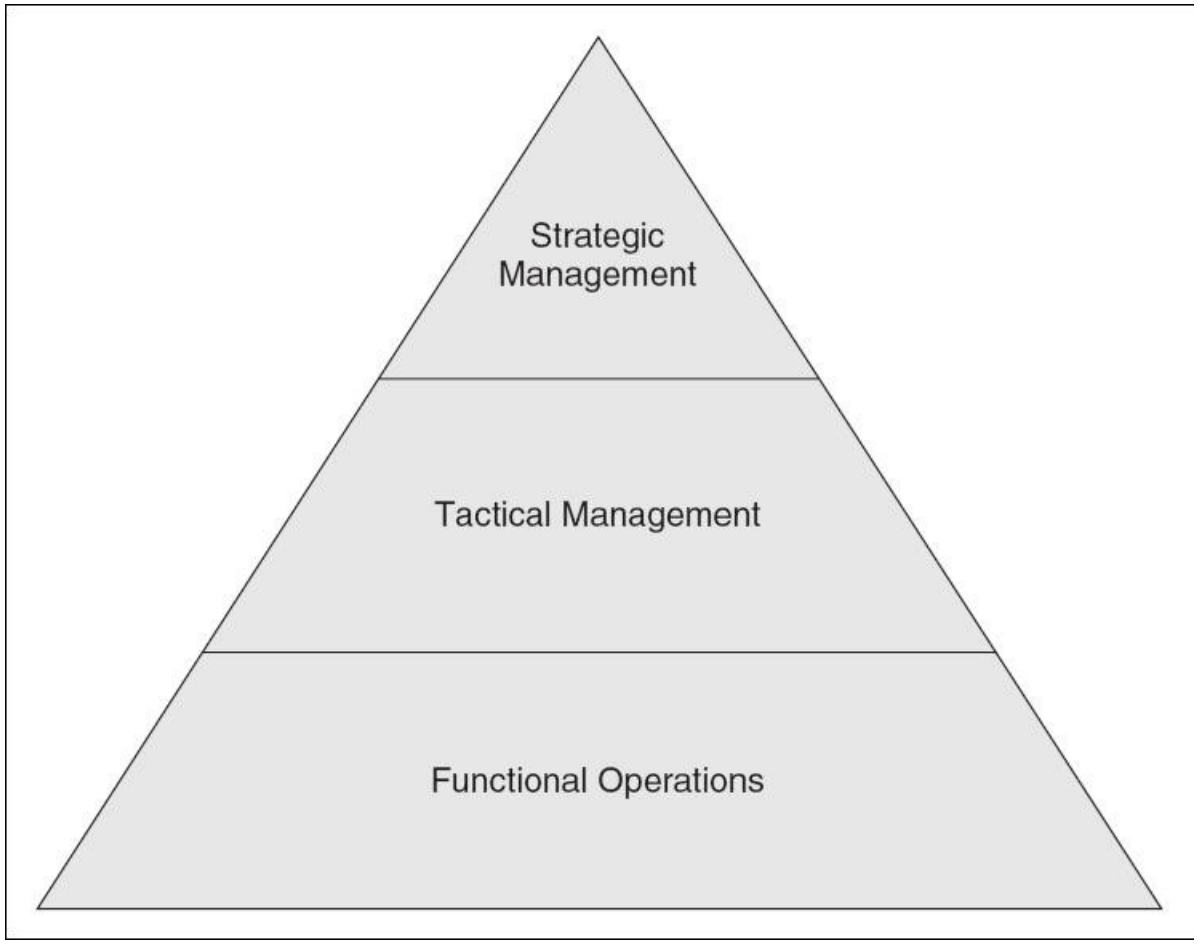
Vertical Silos

- Organizations also divided roles in hierarchical layers from strategic planning to management control and operation control.

- CEOs and Presidents plan long-term strategy, midlevel management focuses on tactical issues and on the execution of organizational policy whereas the lower-level management task is to focus on the day-to-day operations of the company.

- As organizations get big and complex they tend to break functions into smaller units and assign staff the responsibility for these activities allowing them to manage complexity as well as specialize in activities that enhance productivity and efficiency.

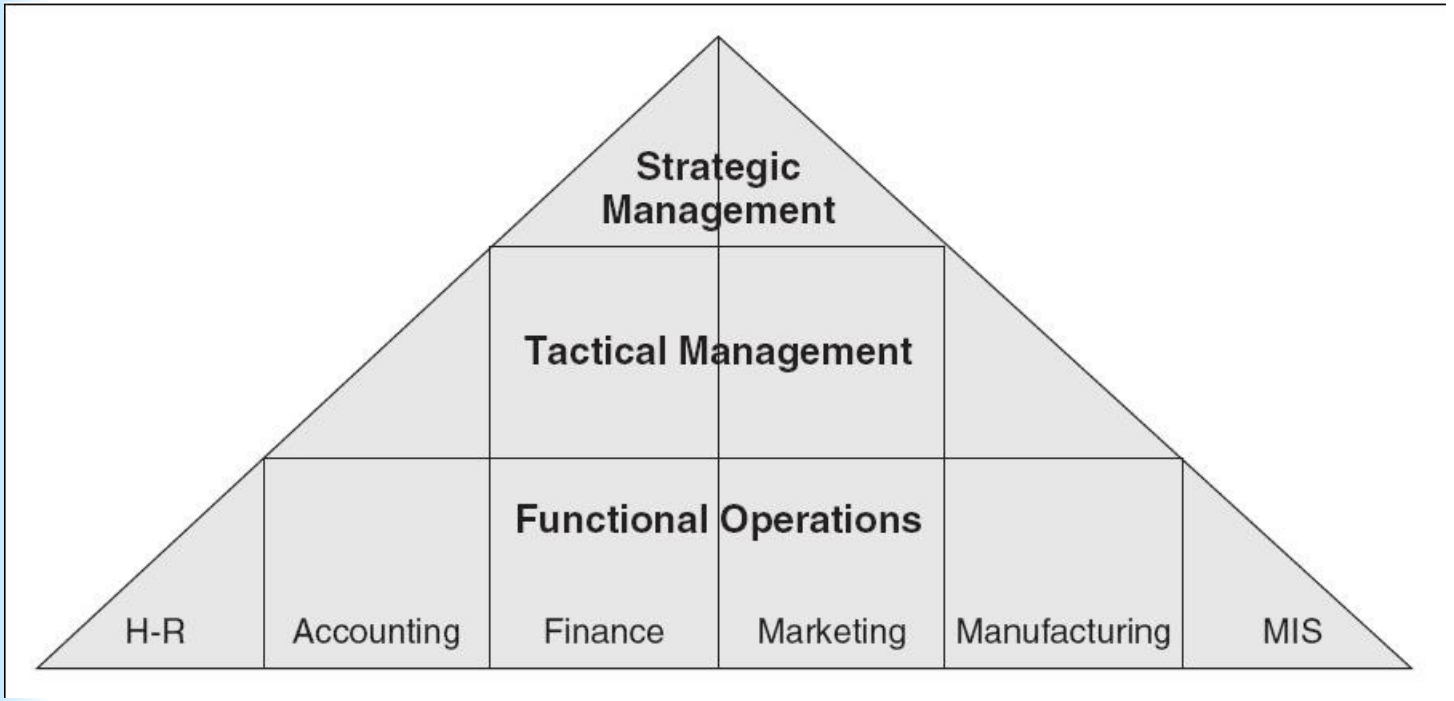
Figure 2-2 Hierarchical Model of Organization



Business Process and Silos

- The problem of functional silos gave birth to business process re-engineering (BPR).
- The cross-functional business process can involve people and resources from various functional departments working together, sharing information at any level of the organization.
- The cross-functional organizational structure breaks the functional silos by opening up the informational flows from one department to another.

Figure 2-3 Matrix Structure of Organization



Evolution of Information Systems in Organizations

- The functions of an organization (e.g., sales, manufacturing, and HR) are important as they provide a structure by which an organization functions smoothly.

- A silo information system is inefficient, inaccurate, and expensive.
  - The system creates bottlenecks for everyone and information is not available in real-time.

- The evolution of IS suggests that its role has generally been to support evolving information needs of the organization.

IS Architectures

- Rapid advances in computer and networking technologies and changing organizational dynamics, drive the emergence of new information system models.

- Web-based systems today use a distributed architecture which allows the sharing of applications and data resources between the client and the server computers.

- In this configuration, personal computers are connected via a network to a Web server that provides a window to an application and database server, which could be a mainframe or another type of computer.

IS Functionalization

- In addition to serving the different management levels, IS also supports major business functions, such as manufacturing, marketing, accounting, finance, and HR.
- Each functional area has different information needs and report requirements.
- Each functional area in an organization also has multiple levels of management, each requiring different levels of analysis and details of information.

Figure 2-5 Destributed Systems Architectures

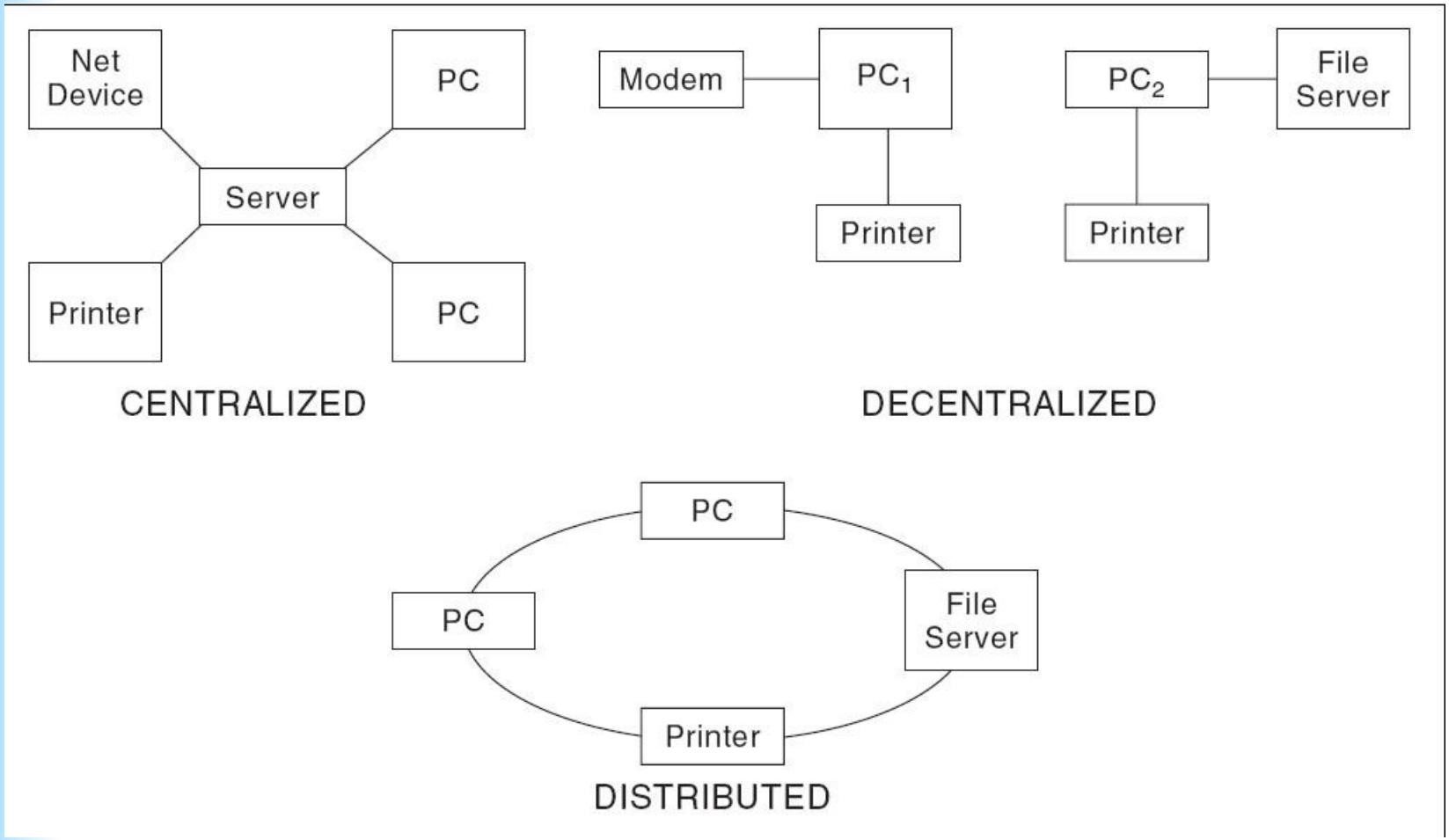
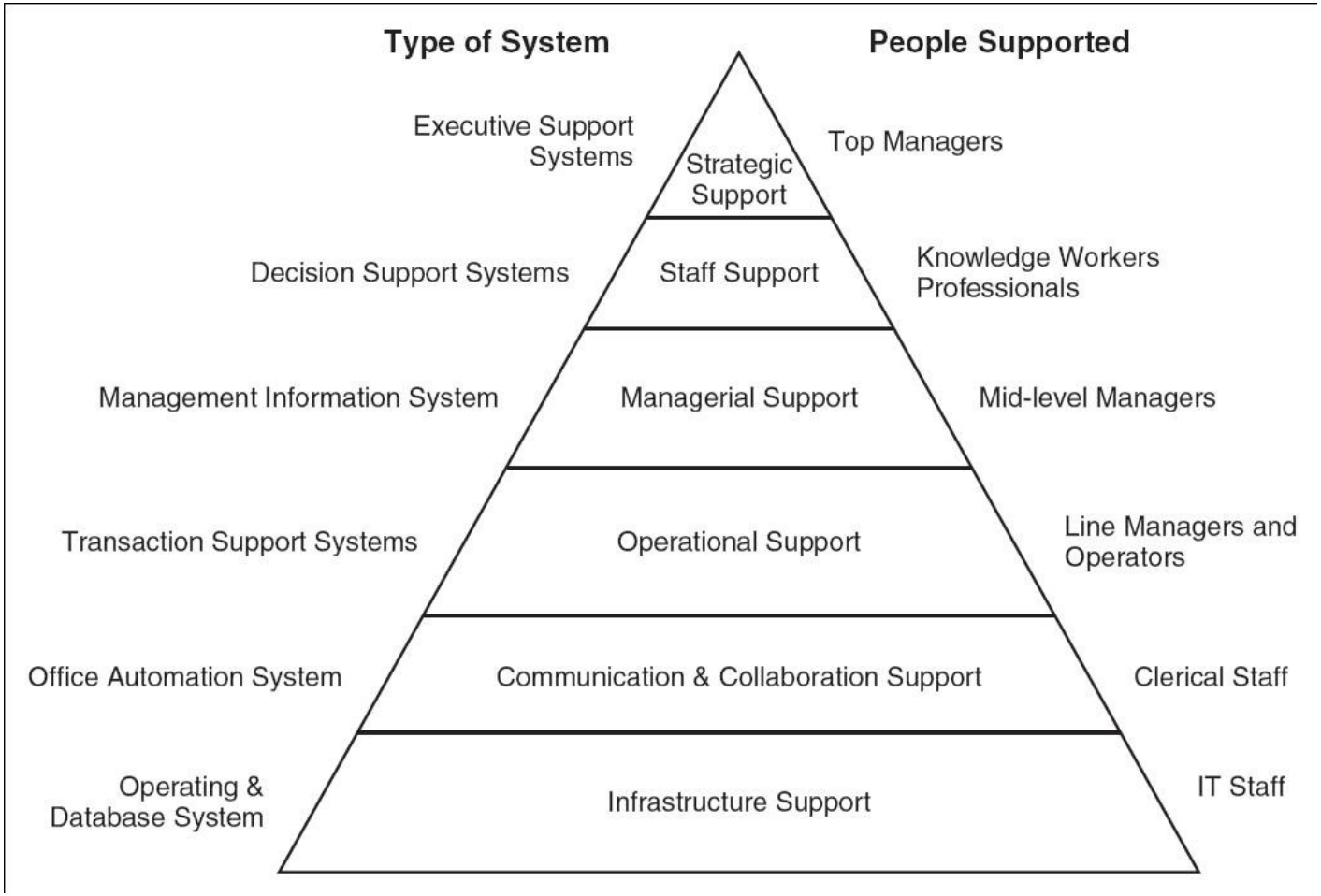


Figure 2-6 IS as Categorized by Functional and Hierarchical Models



Systems Integration

Logical

- Develop information systems that allow organizations to share data with all of its stakeholders based on need and authorization.
- Management needs to change organizational structures, processes, and employee roles and responsibilities.

Physical

- Provide seamless connectivity between heterogeneous systems.
- Business process reengineering involves changing the mindset of the employees in the organization, encouraging and enabling them to do their tasks in a new way.

Steps in Integrating Systems

Step 1	Resource categorization	Instituting IT support for an integrated systems environment is necessary to avoid support and maintenance problems with the integrated system.
Step 2	Compliance and standards	Develop a single sign-on policy because all employees and external partners will need access to an integrated system from anywhere, anytime.
Step 3	Legacy systems support	Develop a policy in support of older legacy applications.
Step 4	Middleware tools	Middleware tools are essential for integration in the short term if existing applications must be used by the organization.

Steps in Integrating Systems (Cont'd)

Step 5	Authentication and authorization policies	Single sign-on policy for application and data access because all employees and external partners will need access to an integrated system from anywhere, anytime.
Step 6	Centralized IT services and support	The IT staff needs to be able to support all applications and platforms with a centralized IT help desk support.
Step 7	Back-up, recovery, and security	A good back-up and recovery system is essential if there is a system failure or a major disaster.
Step 8	Hardware and software standardization	Develop organization standards and policy on acquisition of new hardware and software which is aligned with organization IT strategy.

Benefits and Limitations of Systems Integration

Benefits	Limitations
Increased Revenue and Growth	High Initial Set-up Costs
Leveling the Competitive Environment	Power and Interdepartmental Conflicts (due to the sharing of information)
Enhanced Information Visibility	Long-term and Intangible ROI (Usually several years)
Increased Standardization	Creativity Limitations (Restricts Creativity and Independence)

ERP and Systems Integration

- Enterprise Resource Planning (ERP) systems are integrated, multi-module application software packages designed to serve and support several business functions across an organization.

- ERP systems are typically commercial software packages that facilitate collection and integration of information related to various areas of an organization.

- ERP systems enable the organization to standardize and improve its business processes to implement best practices for its industry.

ERP's Role in Logical Integration

- ERP systems require organizations to focus on business process rather than on functions.
- ERP systems come with built-in processes for a wide variety of common business functions.
- An ERP system implements best practices via specific built-in steps for processing a customer order in terms of:
  - order entry.
  - routing through departments.
  - communication of output to various parties.

ERP's Role in Physical Integration

- Before installing the ERP system, an organization may have to upgrade or install middleware or get rid of their legacy system's hardware and software.

- Integration is also required at the Data level, Client level, and at the Application level.

- A good ERP implementation improves operational efficiency with better business processes that focuses on organizational goals rather than on individual departmental goals.

- Improved efficiency with a paperless flow and electronic data interchange (EDI) or business-to-business (B2B) commerce environment with partners.

Implications for Management

- **Silos do not work.**
  - Most organizations lose out in the long-term when information is not shared in real time across the functional boundaries within the company.
- **System integration has many hidden benefits.**
  - Allows decision making to be cascaded to all departments
  - Allows employees at lower-levels to make better decisions while interacting with clients or partners.
- **System integration has many challenges.**
  - Replacing old hardware and software
  - Working with IT consultants
  - Human challenges, such as impact on IT staff, department heads losing control of data, and rumors of layoffs

Implications for Management (Cont'd)

- **Systems integration raises many new ethical issues.**
  - Possibility of some employees exploiting information for personal advantage and illegal access of information.

- Remedies can consist of:
  - Develop policies on ethical usage of information.
  - Install proper security software and hardware (like firewalls).
  - Allocate resources for training and education on accessing information.

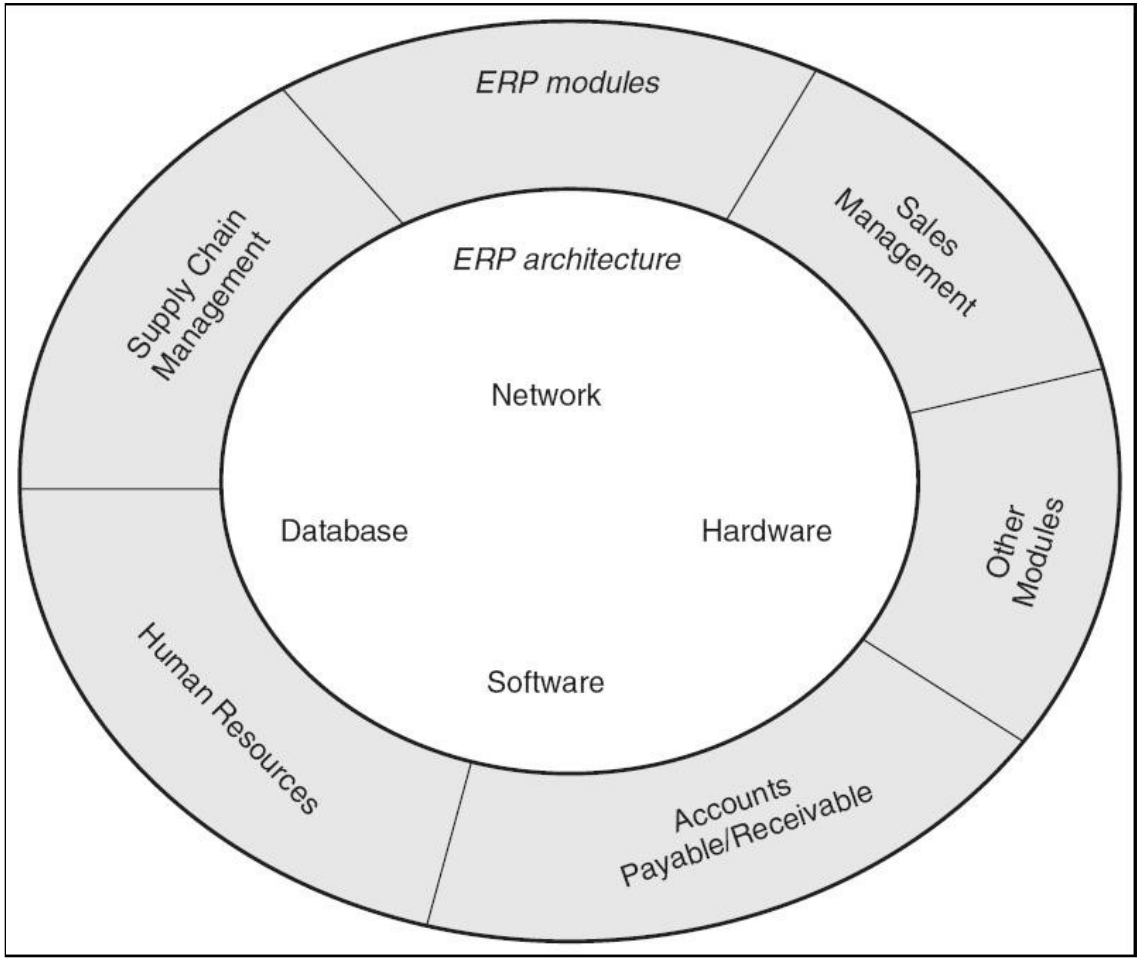
CHAPTER 2

DISTRUBTED SYSTEMS ARCHITECTURE FOR ENTERPRISES

Components of the Enterprise Systems Architecture

- Functional
  - Defines the *ERP modules that support the various business* functions of the organization. Examples include:
    - Accounting
    - Human Resources
    - Procurement
    - Fulfillment
    - Etc.
- System
  - Defines the *ERP architecture through* the physical components of hardware, software, and networking angle.

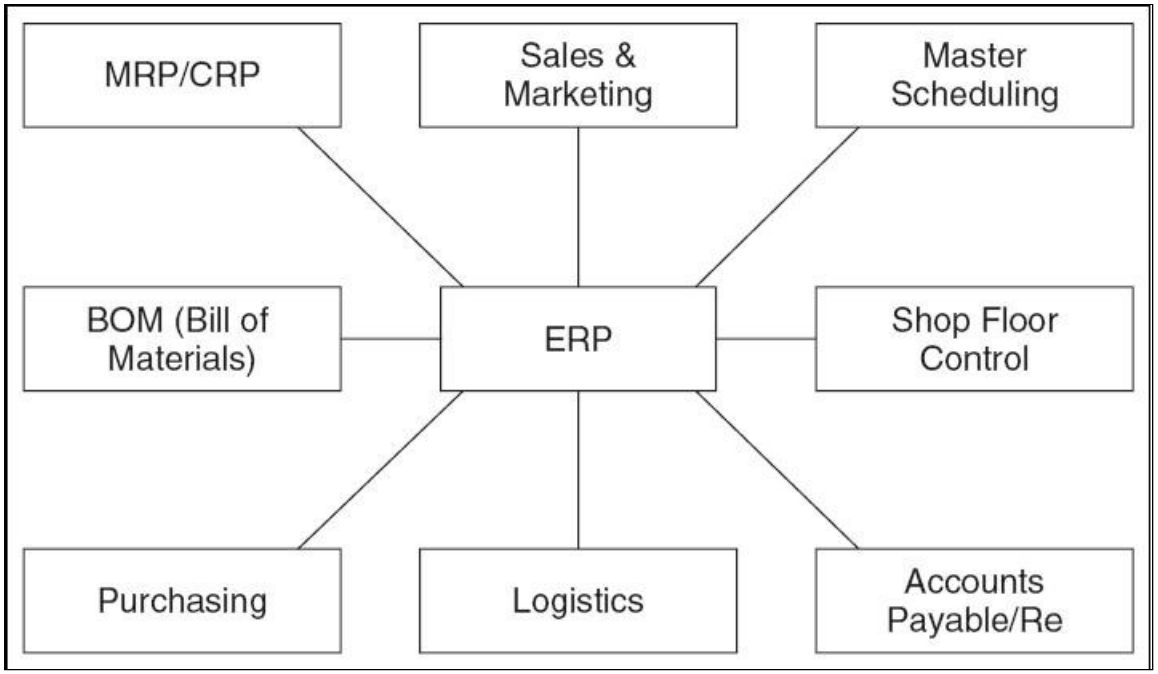
Figure 3-1 Enterprise Systems Architecture (ESA) Model



ERP Modules

- The key role of an ERP system is to provide support for such business functions as accounting, sales, inventory control, and production.
- ERP vendors, including SAP, Oracle, and Microsoft, etc. provide modules that support the major functional areas of a business.
- The ERP software embeds best business practices that implement the organization’s policy and procedure via business rules.

Figure 3-2 Typical ERP Modules



ERP Modules From Three Vendors

Function	SAP	Oracle/ PeopleSoft	Microsoft Dynamics
Sales	Sales and Distribution, Sales Opportunity	Marketing and Sales, Supply Chain Management	Retail POS, Field Service Management
Procurement	Purchasing, Supplier Relationship Management	Procurement and Supplier Relationship Management	Supply Chain Management
Production	MRP, Product Life Cycle Management	Manufacturing	Manufacturing
Accounting	Financial Accounting	Financial Management	Financial Management

ERP Modules From Three Vendors (Cont'd)

Function	SAP	Oracle/ PeopleSoft	Microsoft Dynamics
Distribution	Warehouse Management	Supply Chain Management	Distribution Management
Customer Service	CRM	CRM	CRM
Corporate Performance & Governance	Governance, Risk, and Compliance Management	Corporate Performance Management	Analytics
Human Resources	Human Capital Management	Human Capital Management	HR Management
Miscellaneous	Banking	Campus Solutions	E-commerce, Portals

Overview of Modules

- **Production**
  - Helps in the planning and optimizing of the manufacturing capacity, parts, components, and material resources using historical production data and sales forecasting.
- **Purchasing**
  - Streamlines the procurement process of required raw materials and other supplies.
- **Inventory Management**
  - Facilitates the processes of maintaining the appropriate level of stock in a warehouse.

Overview of Modules (Cont'd)

- **Sales and Marketing**
  - Implements functions of order placement, order scheduling, shipping, and invoicing.
- **Finance**
  - Can gather financial data from various functional departments and generate valuable financial reports.
- **Human Resource**
  - Streamlines the management of human resources and human capitals.
- **Miscellaneous Modules**
  - Nontraditional modules such as business intelligence, self-service, project management, and e-commerce.

Benefits of Key ERP Modules

- **Self Services**
  - Flexible support for employees' business functions.
  - Simplified access to relevant information.

- **Performance Management**
  - Delivery of real-time, personalized measurements and metrics.
  - Provides executives with access to such information as business statistics and key performance measurements.

- **Financials**
  - Ensure compliance and predictability of business performance.
  - Gain deeper financial insight and control across the enterprise.
  - Automate accounting and financial SCM.
  - Rigorous support for financial reporting—SOX Act.

Benefits of Key ERP Modules (Cont'd)

- **HR Management**
  - Attract the right people, develop and leverage talents, align efforts with corporate objectives, and retain top performers.
  - Increase efficiency and help ensure compliance with changing global and local regulations by using standardized and automated workforce processes.
  - Enable creation of project teams based on skills and availability, monitor progress on projects, track time, and analyze results.
  - Manage human capital investments by analyzing business outcomes, workforce trends and demographics, and workforce planning.

Benefits of Key ERP Modules (Cont'd)

- **Procurement and Logistics Execution**
  - Sustain cost savings for all spending categories by automating such routine tasks as converting requisitions into purchase orders and by allowing employees to use electronic catalogs to order products and services.
  - Reduce costs through process automation, integration of suppliers, and better collaboration.
  - Improve resource utilization with support for cross-docking processes and data collection technologies. (RFID and bar codes).
  - Enhance productivity of incoming and outgoing physical goods movements.
  - Reduce transportation costs through better consolidation and collaboration.

Benefits of Key ERP Modules (Cont'd)

- **Product Development and Production**
  - Shorten time to market.
  - Deliver higher quality products and ensure timely delivery.
  - Real time visibility and transparency (availability check).

- **Sales and Service**
  - Higher number of sales orders processed and reduction in administrative costs.
  - Easy access to accurate, timely customer information.
  - Cost-effective mobile access for field employees.
  - Reduce travel costs by using online functions.
  - Adhere to environmental, health, and safety reporting requirements.
  - Improve the management of incentives and commissions.
  - Realize more effective real estate management.

ERP Architecture

- ERP system architecture is organized in layers or tiers to manage system complexity in order to provide scalability and flexibility.

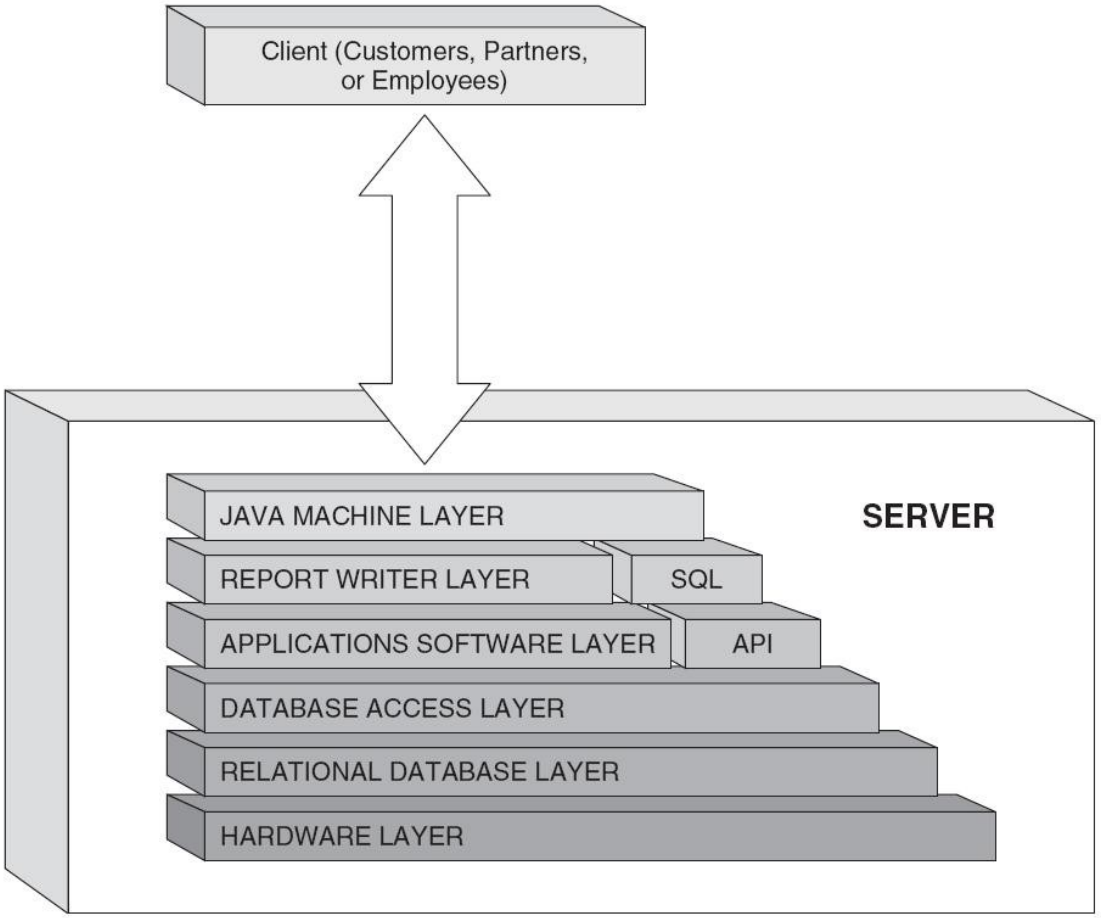
- Three-layer architecture (*the most reliable, flexible, and scalable architecture*) is the most prevalent today and includes:
  - Web Servers.
  - Application Servers.
  - Database Servers.

Layered Architecture Example (Info.Net)

- The Layered ERP architecture generalizes the functional layers to allow it to change with newer technologies.

- A Web-based user interface is provided.
  - Users can access the application via the Internet.
  - The PC needs to be capable of running a Java-enabled Web browser.
  - The PC is connected to both Intranet and Internet to be able to use one of Info.Net's servers.
  - The user interacts with the Java Virtual Machine™ Interface layer to establish a secure connection via a secure socket layer (SSL) connection.
  - The user is then communicating with the server through the applications software layer (ASL).

Figure 3-3 Example of Info.Net Architecture



Infrastructure Requirements

- Traditional networks require upgrading prior to the deployment of ERP systems and must be a component of the overall budget.
- A high-availability network is a requirement for a fully functioning ERP system, one that grows with the user population and supports continued expansion and integration of a supply chain.
- Integration with partner and customer systems allows “a company to manage important parts of the business such as order tracking, inventory management etc.”
- Online analytical processing (OLAP) provides the ability to access, present, and analyze data across dimensions.

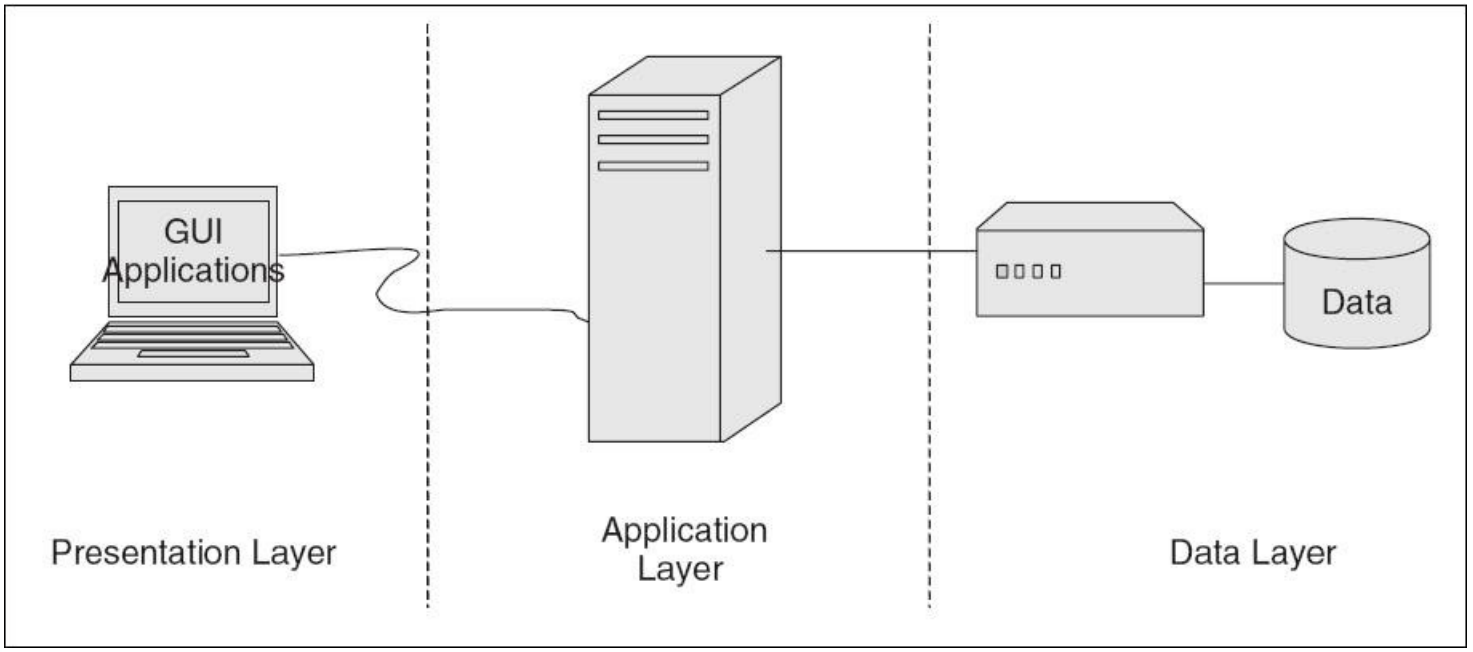
Three-Tier Architecture

- Most of the current ERP implementations follow a *three-tiered architecture, which consists of a Web tier, an application tier, and a data tier.*
- Benefits
  - Scalability - Easier to add, change, and remove applications.
  - Reliability - Implementing multiple levels of redundancy.
  - Flexibility - Flexibility in partitioning is very simple.
  - Maintainability - Support and maintenance costs are less on one server.
  - Reusability - Easier to implement reusable components.
  - Security - IT staff has more control system to provide higher security.
- Limitations
  - Can be very expensive and complex.

Tiers

- The Web Tier
  - Web-based portal allows users the ability to access and analyze information through their Web browser.
- The Application Tier
  - Consists of a Web browser and reporting tool where business processes and end-users interact with the system.
  - It shields the business users from the inner workings of an ERP system, but still provides the information relevant to their job and business process.
- The Data Tier
  - Focus is on structure of all organizational data and its relationships with both internal and external systems.

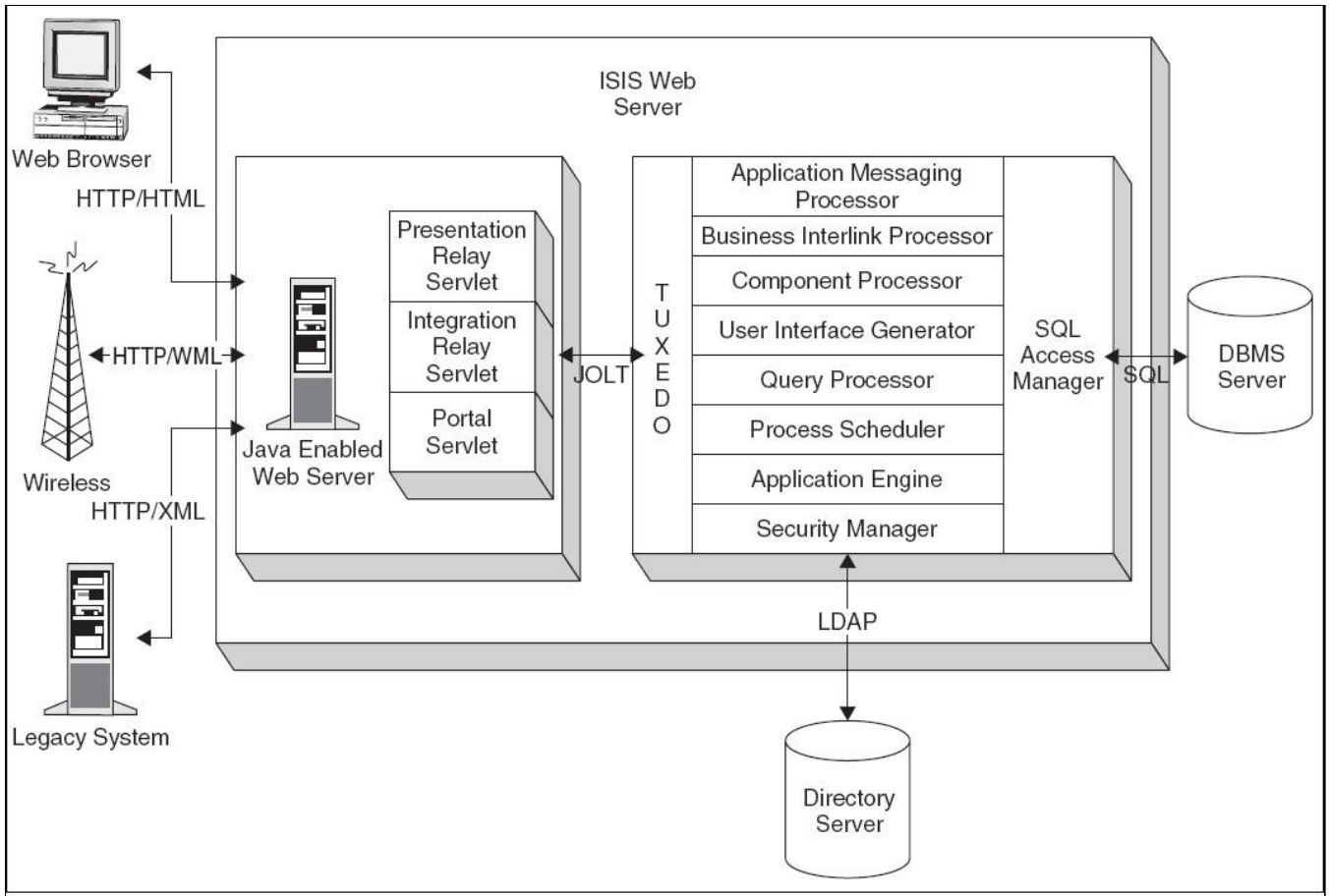
Figure 3-4 A Three-Tier ERP Architecture



Web Services Architectures

- Web-based architecture often described as a fourth tier where the Web tier is split into Web Services tier and Web Browser tier.
- The ERP systems focus on the Internet to provide a powerful new functionality for Internet-based access and integration.
- This functionality is primarily supported through the following Internet access technologies:
  - Web Server.
  - ERP Portal.
  - Back-end Server Integration.
  - Browse Plug-ins or Applets.

Figure 3`-5 Example of PeopleSoft's Server-Centric Internet Architecture



Adapted from: PeopleSoft Application Server Architecture (www.oracle.com)

Benefits and Drawbacks

- Benefits
  - Large numbers of end-users have access to ERP applications over the Web.
  - Easily integrate ERP applications with existing systems.
  - Server-centric—No complex, expensive client software installation.
  - The server-centric architecture enables secure end-user access to ERP application.
  - Client-centric—Architecture has better response time because user requests are mostly processed on the client's computer.
  - Web-based architectures also allow better system-to-system integration.

- Drawbacks
  - Client-centric architectures lack security.
  - Server centric is slower.

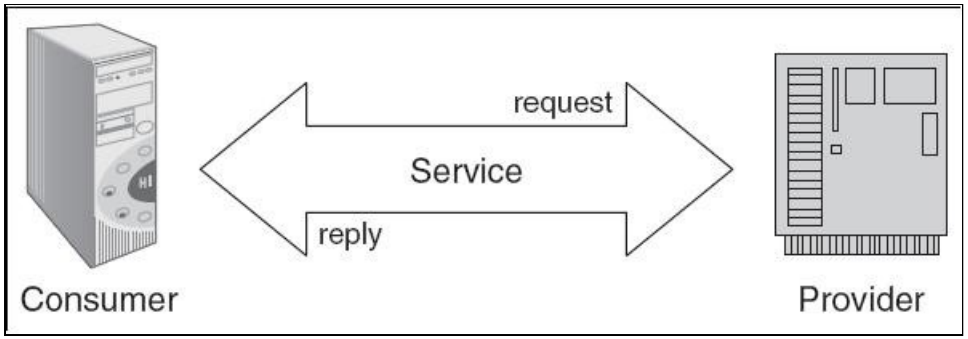
Service-Oriented Architectures

- Also known as object-oriented architectures for Web platforms.
- Breaks the business tier into smaller, distinct units of services, collectively supporting an ERP functional module.
- Allows message interaction between any service consumer and service provider.
- A consumer from a device using any operating system in any language can use this service.

Service-Oriented Architectures

- SOA is a software development model based on a contract between a consumer (client) and a provider (server) that specifies the following:
  - Functional description of the service.
  - Input requirements and output specifications.
  - Precondition environment state before service can be invoked.
  - Post condition environment state after service has been executed.
  - Error handling when there is a breakdown.

Figure 3-6 A SOA Architecture



Benefits of Service-Oriented Architectures

- Business-level software services across heterogeneous platforms.
- Complete location independence of business logic.
- Services can exist anywhere (any system, any network).
- Loose coupling across application services.
- Granular authentication and authorization support.
- Dynamic search and connectivity to other services.
- Enhances reliability of the architecture.
- Reduces hardware acquisition costs.

Benefits of Service-Oriented Architectures (Cont'd)

- Accelerates movement to standards-based server and application consolidation.
- Provides a data bridge between incompatible technologies.
- Provides the ability to build composite applications.
- Creates a self-healing infrastructure that reduces management costs.
- Provides truly real-time decision-making applications.
- Enables the compilation of a unified taxonomy of information across an enterprise.

Business Value Benefits of SOA

- Increases the ability to meet customer demands more quickly.
- Lower costs associated with the acquisition and maintenance of technology.
- Empowers the management of business functionality closer to the business units.
- Leverages existing investments in technology.
- Reduces reliance on expensive custom development.

Drawbacks of Service-Oriented Architectures

- SOA implementations are costly and time-consuming.
- Requires complex security firewalls in place to support communication between services.
- Performance can be inconsistent.
- Requires enterprise-level focus for implementation to be successful.
- Security system needs to be sophisticated.
- Costs can be high because services needs to be junked very often.

SOA and Web Services

- Web services basically are interfaces that allow different software application and components to be operated together.
- According to IT industry standards, different applications can interact with without communication problems.
- The only method of interaction by Web services is by receiving and sending messages.
- Services are developed using open standards such as WSDL (Web Services Description Language), UDDI (Universal Description, Discovery, and Integration), and SOAP (Simple Object Access Protocol).
- The protocols used in Web services are XML-based.

Enterprise Content Management and SOA

- Enterprise content management deals with enterprise software products that usually store, preserve, manage, and deliver content connected to business processes.
- Enterprise content management also about supporting business goals, not just managing content.
- Vendors have come to an understanding that content management takes advantage of technology and information assets across the business and is no longer application specific.

Cloud Architecture

- Cloud computing is basically a software service provided over the Internet, securely, by a service provider on a monthly or yearly lease.

- Companies leasing CC services save money by replacing their purchased software that requires a license fee per seat.

- Some cloud computing providers also let you build your own applications using their engines and then they would host those applications for you as part of the service.

Cloud Architecture

- The cloud computing platform provides a great alternative for organizations that do not want to:
  - Purchase, install, or maintain software applications.
  - Worry about security, privacy and legal issues associated with data storage.
- The cloud computing platform is risky for organizations as it forces them to rely on external vendors for reliability, security, and continuity of enterprise applications.

Benefits of Cloud Computing

- Pay for subscription, not for licenses and upgrades.
- Reduced capital and operating expenditures for IT equipment and support personnel.
- Accessed from everywhere, as long as you have an Internet connection.
- No need to install anything on the user's computer.
- Dynamic scalability available on demand.
- No maintenance fees for software or hardware.
- Promotes green computing environment as servers in cloud run on clean energy.
- Guaranteed reliability.

Drawbacks of Cloud Computing

- Data security.
- Vulnerability.
- Possible conflict of interest, if the company who stores your applications decides to create a similar application to what you created on their servers.
- Not suited for all highly competitive industries like biotech where intellectual property cannot be protected easily.

Implications for Management

- Enterprise architecture is an important technology for the long-term functioning of the organization.
- ERP architecture decisions are complex because their impact goes beyond systems and technology to people, organizational policy, and business processes.
- ERP architecture must be flexible to support a diverse set of hardware and software platforms.
- Management must learn how to filter out the hyped technologies that do not provide value to their organization.

CHAPTER 3

ENTERPRISE SYSTEMS ARCHITECTURE

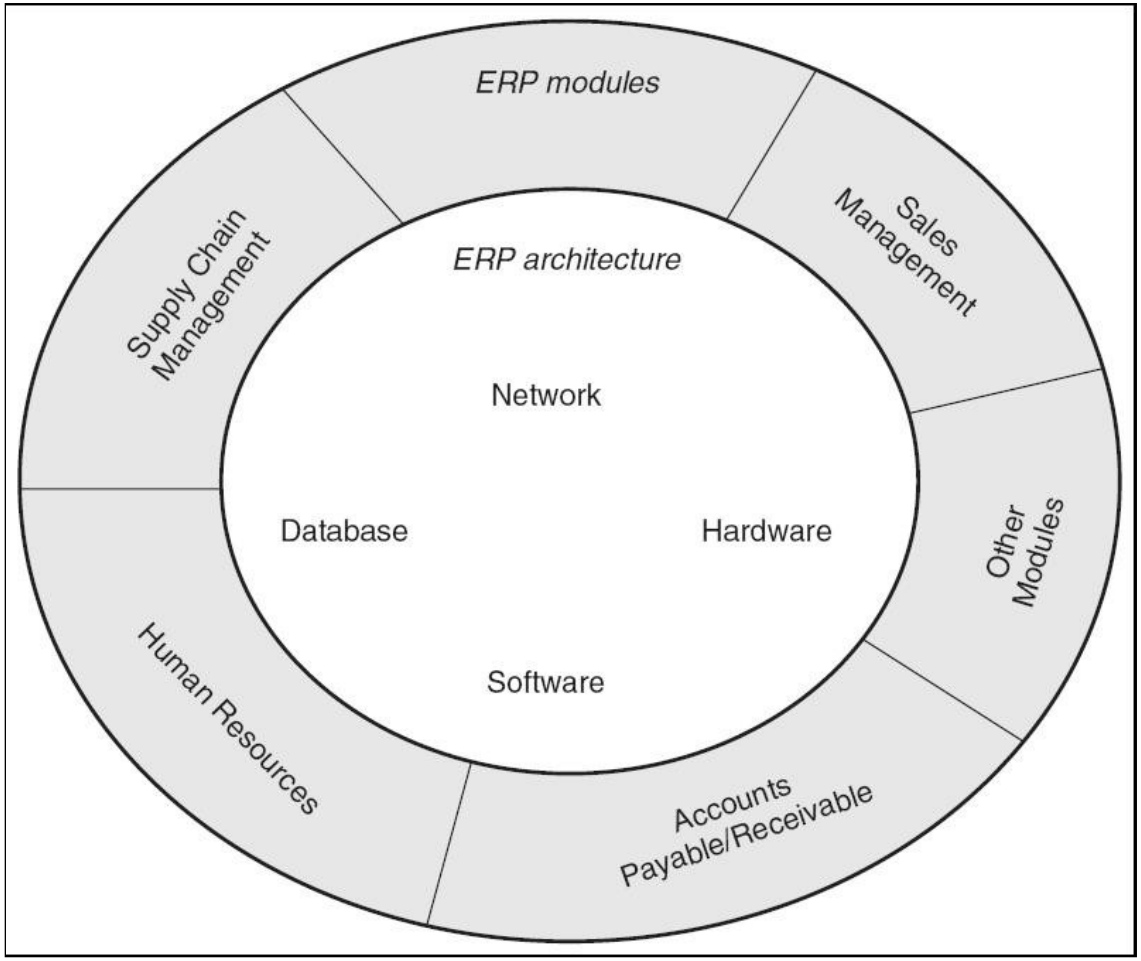
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Components of the Enterprise Systems Architecture

- Functional
  - Defines the *ERP modules that support the various business* functions of the organization. Examples include:
    - Accounting
    - Human Resources
    - Procurement
    - Fulfillment
    - Etc.
- System
  - Defines the *ERP architecture through* the physical components of hardware, software, and networking angle.

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Figure 3-1 Enterprise Systems Architecture (ESA) Model



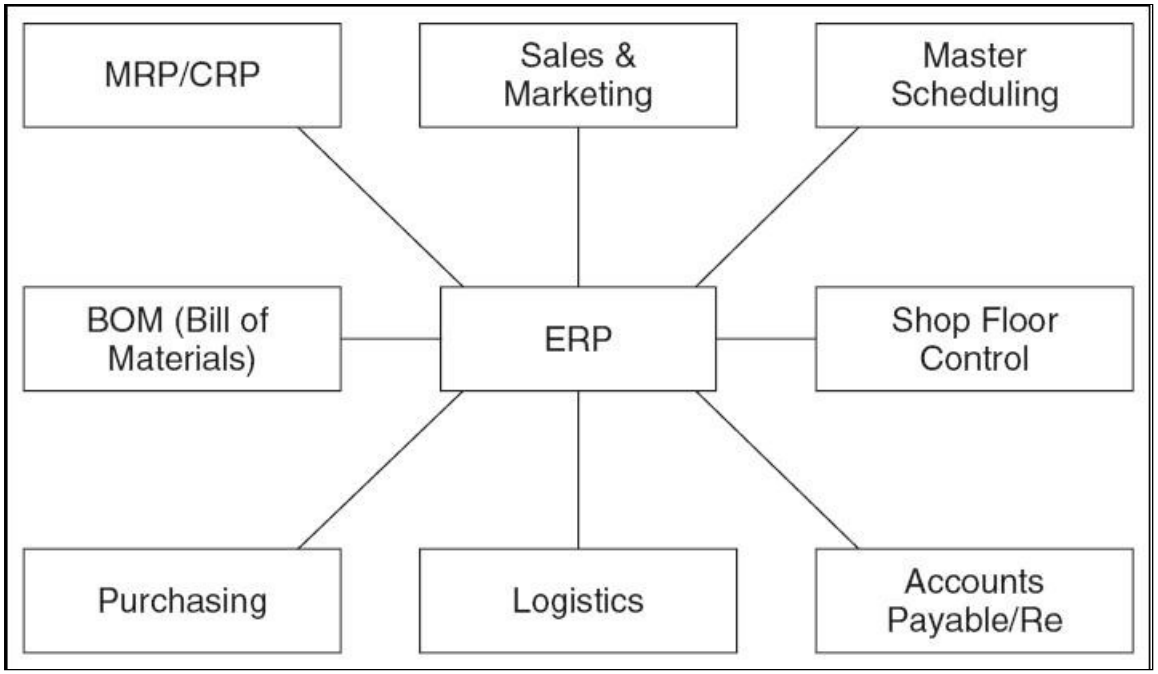
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ERP Modules

- The key role of an ERP system is to provide support for such business functions as accounting, sales, inventory control, and production.
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- The ERP software embeds best business practices that implement the organization’s policy and procedure via business rules.

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Figure 3-2 Typical ERP Modules



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ERP Modules From Three Vendors

Function	SAP	Oracle/ PeopleSoft	Microsoft Dynamics
Sales	Sales and Distribution, Sales Opportunity	Marketing and Sales, Supply Chain Management	Retail POS, Field Service Management
Procurement	Purchasing, Supplier Relationship Management	Procurement and Supplier Relationship Management	Supply Chain Management
Production	MRP, Product Life Cycle Management	Manufacturing	Manufacturing
Accounting	Financial Accounting	Financial Management	Financial Management

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ERP Modules From Three Vendors (Cont'd)

Function	SAP	Oracle/ PeopleSoft	Microsoft Dynamics
Distribution	Warehouse Management	Supply Chain Management	Distribution Management
Customer Service	CRM	CRM	CRM
Corporate Performance & Governance	Governance, Risk, and Compliance Management	Corporate Performance Management	Analytics
Human Resources	Human Capital Management	Human Capital Management	HR Management
Miscellaneous	Banking	Campus Solutions	E-commerce, Portals

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Overview of Modules

- **Production**
  - Helps in the planning and optimizing of the manufacturing capacity, parts, components, and material resources using historical production data and sales forecasting.
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  - Streamlines the procurement process of required raw materials and other supplies.
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  - Facilitates the processes of maintaining the appropriate level of stock in a warehouse.

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Overview of Modules (Cont'd)

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  - Can gather financial data from various functional departments and generate valuable financial reports.
- **Human Resource**
  - Streamlines the management of human resources and human capitals.
- **Miscellaneous Modules**
  - Nontraditional modules such as business intelligence, self-service, project management, and e-commerce.

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Benefits of Key ERP Modules

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  - Flexible support for employees' business functions.
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  - Delivery of real-time, personalized measurements and metrics.
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  - Attract the right people, develop and leverage talents, align efforts with corporate objectives, and retain top performers.
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Benefits of Key ERP Modules (Cont'd)

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Benefits of Key ERP Modules (Cont'd)

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  - Shorten time to market.
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  - Realize more effective real estate management.

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ERP Architecture

- ERP system architecture is organized in layers or tiers to manage system complexity in order to provide scalability and flexibility.

- Three-layer architecture (*the most reliable, flexible, and scalable architecture*) is the most prevalent today and includes:
  - Web Servers.
  - Application Servers.
  - Database Servers.

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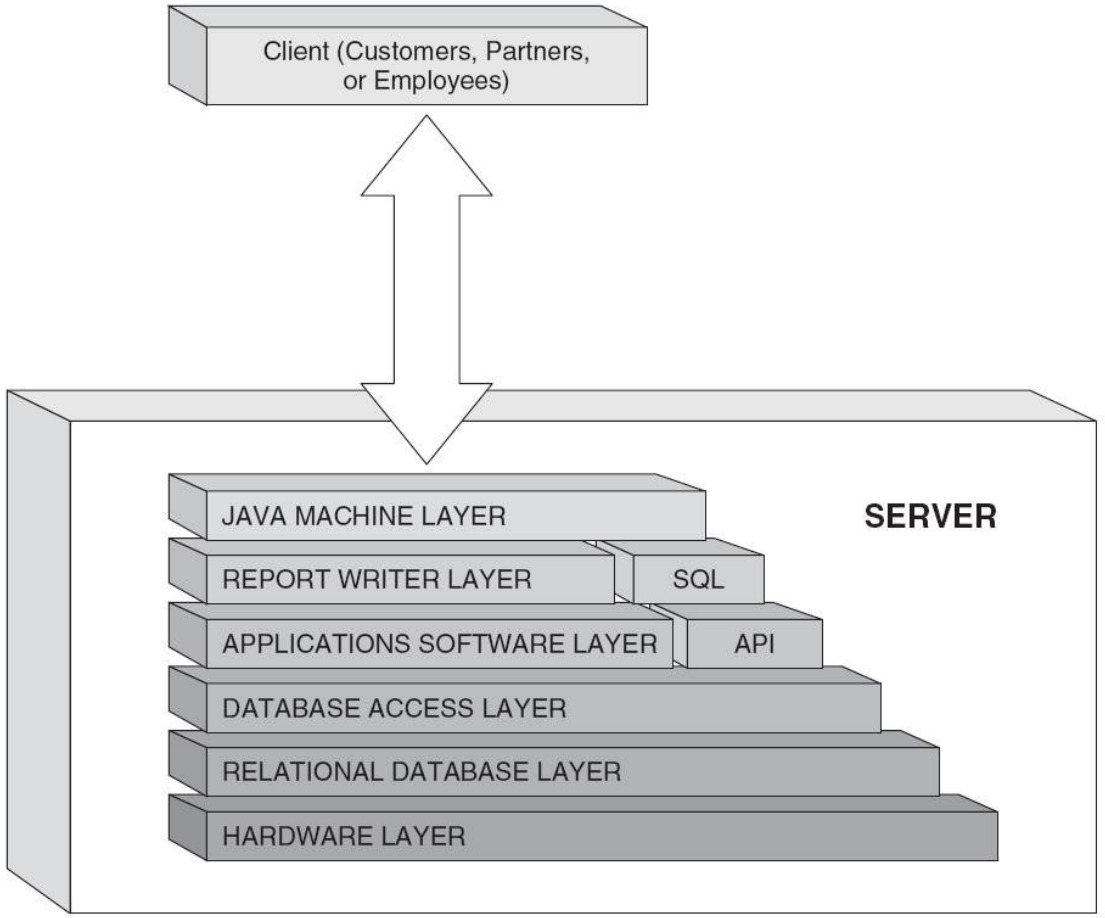
Layered Architecture Example (Info.Net)

- The Layered ERP architecture generalizes the functional layers to allow it to change with newer technologies.

- A Web-based user interface is provided.
  - Users can access the application via the Internet.
  - The PC needs to be capable of running a Java-enabled Web browser.
  - The PC is connected to both Intranet and Internet to be able to use one of Info.Net's servers.
  - The user interacts with the Java Virtual Machine™ Interface layer to establish a secure connection via a secure socket layer (SSL) connection.
  - The user is then communicating with the server through the applications software layer (ASL).

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Figure 3-3 Example of Info.Net Architecture



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Infrastructure Requirements

- Traditional networks require upgrading prior to the deployment of ERP systems and must be a component of the overall budget.
- A high-availability network is a requirement for a fully functioning ERP system, one that grows with the user population and supports continued expansion and integration of a supply chain.
- Integration with partner and customer systems allows “a company to manage important parts of the business such as order tracking, inventory management etc.”
- Online analytical processing (OLAP) provides the ability to access, present, and analyze data across dimensions.

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Three-Tier Architecture

- Most of the current ERP implementations follow a *three-tiered architecture, which consists of a Web tier, an application tier, and a data tier.*
- Benefits
  - Scalability - Easier to add, change, and remove applications.
  - Reliability - Implementing multiple levels of redundancy.
  - Flexibility - Flexibility in partitioning is very simple.
  - Maintainability - Support and maintenance costs are less on one server.
  - Reusability - Easier to implement reusable components.
  - Security - IT staff has more control system to provide higher security.
- Limitations
  - Can be very expensive and complex.

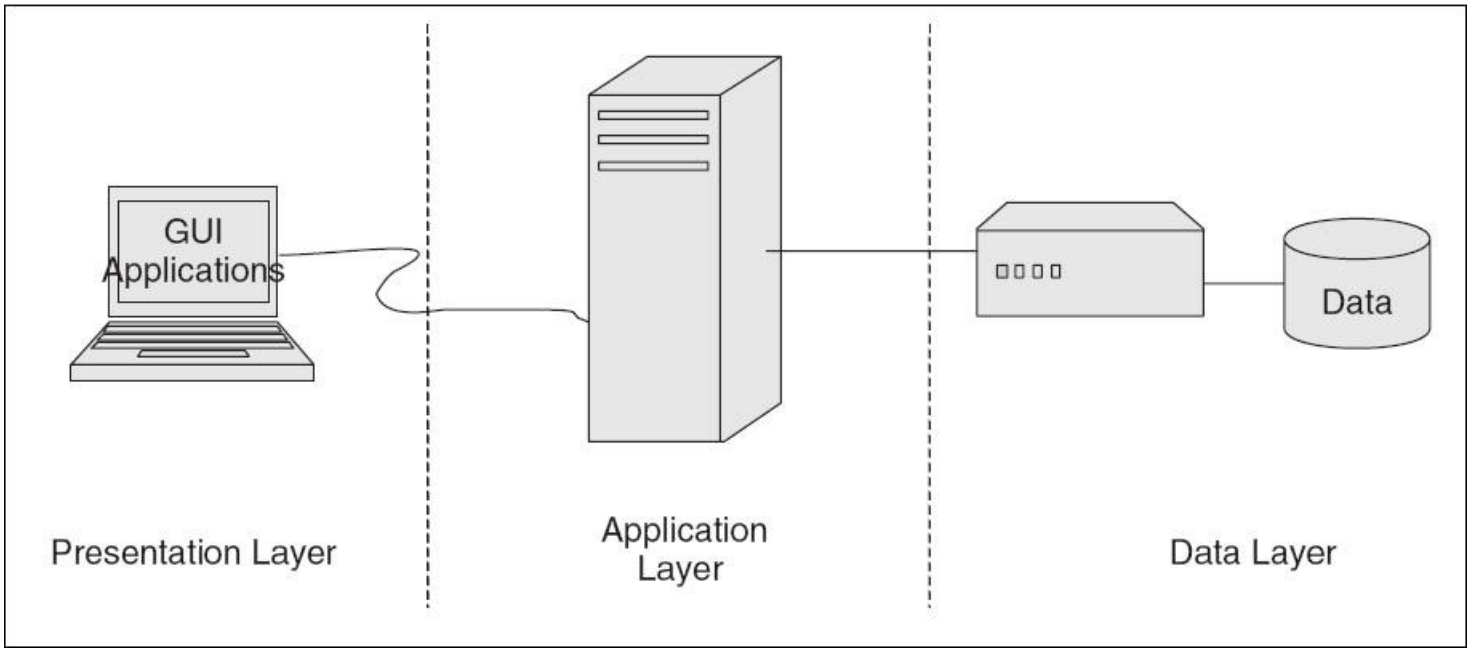
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Tiers

- The Web Tier
  - Web-based portal allows users the ability to access and analyze information through their Web browser.
- The Application Tier
  - Consists of a Web browser and reporting tool where business processes and end-users interact with the system.
  - It shields the business users from the inner workings of an ERP system, but still provides the information relevant to their job and business process.
- The Data Tier
  - Focus is on structure of all organizational data and its relationships with both internal and external systems.

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Figure 3-4 A Three-Tier ERP Architecture



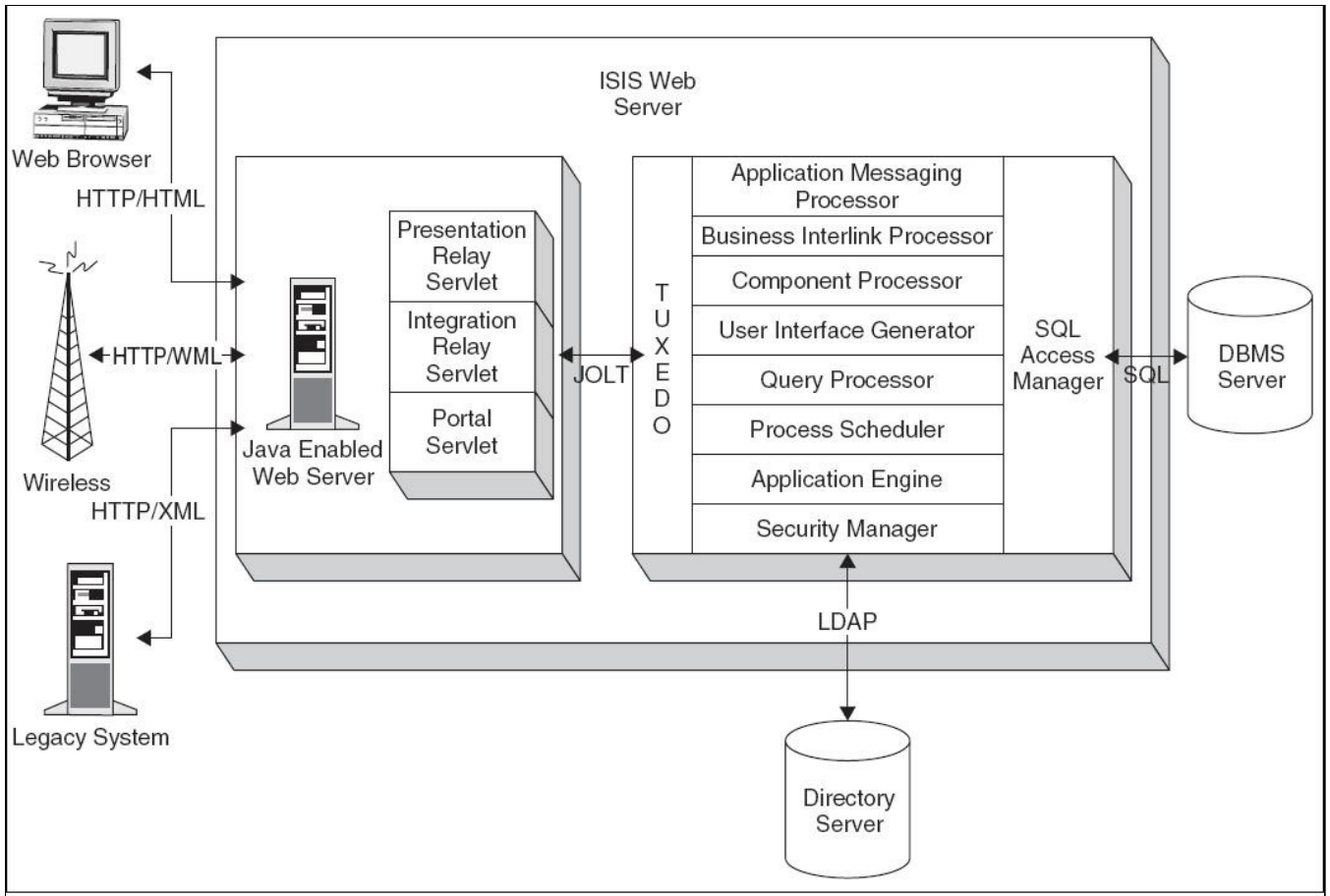
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Web Services Architectures

- Web-based architecture often described as a fourth tier where the Web tier is split into Web Services tier and Web Browser tier.
- The ERP systems focus on the Internet to provide a powerful new functionality for Internet-based access and integration.
- This functionality is primarily supported through the following Internet access technologies:
  - Web Server.
  - ERP Portal.
  - Back-end Server Integration.
  - Browse Plug-ins or Applets.

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Figure 3-5 Example of PeopleSoft's Server-Centric Internet Architecture



Adapted from: PeopleSoft Application Server Architecture (www.oracle.com)

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Benefits and Drawbacks

- Benefits
  - Large numbers of end-users have access to ERP applications over the Web.
  - Easily integrate ERP applications with existing systems.
  - Server-centric—No complex, expensive client software installation.
  - The server-centric architecture enables secure end-user access to ERP application.
  - Client-centric—Architecture has better response time because user requests are mostly processed on the client's computer.
  - Web-based architectures also allow better system-to-system integration.
- Drawbacks
  - Client-centric architectures lack security.
  - Server centric is slower.

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Service-Oriented Architectures

- Also known as object-oriented architectures for Web platforms.
- Breaks the business tier into smaller, distinct units of services, collectively supporting an ERP functional module.
- Allows message interaction between any service consumer and service provider.
- A consumer from a device using any operating system in any language can use this service.

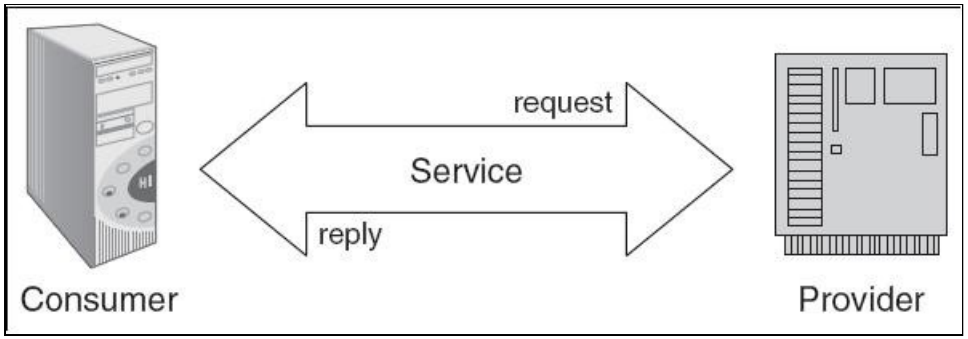
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Service-Oriented Architectures

- SOA is a software development model based on a contract between a consumer (client) and a provider (server) that specifies the following:
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  - Input requirements and output specifications.
  - Precondition environment state before service can be invoked.
  - Post condition environment state after service has been executed.
  - Error handling when there is a breakdown.

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Figure 3-6 A SOA Architecture



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Benefits of Service-Oriented Architectures

- Business-level software services across heterogeneous platforms.
- Complete location independence of business logic.
- Services can exist anywhere (any system, any network).
- Loose coupling across application services.
- Granular authentication and authorization support.
- Dynamic search and connectivity to other services.
- Enhances reliability of the architecture.
- Reduces hardware acquisition costs.

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Benefits of Service-Oriented Architectures (Cont'd)

- Accelerates movement to standards-based server and application consolidation.
- Provides a data bridge between incompatible technologies.
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- Performance can be inconsistent.
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- Security system needs to be sophisticated.
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- According to IT industry standards, different applications can interact with without communication problems.
- The only method of interaction by Web services is by receiving and sending messages.
- Services are developed using open standards such as WSDL (Web Services Description Language), UDDI (Universal Description, Discovery, and Integration), and SOAP (Simple Object Access Protocol).
- The protocols used in Web services are XML-based.

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Enterprise Content Management and SOA

- Enterprise content management deals with enterprise software products that usually store, preserve, manage, and deliver content connected to business processes.
- Enterprise content management also about supporting business goals, not just managing content.
- Vendors have come to an understanding that content management takes advantage of technology and information assets across the business and is no longer application specific.

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Cloud Architecture

- Cloud computing is basically a software service provided over the Internet, securely, by a service provider on a monthly or yearly lease.

- Companies leasing CC services save money by replacing their purchased software that requires a license fee per seat.

- Some cloud computing providers also let you build your own applications using their engines and then they would host those applications for you as part of the service.

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Cloud Architecture

- The cloud computing platform provides a great alternative for organizations that do not want to:
  - Purchase, install, or maintain software applications.
  - Worry about security, privacy and legal issues associated with data storage.
- The cloud computing platform is risky for organizations as it forces them to rely on external vendors for reliability, security, and continuity of enterprise applications.

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Benefits of Cloud Computing

- Pay for subscription, not for licenses and upgrades.
- Reduced capital and operating expenditures for IT equipment and support personnel.
- Accessed from everywhere, as long as you have an Internet connection.
- No need to install anything on the user's computer.
- Dynamic scalability available on demand.
- No maintenance fees for software or hardware.
- Promotes green computing environment as servers in cloud run on clean energy.
- Guaranteed reliability.

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Drawbacks of Cloud Computing

- Data security.
- Vulnerability.
- Possible conflict of interest, if the company who stores your applications decides to create a similar application to what you created on their servers.
- Not suited for all highly competitive industries like biotech where intellectual property cannot be protected easily.

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Implications for Management

- Enterprise architecture is an important technology for the long-term functioning of the organization.
- ERP architecture decisions are complex because their impact goes beyond systems and technology to people, organizational policy, and business processes.
- ERP architecture must be flexible to support a diverse set of hardware and software platforms.
- Management must learn how to filter out the hyped technologies that do not provide value to their organization.

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CHAPTER 3

What is Cloud Computing?



Prepared BY Assoc.Prof. James Rwigema Through “Claude AI”

OVERVIEW

What We'll Cover



01

What is Cloud Computing?



02

NIST Definition of Cloud Computing



03

Essential Characteristics of Cloud Computing



04

Cloud Deployment Models



05

Cloud Service Models



06

Business Drivers for Cloud Computing



07

Considerations for Cloud Adoption



08

What is Amazon Web Services?

KEY TERMS · 1 OF 4

Key Terminology: The Players of Cloud Computing



Data Center

A facility that houses and manages data: hardware (compute, storage, networks) plus software (applications, operating systems).



Cloud User

A person or organization requesting cloud resources and services (also called user, client, customer, or consumer).



Cloud Service Provider (CSP)

An organization that provides a cloud platform, infrastructure, applications, and services.



Multitenancy

An architecture where multiple customers (tenants) of a CSP share resources without the knowledge of one another.

KEY TERMS · 2 OF 4

Key Terminology: Business Words



Economies of Scale

The relationship between per-unit cost and production volume: more production spreads costs over more units, so per-unit cost falls.



Organizational Agility

The ability of an organization to quickly adapt to changes in the market or industry.



Vendor Lock-in

Reliance on proprietary software or technology that makes it hard and costly to switch to alternatives.



Barrier to Entry

An economics or business term for the obstacles a new company faces when entering a given market.

KEY TERMS · 3 OF 4

Key Terminology: Money Matters

CAPEX

Capital Expenses

Costs of fixed assets: their purchase, maintenance, and improvement.

IT example: land, computing equipment, software

OPEX

Operational Expenses

Costs of the day-to-day running of a business.

IT example: Internet costs, licensing fees

TCO

Total Cost of Ownership

The complete cost of an object or service over its lifetime, from purchase to disposal, direct and indirect.

Think: every cost, start to finish

KEY TERMS · 4 OF 4

Key Terminology: Scaling & Design

Cloud-native

Specifically designed for the cloud environment.

Vertical scaling

Adding resources such as memory and processing power to the same node.

Horizontal scaling

Adding more nodes to a distributed system.

Monolithic application

An application designed as a single unit.

Microservices-based application

An application broken into many loosely-coupled services that interact with each other.

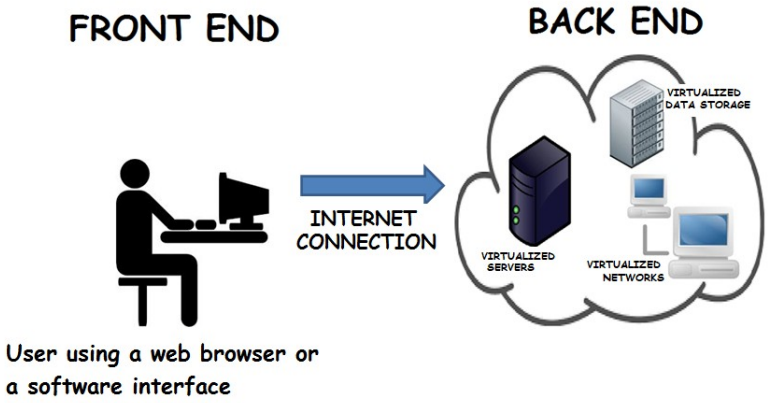
Serverless

Independent of the need to provision or manage servers.

CLOUD BASICS

What is Cloud Computing?

An on-demand delivery of servers, storage, databases, applications and other IT resources as services over the Internet.



Resources are typically hosted in large data centers by CSPs such as AWS, Microsoft Azure, Google, etc., in locations across the globe. Implementation details are abstracted (hidden) from the user.

CLOUD BASICS

Think of It Like Electricity



Source: AWS

Plug in on demand

A vast grid managed by experts: low cost and reliable.

Pay only for what you use

Metered, just like electric or gas utilities.

Use as much or as little

Consume as many or as few resources as you need.

CLOUD BASICS

The NIST Cloud Model

The US National Institute of Standards and Technology (NIST) ***Special Publication 800-145*** is a widely accepted standard, and the premise of our next discussion. It includes the definition of Cloud Computing and describes:

5

Essential characteristics

4

Deployment models

3

Service models

NIST Definition of Cloud Computing

> *Cloud computing is a model for enabling* ***ubiquitous, convenient, on-demand network access*** *to a* ***shared pool of configurable computing resources*** *(e.g., networks, servers, storage, applications and services) that can be* ***rapidly provisioned and released*** *with* ***minimal management effort*** *or service provider interaction.*

NIST



Common misconception

Cloud computing is not a technology. It is a business or service model that relies on enabling technologies such as virtualization and storage.

NIST · 1–3 OF 5

Essential Characteristics



1

On-demand Self Service

A user can provision resources as needed, automatically, without any human interaction with the service provider.



2

Broad Network Access

Resources are available via different client platforms such as phones, tablets, laptops, workstations, etc.



3

Resource Pooling

The provider creates pools of resources (servers, storage, etc.) to serve multiple consumers (multitenancy).

Essential Characteristics



4

Rapid Elasticity

Resources can be provisioned or released to match demand. For example, virtual machines can be added when demand grows and terminated when demand drops.



5

Measured Service

Resource usage is monitored, controlled and reported, giving transparency to both provider and consumer. Billing is based on actual consumption: a “pay as you go” model (similar to consuming electricity).

Good to know: other terms with a similar meaning to “pay as you go” are **pay-per-use**.

DEPLOYMENT MODELS

Cloud Deployment Models

A **deployment model** defines who the user of that cloud is and where the cloud infrastructure resides.



Public

Open to the general public



Private

Exclusive to a single organization



Hybrid

Public and private (or other) bridged together



Community

Shared by organizations with shared concerns

DEPLOYMENT MODELS

Public Cloud



Infrastructure is provisioned for the **general public** to use.



Exists on the premises of the CSP.



Consumers use services via the Internet and pay **metered usage charges** (closest to the utility model).

STRENGTH

Most cost-effective through multitenancy and enormous scalability.

WATCH OUT

You don't control the location of data and equipment: a concern under regulatory mandates.

For example: AWS · Google · Salesforce

DEPLOYMENT MODELS

Private Cloud

For **exclusive use** by a single organization with multiple consumers (e.g., departments or business units).

Infrastructure is owned and managed by the organization on-premise[1] (cost-intensive).

Requires in-house IT skills and expertise.

STRENGTH

Better control over data location and security.

WATCH OUT

Limited scalability.

Preferred by companies that already own a data center and developed IT infrastructure, or are driven by legal and compliance requirements.

[1] *If the organization lacks the expertise, or for cost reasons, a private cloud can also be managed by a third party and hosted off-premise.*

DEPLOYMENT MODELS

Hybrid Cloud

A **combination** of public and private (or other) cloud models.

The models stay unique entities but are bridged by standardized or proprietary technology that enables data and application portability.

STRENGTH

Better alignment to business needs.

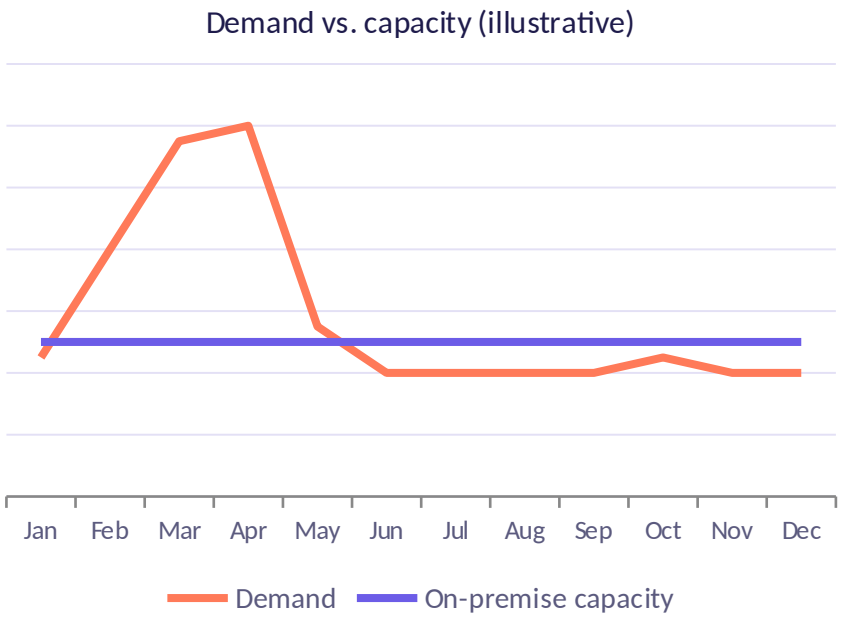
WATCH OUT

Harder to manage.

For example: an organization with its own data center may use a public cloud for inexpensive data backup.

DEPLOYMENT MODELS · HYBRID

Cloud Bursting



HOW IT WORKS

Use private cloud resources to run applications. When a spike in demand exceeds local limits, the **public cloud** handles the burst.

EXAMPLE

A tax preparation company sees 2–3× its usual volume in tax season. Sizing on-premise for the peak leaves resources idle all year. Better: size for typical load and burst to the public cloud at peak.

DEPLOYMENT MODELS

Community Cloud



Provisioned for use by **multiple organizations** with shared concerns (e.g., mission, security requirements, policy, compliance considerations).

RUN BY

The organizations themselves or a third party.

LOCATED

On- or off-premise.

For example: US Government agencies with similar security, privacy, and audit requirements can use a community cloud.

SERVICE MODELS

Cloud Service Models

A **service model** defines the type of service that the cloud provider is offering.

SaaS **Software as a Service**	Ready-to-use applications hosted in the cloud for end-users and organizations.	**e.g.,** Salesforce CRM, Microsoft Office 365
PaaS **Platform as a Service**	A software platform (languages, libraries, services, tools) for developers to quickly create and deploy custom applications.	**e.g.,** Microsoft Azure, Google App Engine
IaaS **Infrastructure as a Service**	Processing, storage, networks, and other fundamental computing resources for system administrators (IT Operations).	**e.g.,** Amazon EC2 (Elastic Compute Cloud)

"As a service" implies that SaaS, PaaS, and IaaS are delivered by way of software.

Characteristics of IaaS



IaaS

Infrastructure as a Service

Provision of hardware

SaaS

PaaS

IaaS



Reduces or eliminates the **capital costs** and complexity of owning hardware. Why buy servers when you can rent them?



The **CSP manages the infrastructure**; the client is responsible for other aspects of the deployment, such as the operating system and applications.



IaaS is the **base layer** of the cloud services stack and the foundation for both PaaS and SaaS.

Characteristics of PaaS

PaaS

Platform as a Service

A platform to build on

- SaaS
- **PaaS**
- IaaS

An environment for **low-cost, rapid development** of new applications (e.g., web and mobile).

Most platforms are accessed through an **Application Programming Interface (API)**.

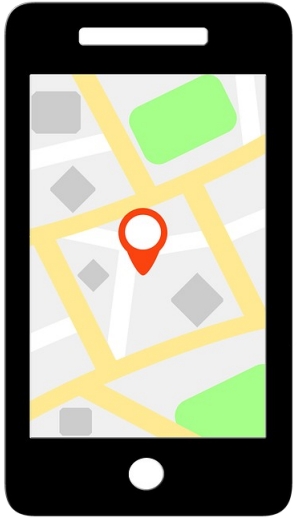
Developers fully control deployed applications and hosting settings, but don't manage the underlying networks, servers, storage, or operating systems.

High risk of vendor lock-in

What is an API?

An API allows one piece of software to talk to another.

An API defines how a programmer can write software to extend an existing application's features, or even build entirely new applications.



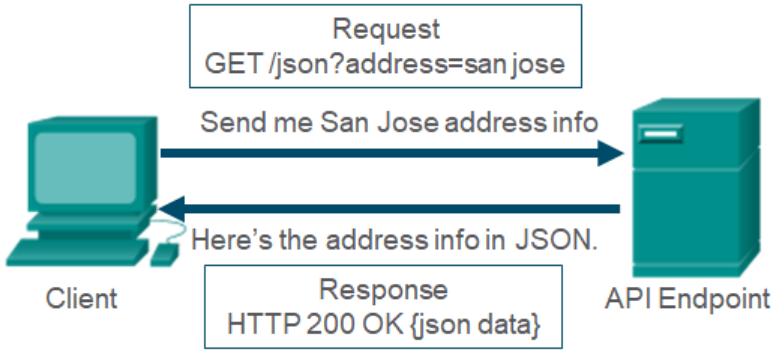
Source: Cisco

API Using HTTP

1. Developers construct API requests using **HyperText Transfer Protocol (HTTP)**.

2. The HTTP request asks for **JSON-formatted** data.

3. If correctly formatted per the API documentation, the server responds with JSON data.



Source: Cisco

JSON = JavaScript Object Notation

Characteristics of SaaS

Most popular service

Cloud-native & multitenant

SaaS applications are cloud-native and multitenant.

Prebuilt

Applications are prebuilt and consumed without significant customization.

CSP does the upkeep

Deployment, maintenance, patches, and updates are handled by the CSP.

Just use it

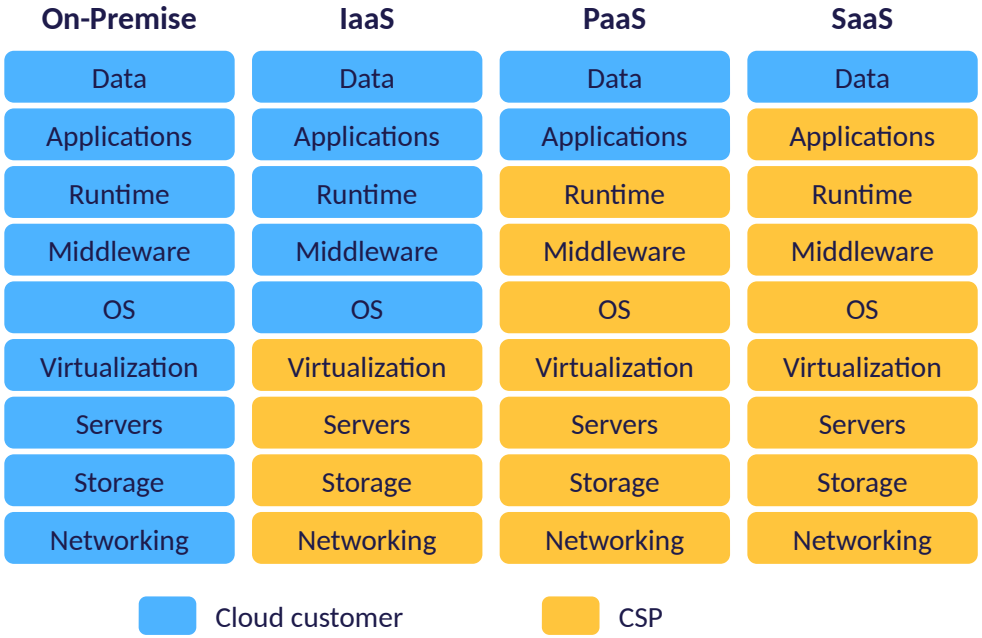
Users simply use the software; they don't manage the underlying infrastructure.

Anytime, anywhere

Access from any client device (device and location independence).

SERVICE MODELS

Shared Responsibility Model



On-premise: the organization manages everything.

IaaS: the CSP manages servers, storage, networking and virtualization. You manage the OS (patching and upgrades), applications, and data.

Your duties shrink from IaaS to PaaS to SaaS, but you are always responsible for your **data**.

BUSINESS DRIVERS

Business Drivers for Cloud



For most businesses, IT has become a necessity. Hosting IT services in-house can be costly and distract businesses from focusing on their core competencies.

Two major business drivers for the cloud:



Lower costs



Increased agility

BUSINESS DRIVERS

Reducing Costs

Lower costs from economies of scale

- Public CSPs support thousands of customers.
- Economies of scale let them offer services at reasonable prices.
- Customers can choose from multiple pricing options based on workload requirements.

Shift from CAPEX to OPEX (variable)



- No upfront cost of purchasing and maintaining hardware (except when on-premise).
- Pay only for resources or services consumed (pay-per-use).

BUSINESS DRIVERS

Increasing Agility



Scalability & elasticity

Scale resources up[1] or down and out[2] or in as needed. No more guessing, planning, or provisioning capacity.



Faster time to market

Quickly deploy development and testing environments, and rapidly experiment, innovate, and build new products.



More focus on core business

The CSP takes over routine administrative duties, so IT staff can focus on bigger business goals.



Global access

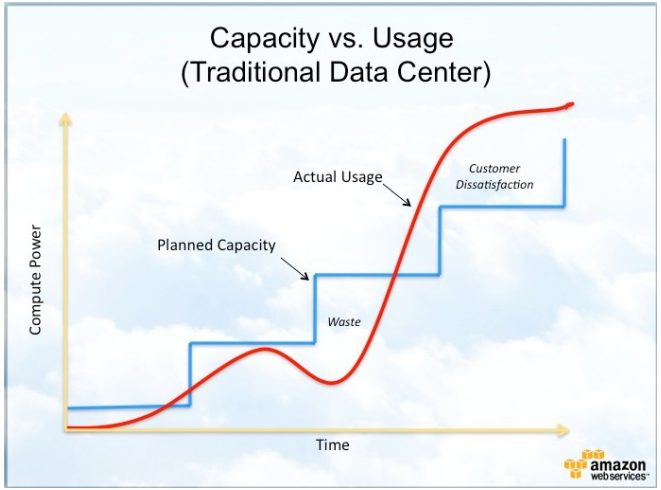
Deploy applications in multiple regions worldwide for lower latency and a better user experience.

[1] *Scaling up = vertical scaling.* [2] *Scaling out = horizontal scaling. Using both is called diagonal scaling.*

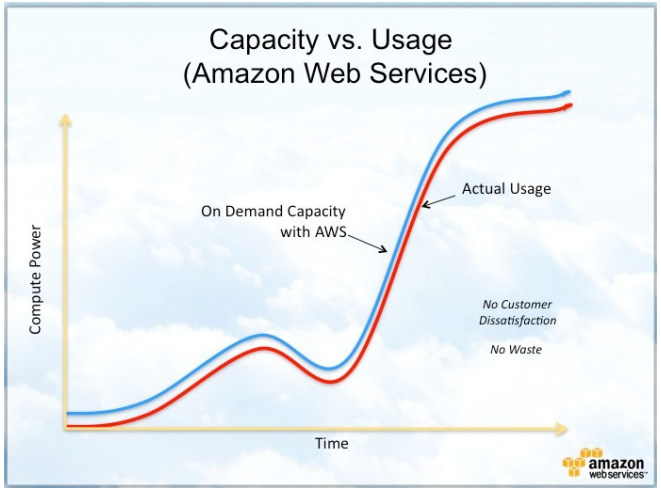
BUSINESS DRIVERS

Capacity Provisioning

Traditional data center



Cloud (AWS)



AWS closely follows the demand curve: no waste, no customer dissatisfaction.

Source: AWS

CONSIDERATIONS FOR ADOPTION

Choosing a Deployment Model

Organizations must consider various factors before moving their business processes to the cloud.



PUBLIC

Startups

The greatest barrier to entry for startups is capital. Public cloud's reduced costs outweigh other risks.



HYBRID

Small-to-medium businesses

With a moderate customer base, they may prefer hybrid: mission-critical apps on a private cloud, less critical apps in a public cloud.



PRIVATE

Enterprises

A large global customer base and financial strength: they may prefer private cloud to control data location, security, etc.

CONSIDERATIONS FOR ADOPTION

Migrate or Retain?

KEEP ON-PREMISE / PRIVATE

- Mission-critical applications are best run on-premise or in the private cloud.
- Regulatory and compliance requirements may prevent moving sensitive data to the public cloud.
- Legacy or proprietary-technology applications should run in-house.
- Apps depending on resources in a private data center may suffer latency if migrated.

GOOD FIT: PUBLIC CLOUD

Non-critical, low-risk applications:



Collaboration (email, web conferencing)



Personal productivity (word processing, spreadsheets)



Development and testing

CONSIDERATIONS FOR ADOPTION

Choosing a Cloud Provider

1. How long has the provider been in business?
2. Can the provider meet the current and future requirements of the organization?
3. How easy is it to add or remove services?
4. Can applications and services be easily transferred to another provider at the end of the contract?
5. Does the provider notify clients of maintenance windows, and is that downtime scheduled?

CONSIDERATIONS FOR ADOPTION

Is There a Cost Advantage?

Compare the **TCO** of an on-premise environment versus the public cloud to see whether cloud adoption pays off.

ON-PREMISE TCO

CAPEX: data center facilities, servers, storage devices, network equipment, operating systems, etc.

OPEX: power and cooling, personnel, Internet service, etc.

CLOUD COSTS

- Migration costs
- Usage fees for cloud services
- Costs to ensure security and compliance



Free tools: AWS TCO calculators, and the **AWS Pricing Calculator** to estimate AWS service costs.

Migration Strategies

Review existing applications and pick the strategy that best matches business needs: cost savings, scalability, availability, etc.

Rehosting (Lift-and-shift)

FASTEST

- Redeploy on-premise applications into cloud IaaS.
- Meets business needs quickly, with no optimizations.
- Fastest, but does not use the cloud's full potential.
- Manual or automated, e.g., AWS VM Import/Export (VM images to Amazon EC2).

Re-platforming

- Optimizations without changing the core architecture.
- Example: move an on-premise database to a fully managed service such as Amazon RDS, so the CSP handles most administrative tasks.

Migration Strategies



Repurchasing

- Invest in SaaS (cloud-native) applications, i.e., move to a different product.
- Frees IT staff from maintenance and updates.
- Examples: HR system to Workday, Outlook to Office 365, in-house CRM to Salesforce.



Rebuilding / Re-architecting

MOST EXPENSIVE

- Change the core architecture, e.g., turn a monolithic application into microservices, or go serverless (more in Chapter 3).
- Goal: a considerable increase in performance, scale, agility, or new features.

CONSIDERATIONS FOR ADOPTION

Service Level Agreements



An **SLA** is a contract between a customer and a CSP that specifies the level of **uptime** the provider will support, as well as the expected performance metrics.



Should say how the CSP compensates customers if SLAs are not met (e.g., money or service credits).



Specific to a cloud service: storage and virtual machines can have different SLAs.



Can be uniform for all customers, or a custom agreement between one customer and the CSP.



Helps ensure key resources stay available so organizations avoid excessive losses.

Example: AWS EC2 SLA

For each **Amazon EC2 instance**, AWS will make commercially reasonable efforts to keep it available at least **99.5%** of the time in any monthly billing cycle. If it falls short, you can receive a **Service Credit**.

An **EC2 (Elastic Compute Cloud) instance** is synonymous with a virtual machine deployed in the AWS cloud.

UPTIME THIS MONTH	SERVICE CREDIT
99.0% to < 99.5%	10%
95.0% to < 99.0%	30%
Below 95.0%	100%

Instance-level SLA published by AWS. Source: AWS

CONSIDERATIONS FOR ADOPTION

Current Trend: Multi-cloud

Multi-cloud

Using cloud services from more than one cloud vendor.



Each vendor offers a myriad of services, and one vendor's service may beat the others on performance, ease of use, cost, compliance, etc.

Flexibility to choose the best-fit service from each vendor, and it may also prevent vendor lock-in.

Very appealing, but the obvious pitfall is **management complexity**.

AWS

What is Amazon Web Services?

March 2006

AWS was founded

#1

World's largest public cloud provider today

Core services over its Global Infrastructure of Regions, Availability Zones, and Edge Locations: compute, storage, database, networking, and more.

Three ways to access AWS



Management Console

Think of it as a GUI.



Command Line Interface

The CLI allows scripts.

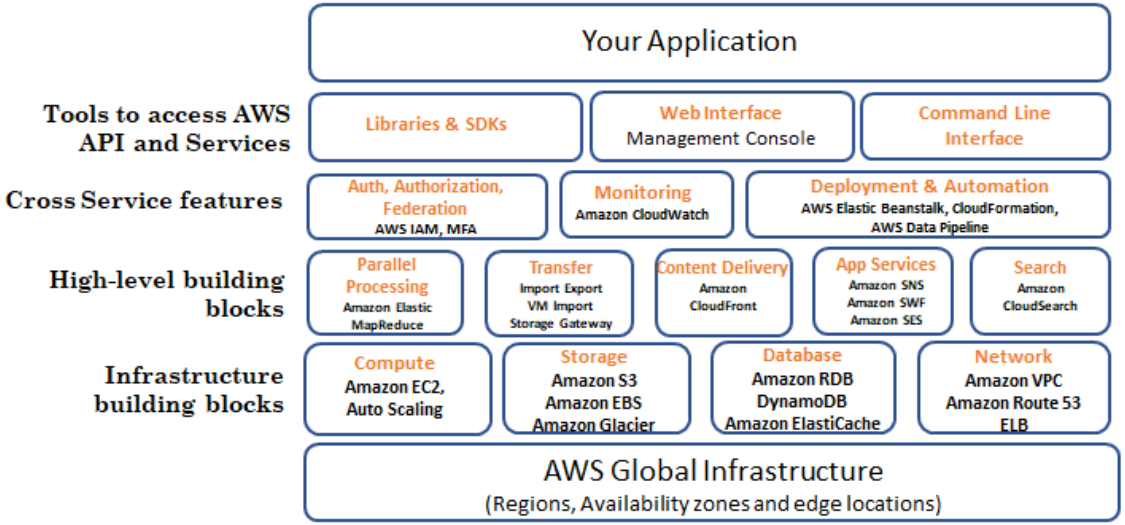


SDKs

Preferred by developers. Python, Java, PHP, Ruby, C++, etc.

AWS

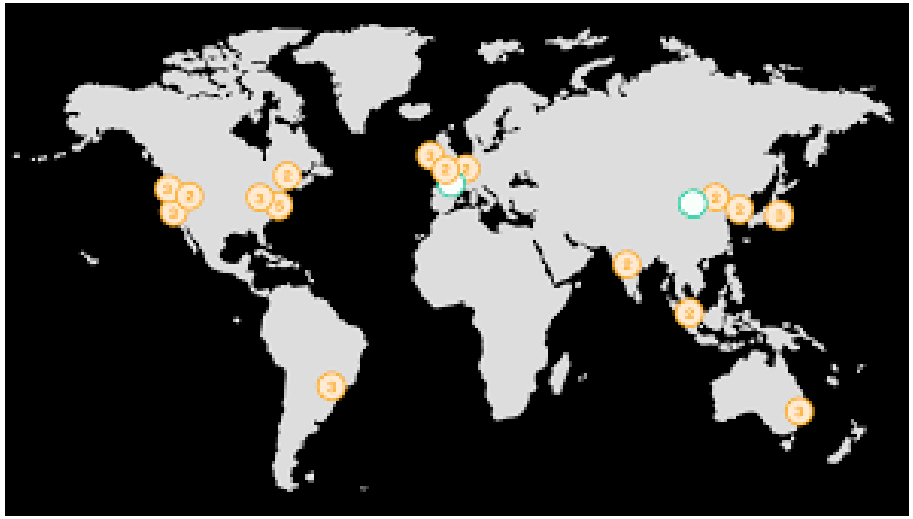
AWS Services at a Glance



Read it bottom-up

Your application sits on top of AWS services, which run on the AWS Global Infrastructure.

AWS Regions



Source: AWS

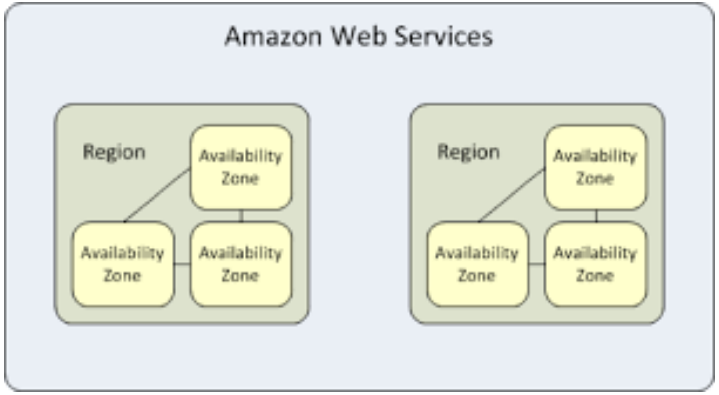
The AWS footprint consists of several geographic Regions spread across the globe.

Each Region has multiple, physically separate **Availability Zones**. The typical number is three.

Learn more:
AWS Regions and Availability Zones

AWS · GLOBAL INFRASTRUCTURE

Availability Zones



Source: AWS

If one AZ is hit by a power outage or a disaster (earthquake, tornado), the **other AZs stay unaffected**.

An Availability Zone is a cluster of one or more discrete data centers.

AZs are powered independently of other AZs in the Region.

They are separated by about 100 km (60 miles).

AZs in a Region are linked by high-throughput metro fiber.

KEY TAKEAWAYS

Chapter Recap

5 Essential characteristics

- On-demand self service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

4 Deployment models

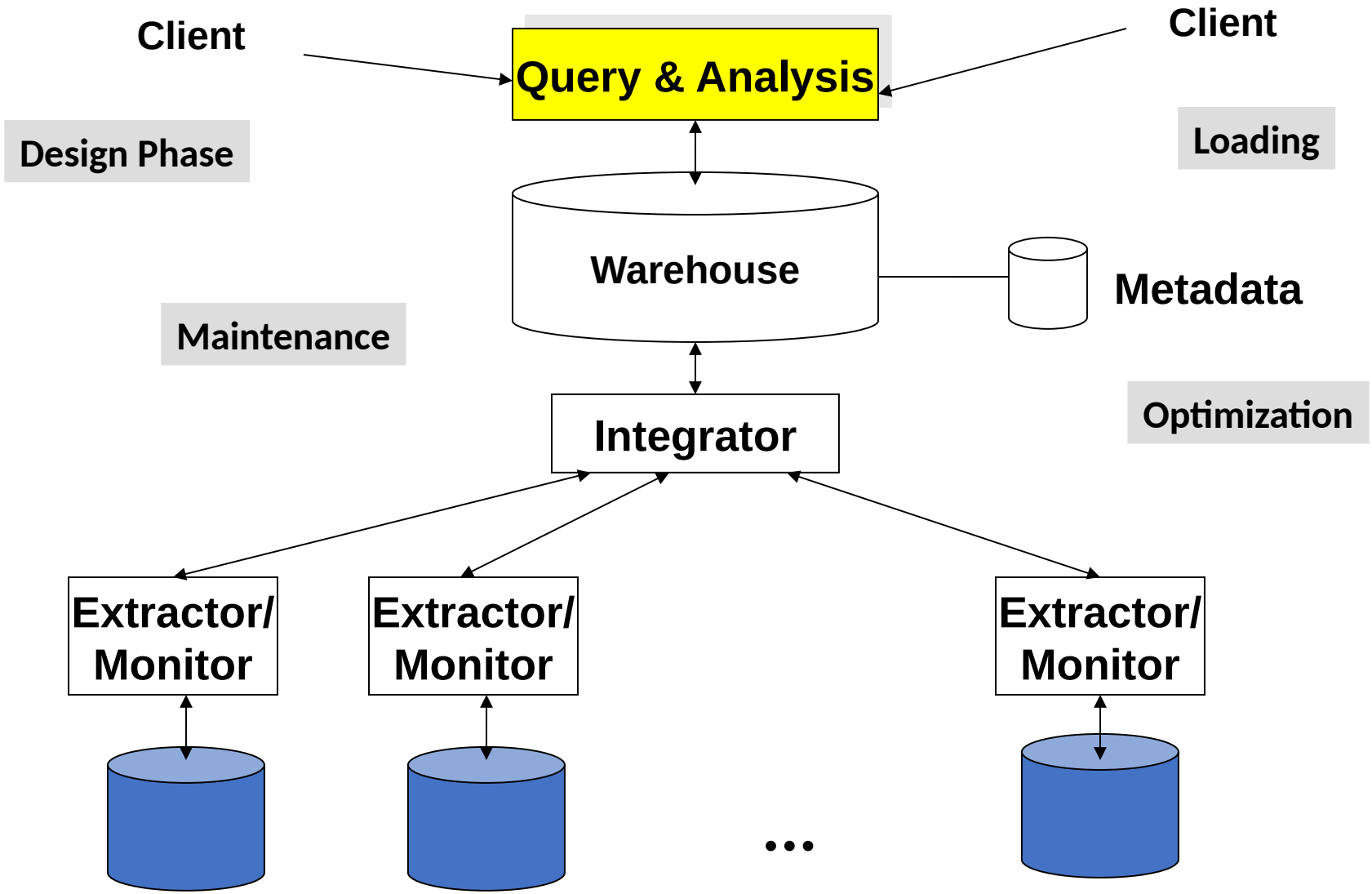
- Public
- Private
- Hybrid
- Community

3 Service models

- SaaS
- PaaS
- IaaS

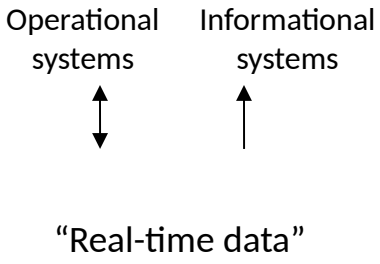
Cloud computing is a business and service model, not just a technology.

Generic Warehouse Architecture

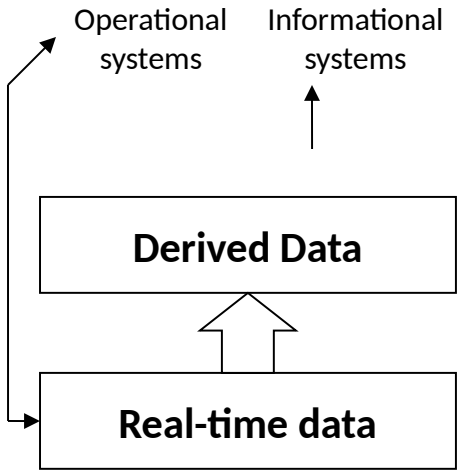

Client
Client
Query & Analysis
Design Phase
Loading
Warehouse
Metadata
Maintenance
Integrator
Optimization
Extractor/
Monitor
Extractor/
Monitor
Extractor/
Monitor
...

Data Warehouse Architectures: Conceptual View

- **Single-layer**
  - Every data element is stored once only
  - **Virtual warehouse**



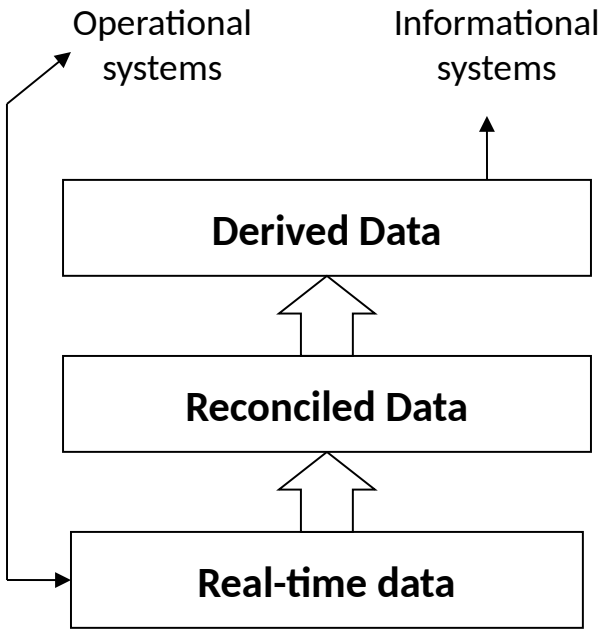
- **Two-layer**
  - **Real-time + derived data**
  - **Most commonly used approach in industry today**



Three-layer Architecture: Conceptual View

- **Transformation of real-time data to derived data** really requires **two steps:**

- **Data reconciliation**: used to **describe a verification** phase during a **data** migration where the target **data** is compared against original source **data** to ensure that the migration architecture has transferred the **data** correctly.



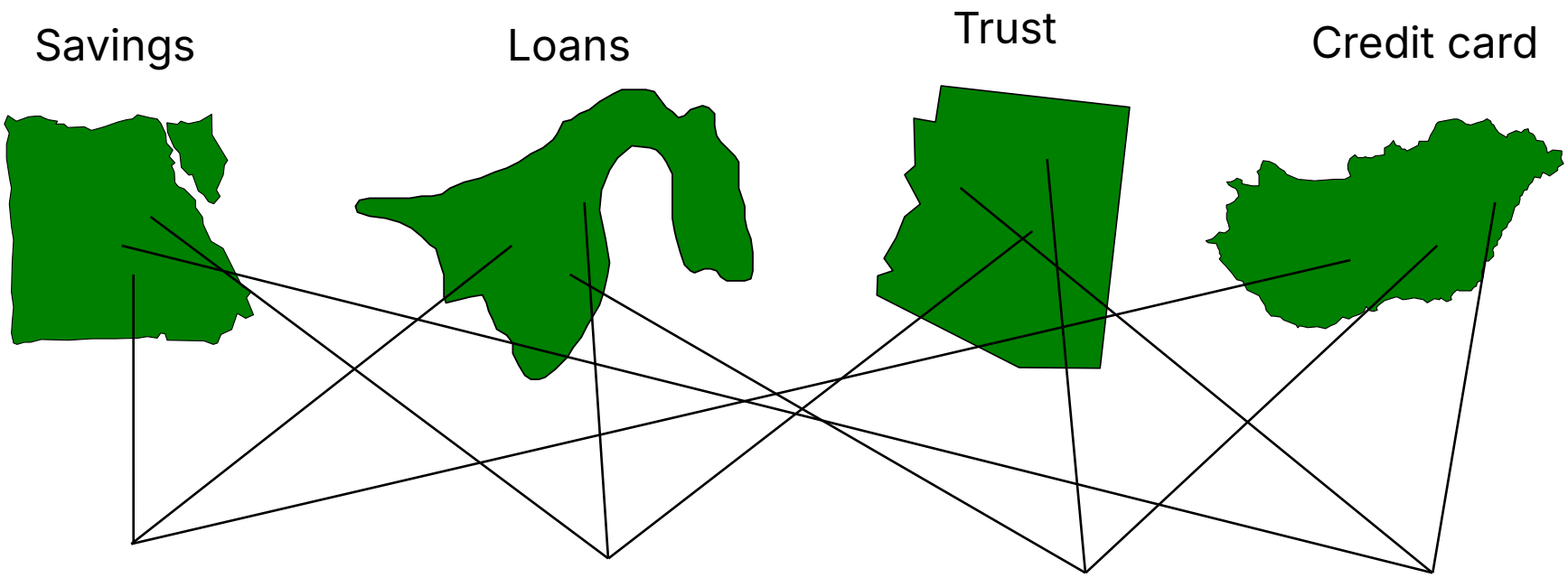
View level
"Particular informational needs"

Physical Implementation of the Data Warehouse

Data Quality - The Reality

- Tempting to think creating a **data warehouse is simply extracting operational data and entering into** a data warehouse
- **Nothing could be farther from the truth**
- Warehouse data comes from disparate questionable sources

Data Integration Across Sources



Same data
different name

Different data
Same name

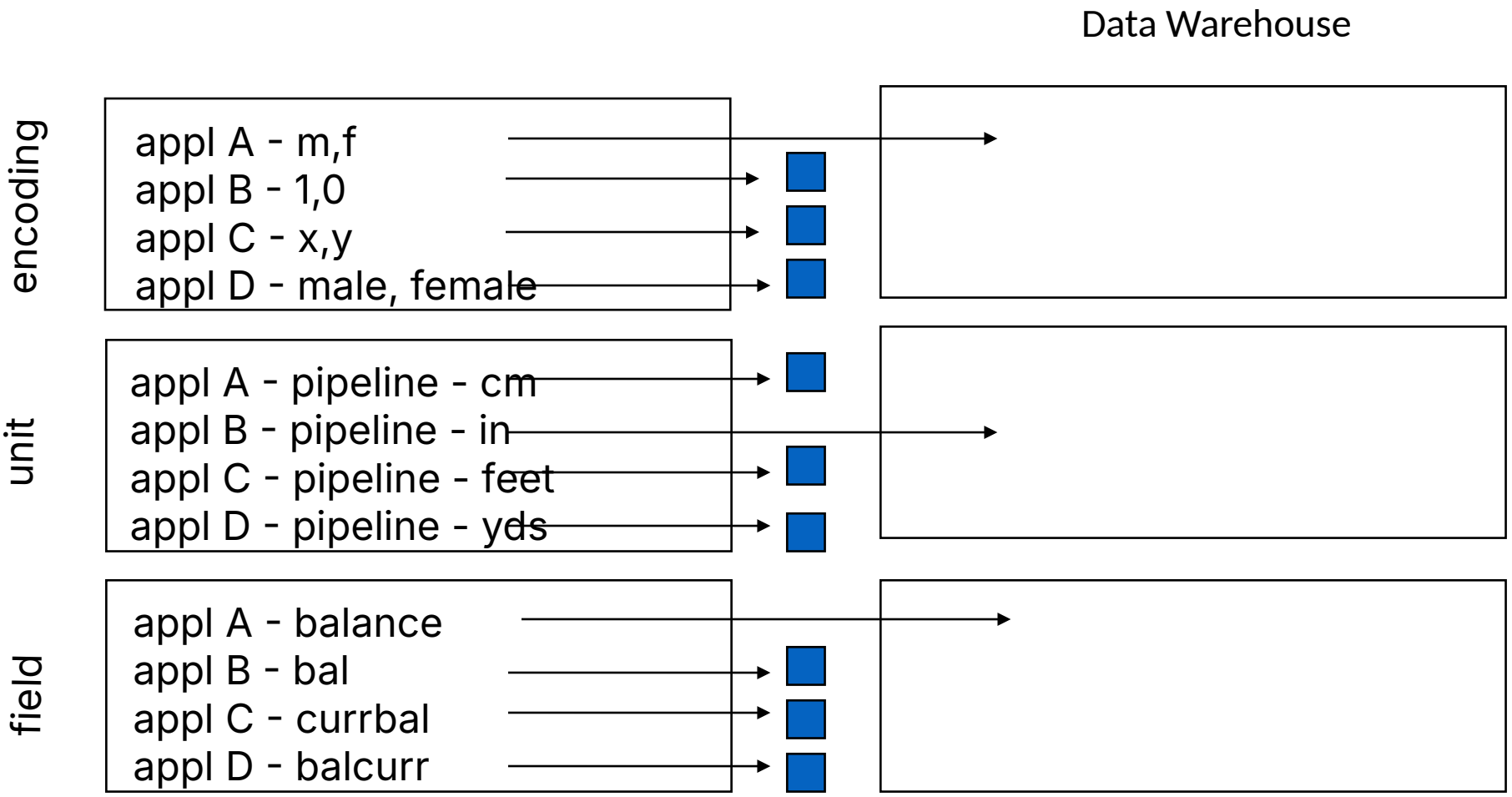
Data found here
nowhere else

Different keys
same data

Data Integrity Problems

- **Same person, different spellings**
  - Agarwal, Agrawal, Aggarwal etc...
- **Multiple ways to denote company name**
  - Persistent Systems, PSPL, Persistent Pvt. LTD.
- **Use of different names**
  - mumbai, bombay
- **Different account numbers generated by different applications for the same customer**
- **Required fields left blank**
- **Invalid product codes collected at point of sale**
  - manual entry leads to mistakes
  - “in case of a problem use 9999999”

Data Transformation Example



Data Transformation Terms

- Extracting
- Conditioning
- Scrubbing
- Merging
- Householding

- Enrichment
- Scoring
- Loading
- Validating

- **Data scrubbing** is an **error correction** technique that uses a background task to periodically inspect **main memory** or **storage** for errors, then corrects detected errors using **redundant data** in the form of different **checksums** or copies of data.

Data Transformation Terms

- **Extracting**
  - ✓ Capture of data from operational source in “as is” status
- **Conditioning**
  - ➢ The **conversion of data types** from **the source to the target data store (warehouse)** -- always a relational database

Data Transformation Terms cont′

- **Householding:**
  - Identifying all members of a household (living at the same address)
  - Ensures only one mail is sent to a household
  - Can result in substantial savings:

- **Data validation** means **checking the accuracy and quality of source data before using, importing or otherwise processing data**.
- Data validation is a form of data cleansing.

Data Transformation Terms

- **Enrichment**
  - Bring data from external sources to augment/enrich operational data.
  - **Scoring**

  Computation of a probability of an event. e.g…chance that a customer is likely to buy a new product.

- **Data merging** is **the process of combining two or more similar records into a single one**.
- Merging is done to add variables to a dataset, append or add cases or observations to a dataset, or remove duplicates and other incorrect information.

Loads

- After extracting, cleaning, validating etc. need to load the data into the warehouse
- Issues
  - ✓huge volumes of data to be loaded
  - ✓small time window available when warehouse can be taken off line (usually nights)
  - ✓when to build index and summary tables
  - ✓allow system administrators to monitor, cancel, resume, change load rates
  - ✓**Recover gracefully** -- restart after failure from where you were and without loss of data integrity

Refresh

- **Propagate updates** on source data to the warehouse
- Issues:
  - ✓ when to refresh
  - ✓ how to refresh -- refresh techniques

When to Refresh?

- **periodically** (e.g., every night, every week) or after significant events
- **on every update**: not warranted unless warehouse data require current data (up to the minute stock quotes)
- **refresh policy set by administrator based on user needs and traffic**
- **possibly different policies for different sources**

Data Extraction and Cleansing

- Extract data from existing operational and legacy data
- Issues:
  - ✓ Sources of data for the warehouse
  - ✓ Data quality at the sources
  - ✓ Merging different data sources
  - ✓ Data Transformation
  - ✓ How to propagate updates (on the sources) to the warehouse
  - ✓ Terabytes of data to be loaded

Granularity in Warehouse

- **Can not answer some questions with summarized data**
  - Did Anand call Seshadri last month? Not possible to answer if total duration of calls by Anand over a month is only maintained and individual call details are not.
- **Detailed data too voluminous**

- **Granularity** tells us about the level (high to low) of information residing in the table.
- The **lower the granularity is**, the **low-level information it contains**.

Derived Data

- **Introduction of derived (calculated data**) may often help
- Have seen this in the context of dual levels of granularity
- **Can keep auxiliary views and indexes to speed up query processing**

- **Derived data** is generated from existing **data** using a mathematical operation or a **data** transformation.
- **OLAP Services uses SQL ROLLUP** to generate aggregate **data** in the **data warehouse**.

Data Warehousing: Two Distinct Issues

(1) How to get information into warehouse

"Data warehousing"

(2) What to do with data once it's in warehouse

"Warehouse DBMS"

Issues in Data Warehousing

- Warehouse Design
- Extraction
  - Wrappers, monitors (change detectors)
- Integration
  - Cleansing & merging
- Warehousing specification & Maintenance
- Optimizations
- Miscellaneous (e.g., evolution)

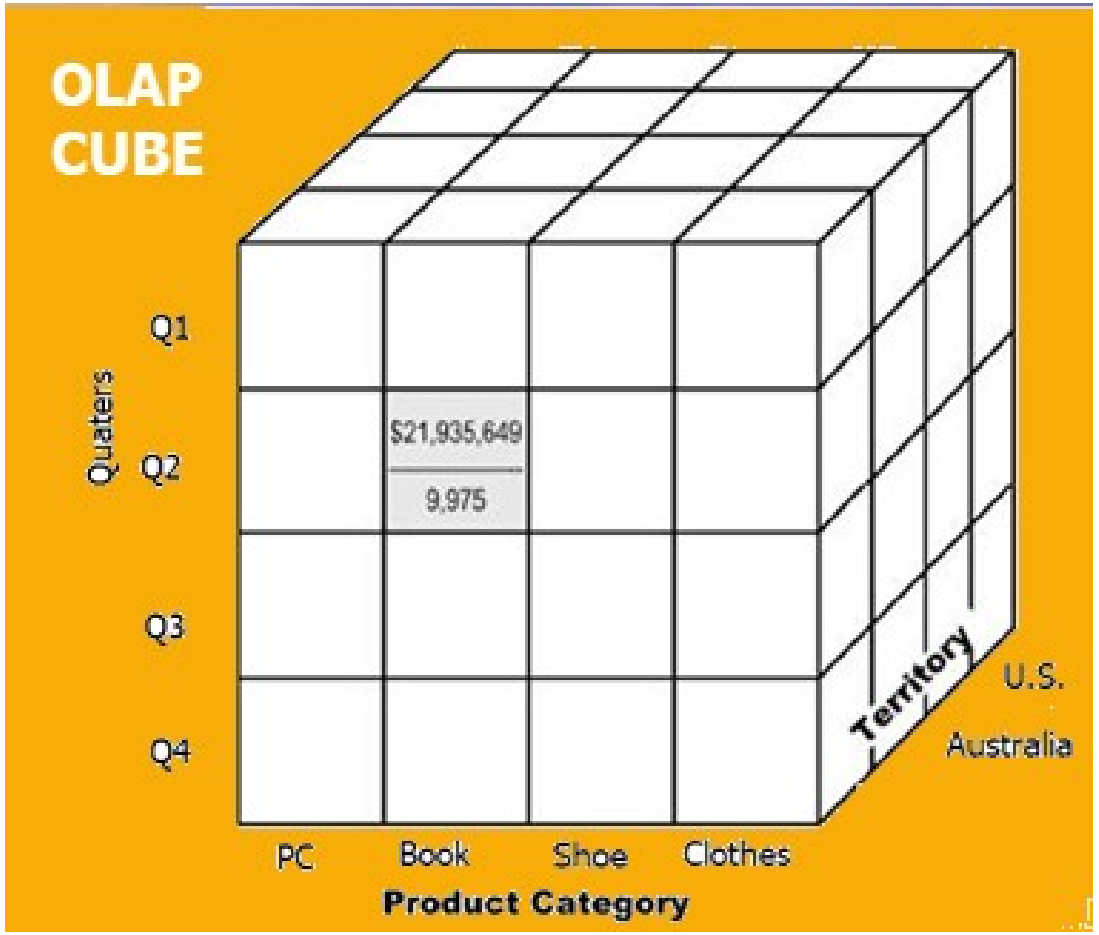
OLAP – Online Analytical Processing

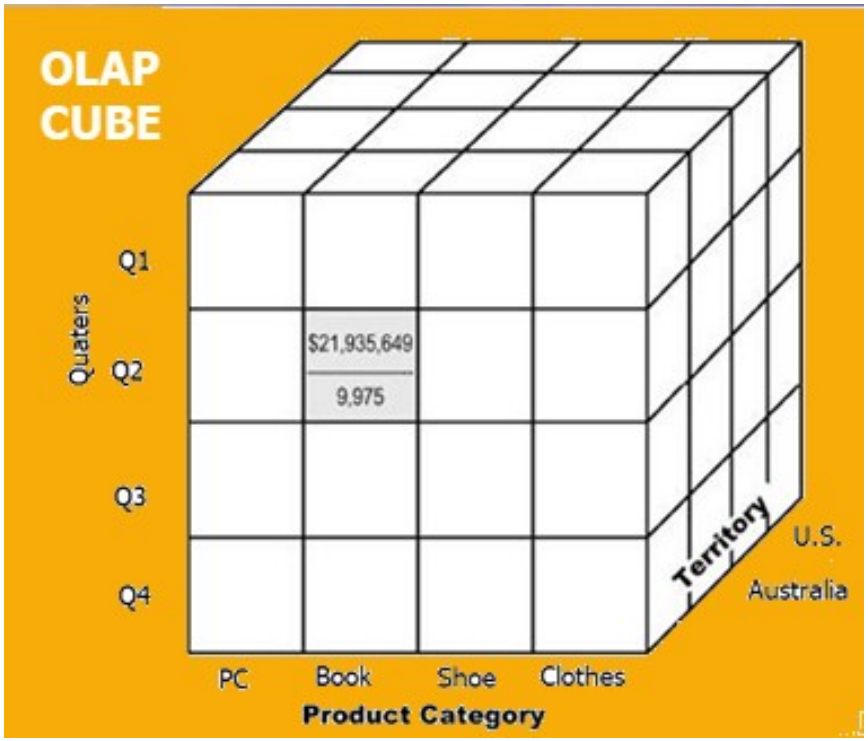
OLAP – Online Analytical Processing

- Online analytical processing, or OLAP is an **approach to answering multi-dimensional analytical (MDA) queries swiftly**.
- OLAP tools enable users **to analyze multidimensional data interactively from multiple perspectives**.
- OLAP consists of **three** basic analytical operations: **consolidation** (roll-up), **drill-down**, and **slicing** and **dicing**.

- **Consolidation** involves the **aggregation of data that can be accumulated and computed in one or more dimensions.**

- OLAP databases are divided into **one or more cubes.**
- **The cubes** are designed in such a way that creating and viewing reports become easy.





•At the core of the OLAP concept, is an **OLAP Cube.**

•The OLAP cube is a **data structure optimized for very quick data analysis**.

•**The OLAP Cube** consists of numeric **facts** called **measures** which are categorized **by dimensions.**

•OLAP Cube is also called the **hypercube**.

How does it work?

- A Data warehouse **would extract information** from **multiple data sources and formats** like **text files, excel sheet, multimedia files, etc.**
- The extracted data is **cleaned and transformed**.
- Data is loaded into an **OLAP server (or OLAP cube)** where information is **pre-calculated** in advance for **further analysis**.

Basic analytical operations of OLAP (we shall the details later)

Four types of analytical operations in OLAP are:

1.Roll-up

2.Drill-down

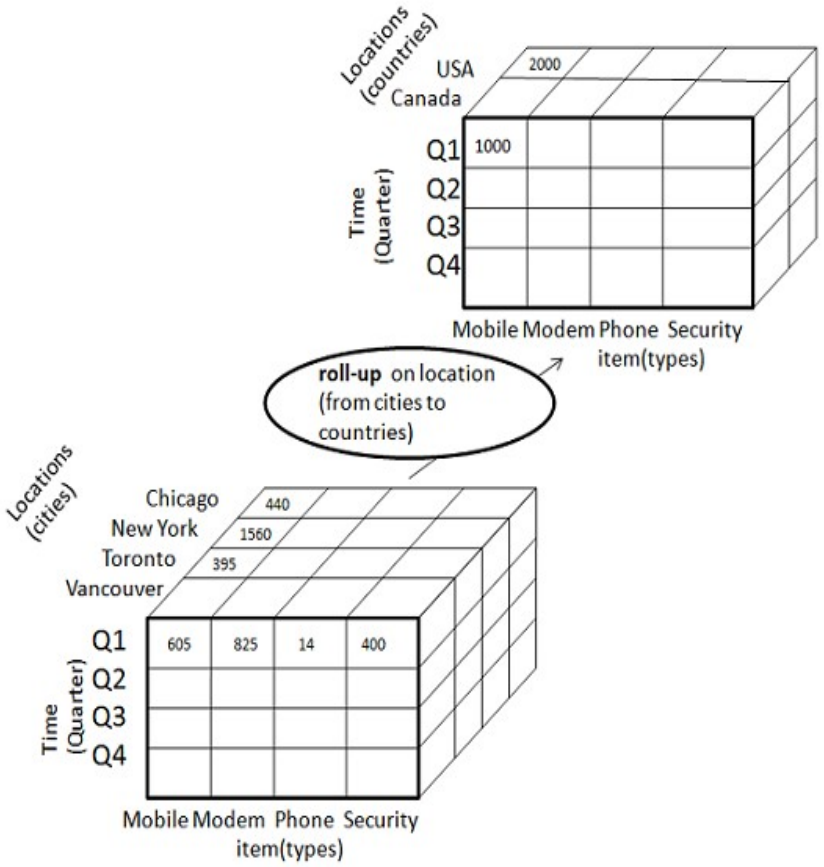
3.Slice and dice

4.Pivot (rotate)

Roll-up

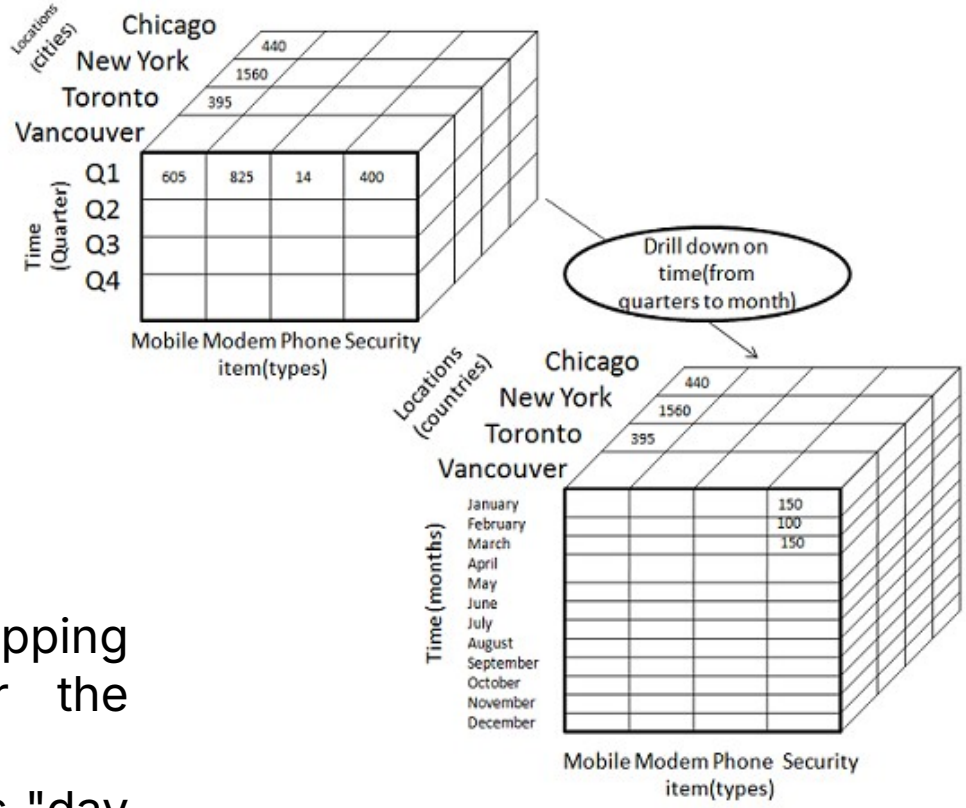
- Roll-up performs aggregation on a data cube in any of the following ways –
- By climbing up a concept hierarchy for a dimension
- By dimension reduction
- The following diagram illustrates how roll-up works.

•Initially the concept hierarchy was "street < city < province < country".
•On rolling up, the data is aggregated by ascending the location hierarchy from the level of city to the level of country.
•The data is grouped into cities rather than countries.



Drill-down

- Drill-down is the reverse operation of roll-up. It is performed by either of the following ways –
- By stepping down a concept hierarchy for a dimension
- By introducing a new dimension.
- The following diagram illustrates how drill-down works –



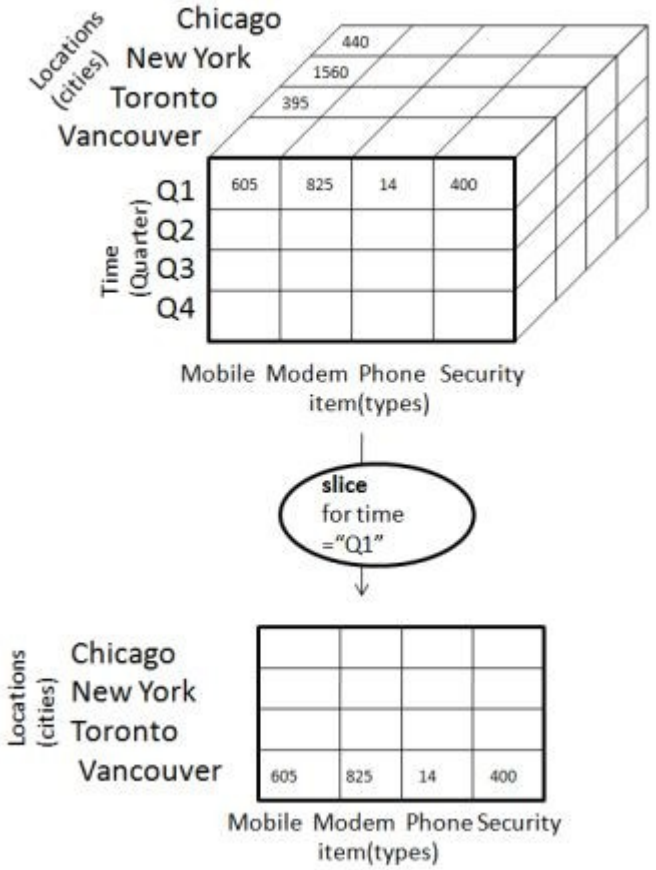
•Drill-down is performed by stepping down a concept hierarchy for the dimension time.

•Initially the concept hierarchy was "day < month < quarter < year."

•On drilling down, the time dimension is descended from the level of quarter to the level of month.

Slice

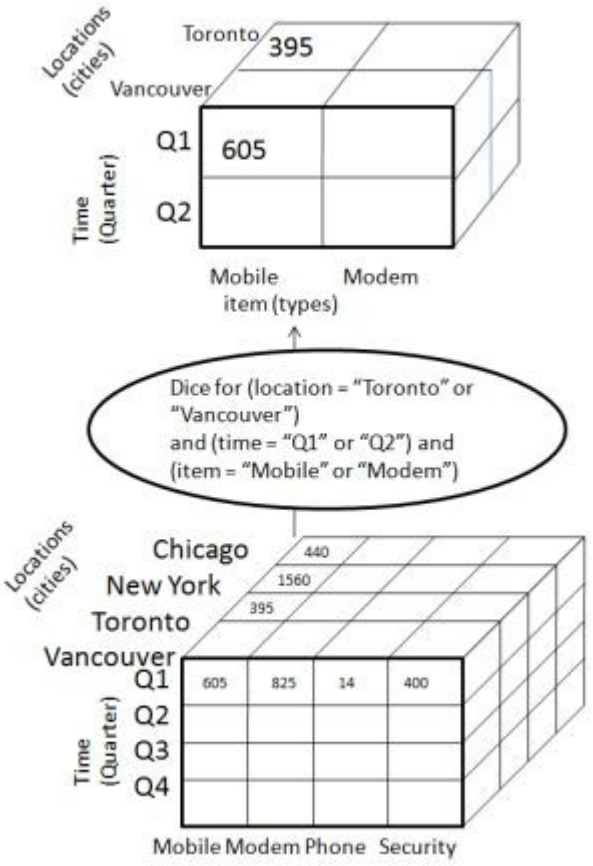
- The slice operation selects one particular dimension from a given cube and provides a new sub-cube. Consider the following diagram that shows how slice works.



•Here Slice is performed for the dimension "time" using the criterion time = "Q1".
•It will form a new sub-cube by selecting one or more dimensions.

Dice

- Dice selects two or more dimensions from a given cube and provides a new sub-cube. Consider the following diagram that shows the dice operation.

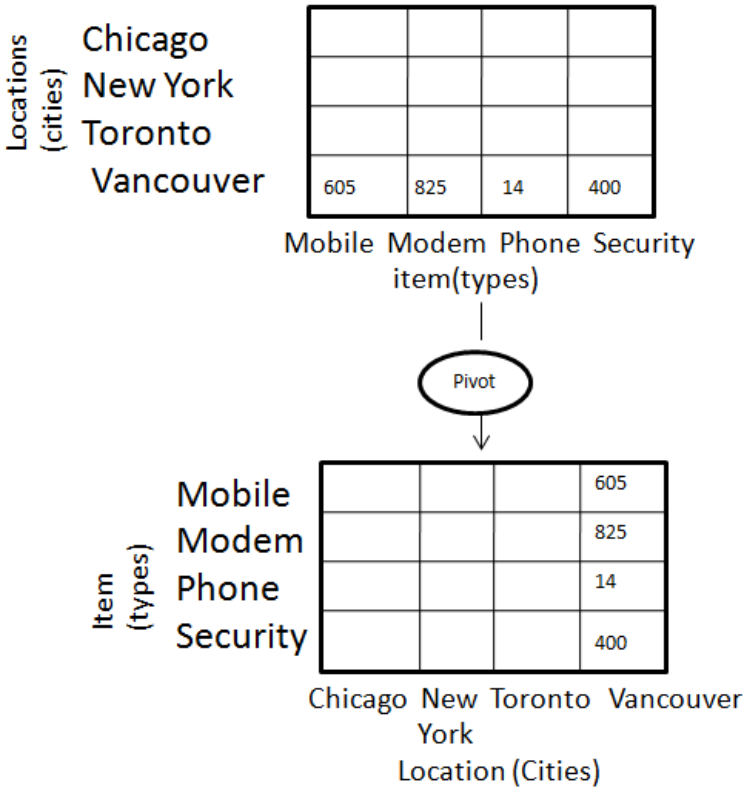


The dice operation on the cube based on the following selection criteria involves three dimensions.

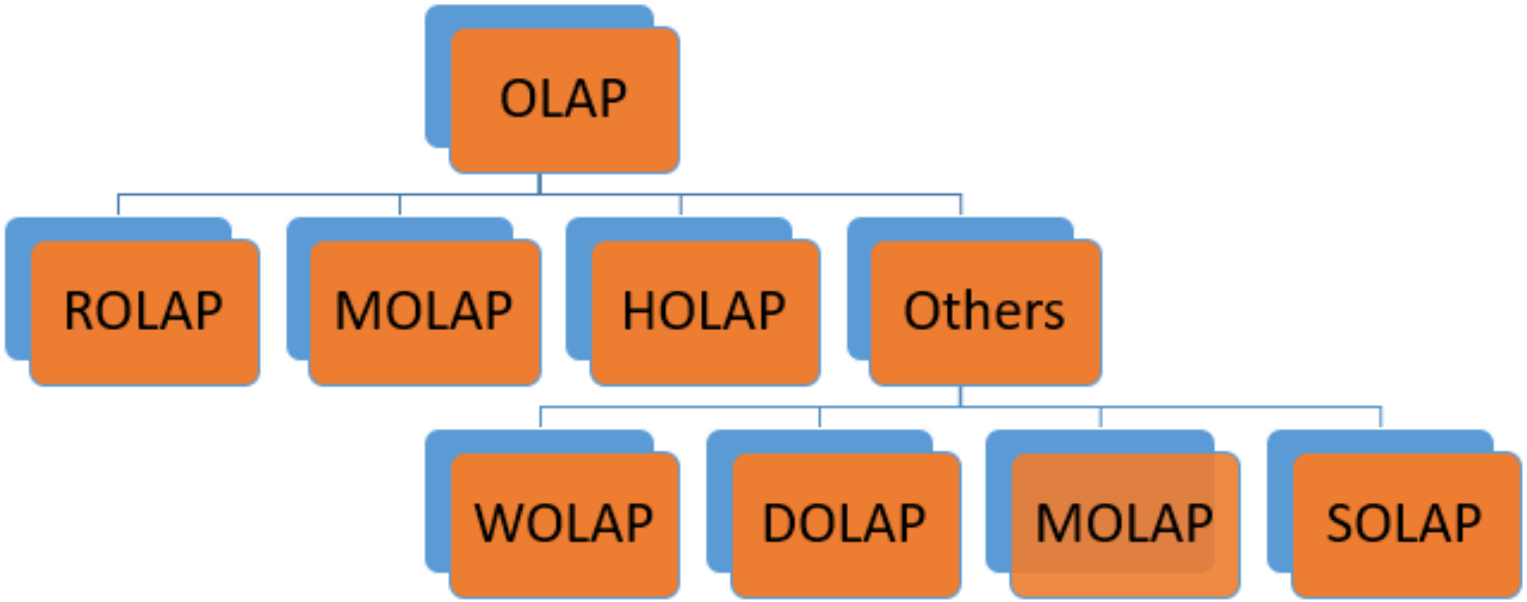
•(location = "Toronto" or "Vancouver")
•(time = "Q1" or "Q2")
•(item =" Mobile" or "Modem")

Pivot

The pivot operation is also known as rotation. It rotates the data axes in view in order to provide an alternative presentation of data. Consider the following diagram that shows the pivot operation.



Types of OLAP systems

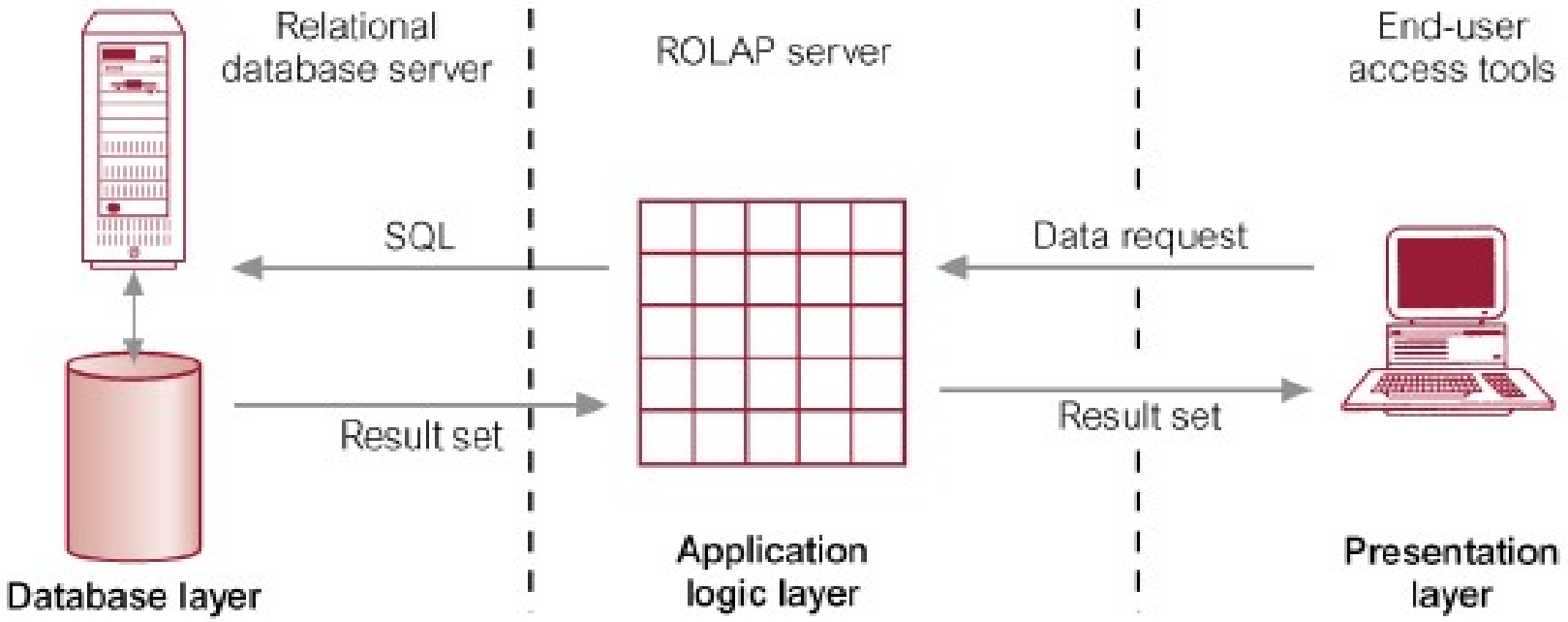


OLAP Hierarchical Structure

Type of OLAP	Explanation
Relational OLAP(ROLAP):	ROLAP is an extended RDBMS along with **multidimensional data** mapping to perform the **standard relational operation**.
Multidimensional OLAP (MOLAP)	MOLAP Implements **operation in multidimensional data**.
Hybrid Online Analytical Processing (HOLAP)	In HOLAP approach the aggregated totals are stored in a multidimensional database while the detailed data is stored in the relational database. This offers **both data efficiency** of the ROLAP model and the **performance of the MOLAP model**.
Desktop OLAP (DOLAP)	In Desktop OLAP, a user downloads a part of the data from the database locally, or on their desktop and analyze it. DOLAP is relatively cheaper to deploy as it offers very few functionalities compares to other OLAP systems.
Web OLAP (WOLAP)	Web OLAP which is OLAP system accessible via the web browser. WOLAP is a three-tiered architecture. It consists of three components: client, middleware, and a database server.
Mobile OLAP:	Mobile OLAP helps users to access and analyze OLAP data using their mobile devices
Spatial OLAP :	SOLAP is created to facilitate management of both spatial and non-spatial data in a Geographic Information system (GIS)

ROLAP

- **ROLAP works with data that exist in a relational database.**
- **Facts** and **dimension tables** are stored as relational tables.
- It **also allows multidimensional analysis of data and is the fastest growing OLAP**.



Relational OLAP(ROLAP)

Advantages of ROLAP model:

- **High data efficiency:** It offers high data efficiency because query performance and access language are optimized particularly for the multidimensional data analysis.
- **Scalability.** This type of OLAP system offers scalability for managing large volumes of data, and even when the data is steadily increasing.

Drawbacks of ROLAP model:

- **Demand for higher resources:** ROLAP needs high utilization of manpower, software, and hardware resources.
- **Aggregately data limitations.** ROLAP tools use SQL for all calculation of aggregate data. However, there are no set limits to the for handling computations.
- **Slow query performance.** Query performance in this model is slow when compared with MOLAP

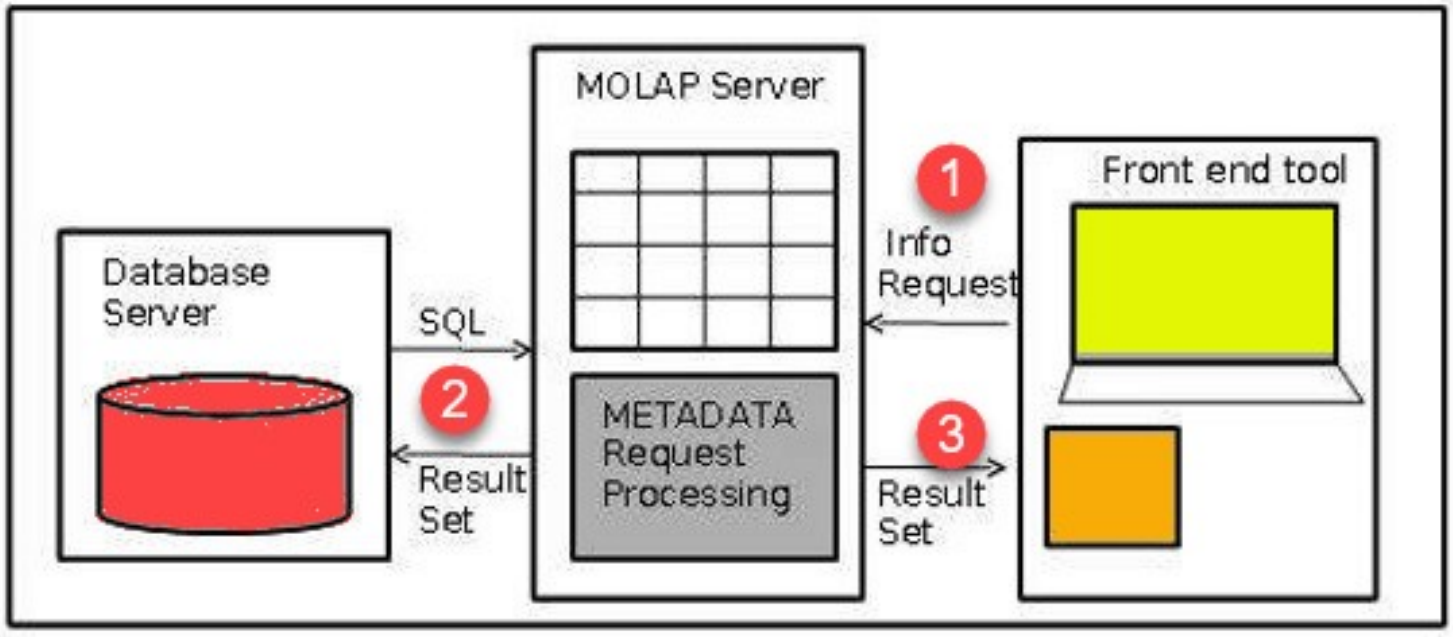
Multidimensional OLAP (MOLAP)

- It is a classical OLAP that facilitates data analysis by using a **multidimensional data cube.**
- Data is **pre-computed, re-summarized, and stored in a MOLAP (a major difference from ROLAP).**
- Using a MOLAP, a user can **use multidimensional view data with different facets.**
- **Facts are stored** in **multi-dimensional array and dimensions used to query them.**
- Basically, they use an **OLAP cube.**

MOLAP Architecture

MOLAP Architecture includes the following components ?

- Database server.
- MOLAP server.
- Front-end tool.



- The user request reports through the interface
- The application logic layer of the MDDB retrieves the stored data from Database
- The application logic layer forwards the result to the client/user.

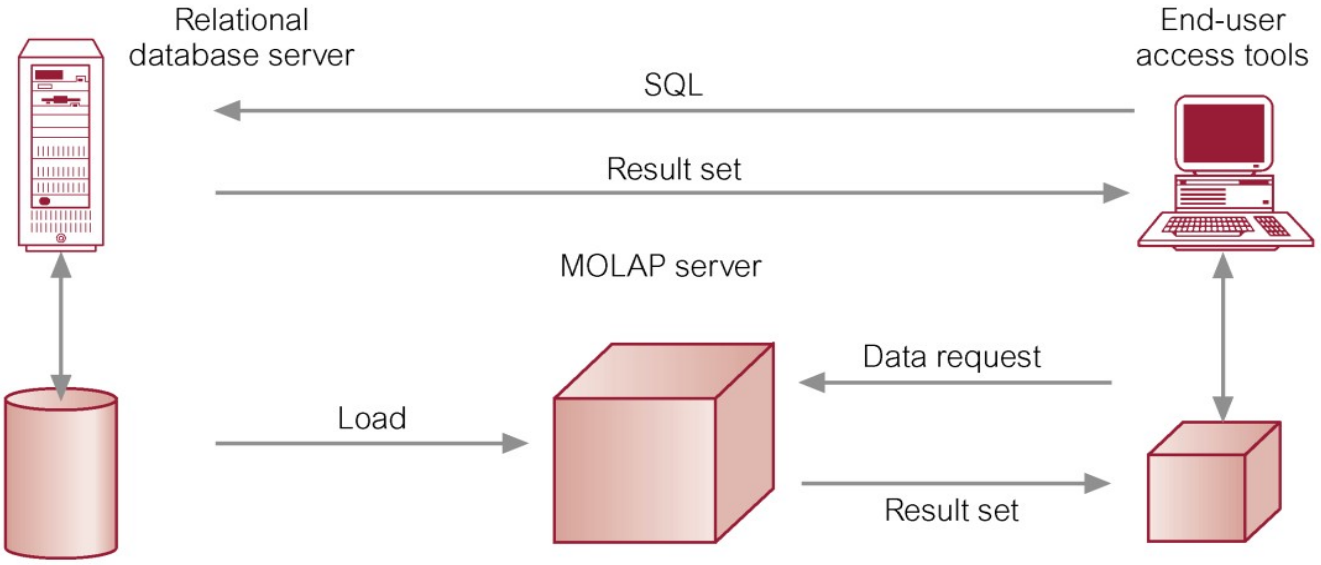
- MOLAP architecture mainly **reads the precompiled data**.
- MOLAP architecture has limited capabilities to dynamically create aggregations or to calculate results that have not been pre-calculated and stored.

- ✓ For example, an accounting head can run a report showing the corporate P/L account or P/L account for a specific subsidiary.
- ✓ The MDDB would retrieve precompiled Profit & Loss figures and display that result to the user.

Hybrid OLAP

- Hybrid OLAP is a mixture of both **ROLAP and MOLAP**.
- It offers fast computation of MOLAP and **higher scalability of ROLAP**. HOLAP uses **two databases**:

1. **Aggregated or computed data** is stored in a **multidimensional OLAP cube**
2. **Detailed information is stored in a relational database.**



Benefits of Hybrid OLAP

- This kind of OLAP helps to economize the disk space, and it also remains compact which helps to avoid issues related to access speed and convenience.
- Hybrid HOLAP's uses cube technology which allows faster performance for all types of data.
- ROLAP are instantly updated and HOLAP users have access to this real-time instantly updated data.
- MOLAP brings cleaning and conversion of data thereby improving data relevance. This brings best of both worlds.

ROLAP vs. MOLAP

- Performance:
  - How fast will the system appear to the end-user?
  - MOLAP vendors believe this is a key point in their favor
- Data volume and scalability:
  - While MOLAP servers can handle up to 100GB of storage, ROLAP servers can handle hundreds of gigabytes and terabytes

Benefits		MOLAP	ROLAP
User Benefits	Multidimensional View	✓	✓
	Excellent Performance	✓	
	Analytical Flexibility	✓	
	Real-time Data Access		✓
	High Data Capacity		✓
MIS Benefits	Leverages Data Warehouse		✓
	Easy Development	✓	
	Low Structure Maintenance		✓
	Low Aggregate Maintenance	✓	

Summary cont′:

- OLAP is a **technology that enables analysts to extract and view business data from different points of view**.
- At the core of the OLAP concept, is an **OLAP Cube.**
- Various business applications and other data operations require the **use of OLAP Cube.**
- There are primary five types of analytical operations in OLAP

  1) Roll-up 2) Drill-down 3) Slice 4) Dice and 5) Pivot
- Three types of widely used OLAP systems are **MOLAP, ROLAP, and Hybrid OLAP.**
- Desktop OLAP, Web OLAP, and Mobile OLAP are some other types of OLAP systems.

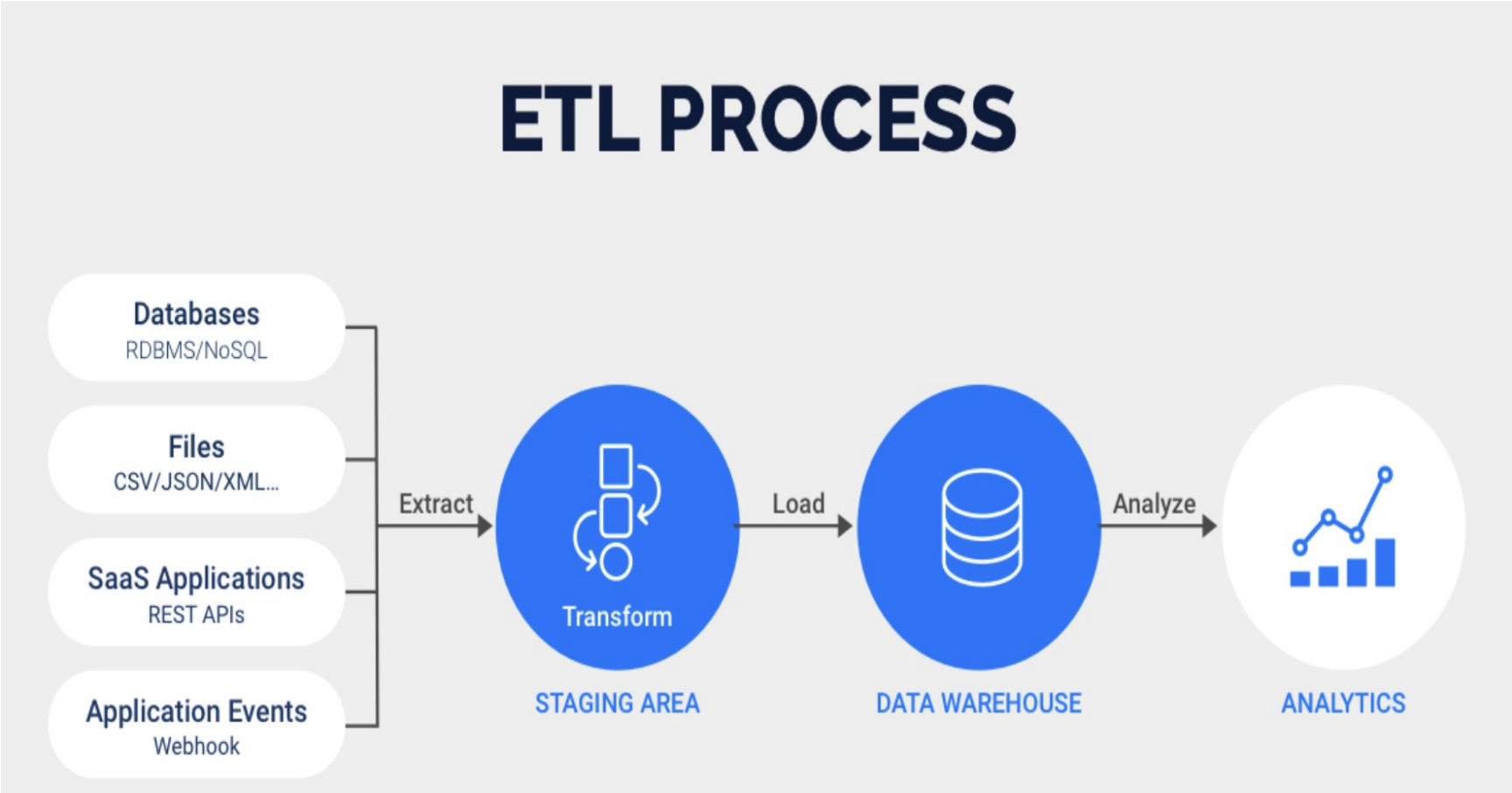
OLTP – Online Transaction Processing

- **OLTP** is an operational system that supports transaction-oriented applications in a **3-tier architecture.**
- It administers the **day-to-day transaction of an organization.**
- OLTP is **basically focused on query processing, maintaining data integrity in multi-access environments as well as effectiveness** that is measured by the total number of transactions per second.
- The full form of OLTP is Online Transaction Processing.

Characteristics of OLTP

- OLTP uses transactions that include **small amounts of data.**
- **Indexed data** in the database can be accessed easily.
- OLTP has a **large number of users.**
- It has **fast response times.**
- Databases are directly **accessible to end-users**
- OLTP uses a **fully normalized schema for database consistency.**
- The **response time of OLTP system is short.**
- It strictly performs only the predefined operations on a small number of records.
- OLTP stores the records of the last few days or a week.
- It supports **complex data models and tables.**

ETL Processes



Example of OLTP Transaction

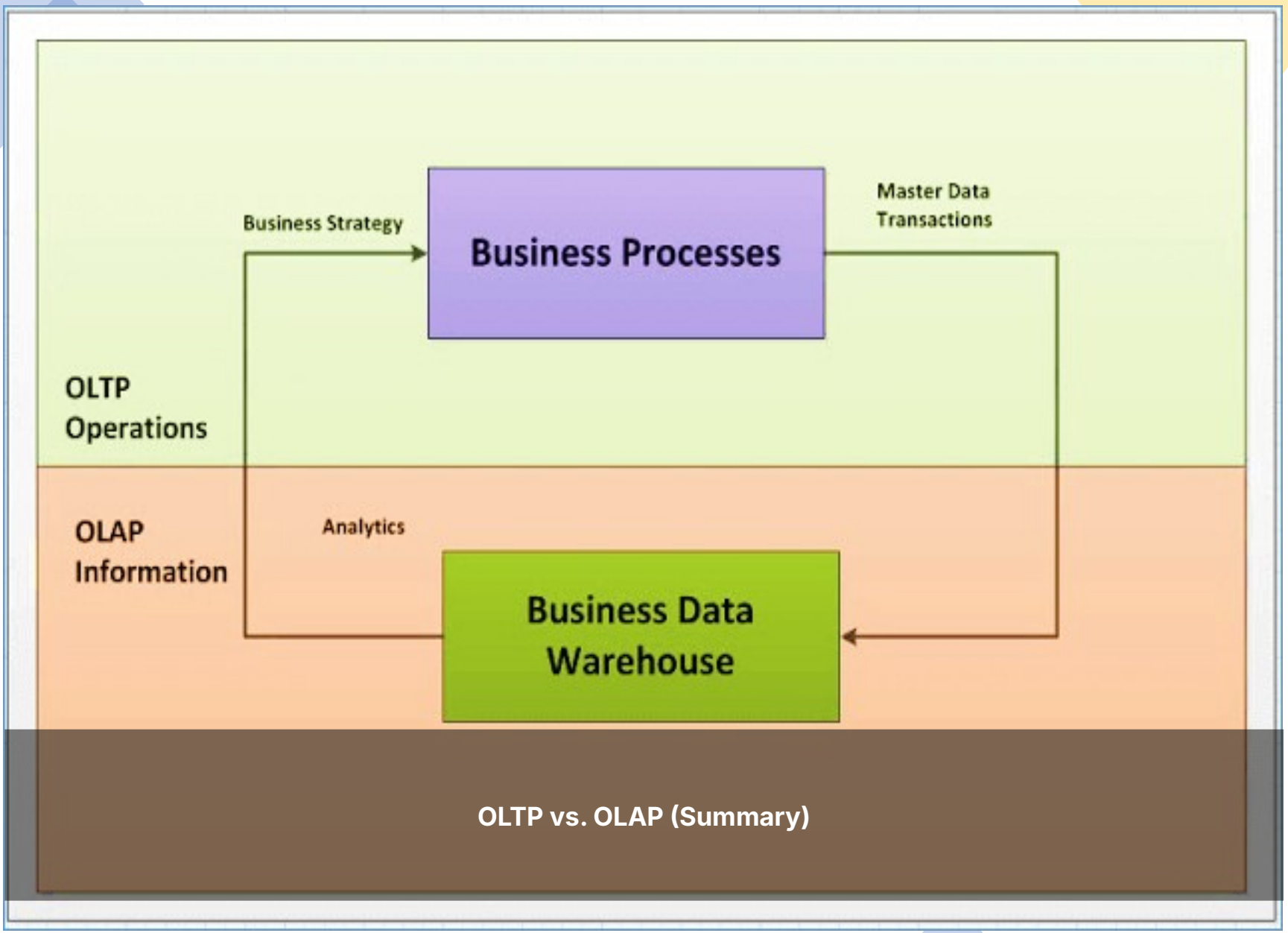


ATM image

•An example of the OLTP system is the **ATM center.**

Other examples of OLTP system are:

- Online banking
- Online airline ticket booking
- Sending a text message
- Order entry
- Add a book to shopping cart

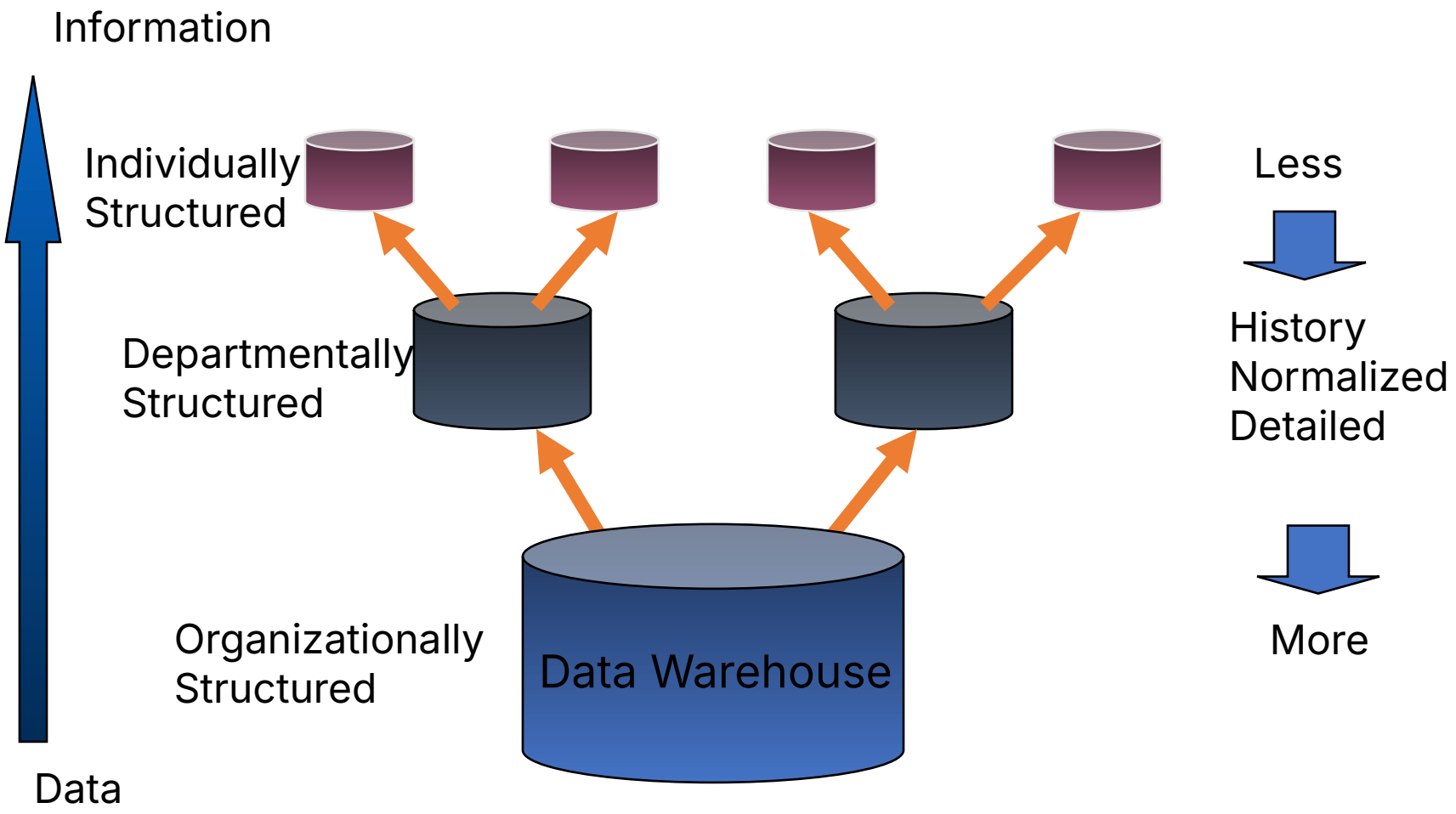


OLTP vs. OLAP (Summary)

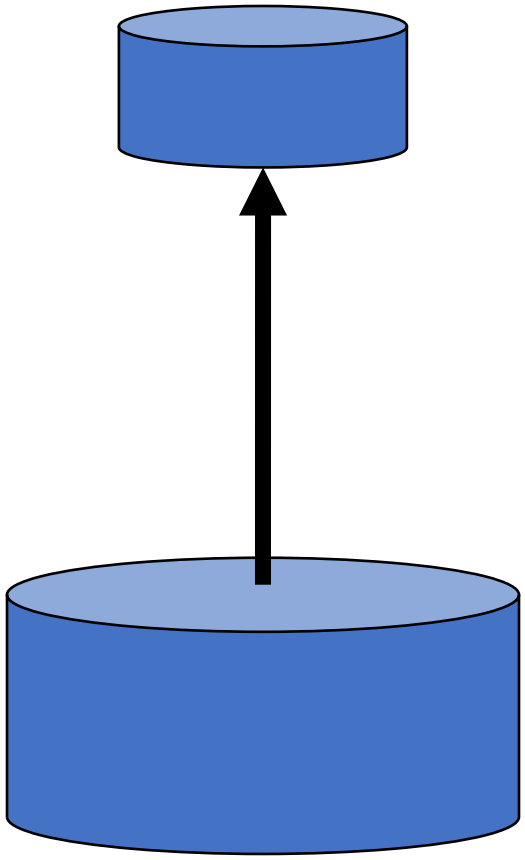
Data Warehouse vs. Data Marts

What comes first?

From the Data Warehouse to Data Marts



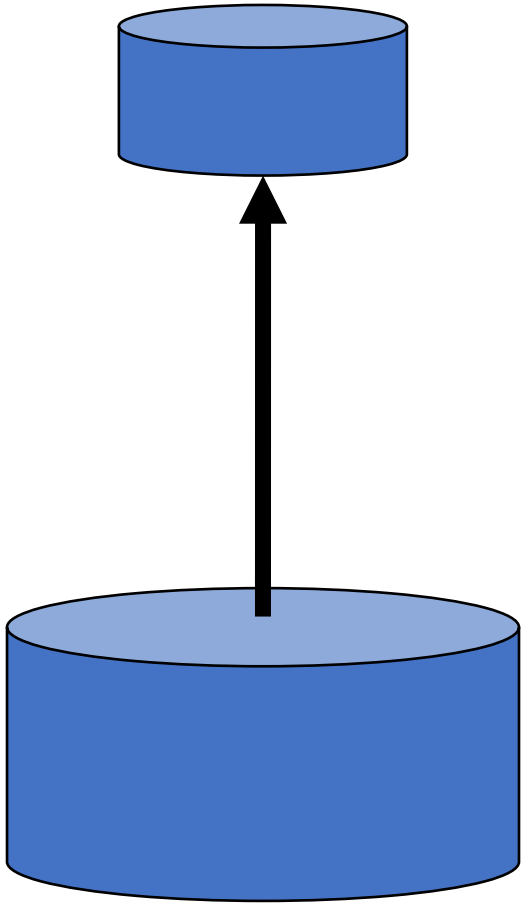
Data Warehouse and Data Marts



OLAP
Data Mart
Lightly summarized
Departmentally structured

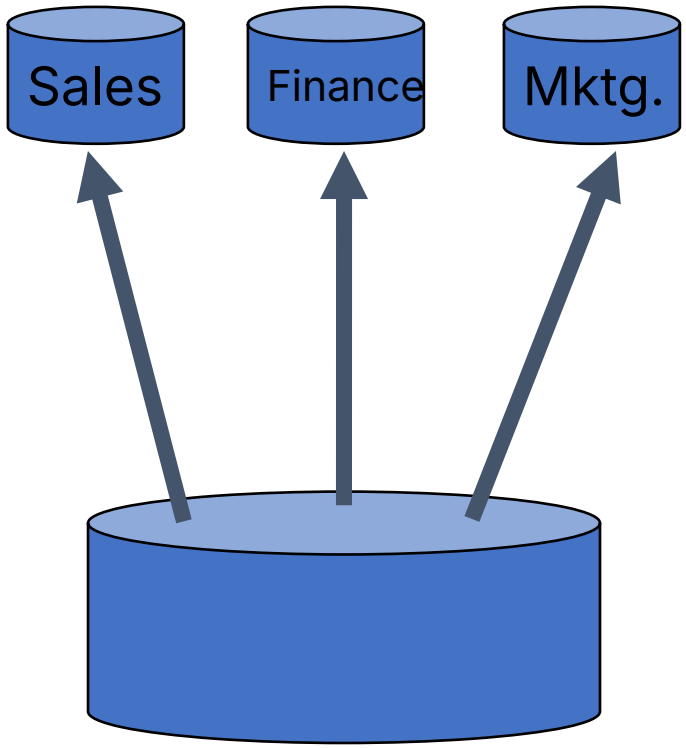
Organizationally structured
Atomic
Detailed Data Warehouse Data

Characteristics of the Departmental Data Mart



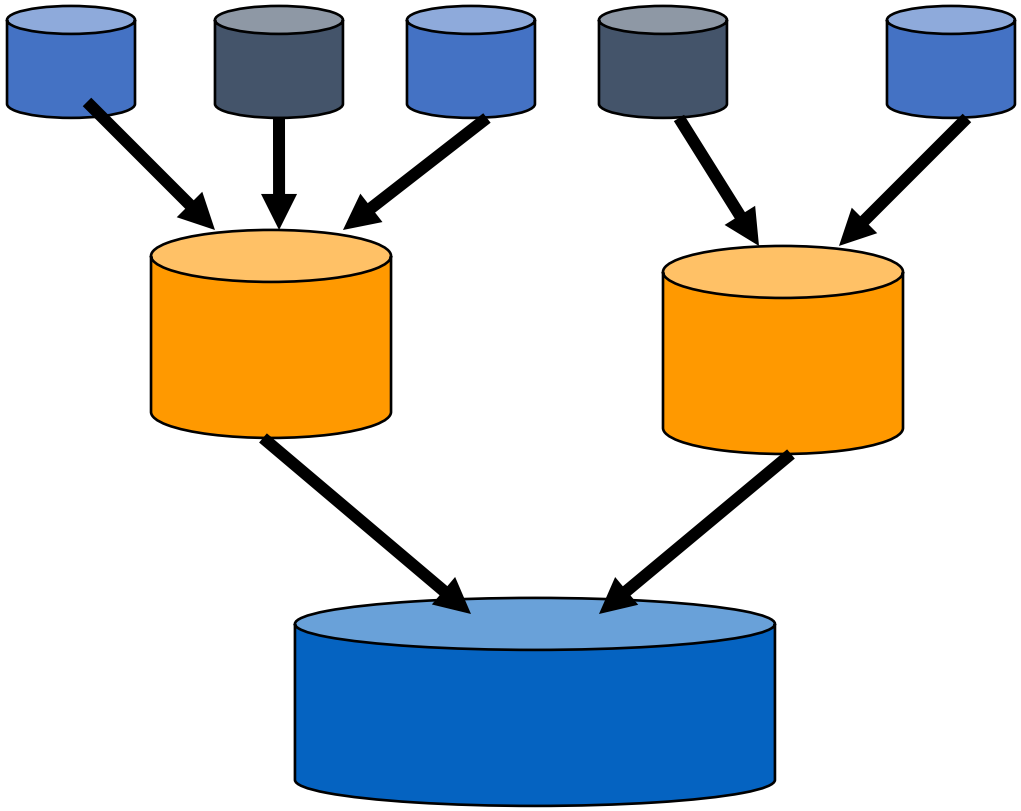
- OLAP
- Small
- Flexible
- Customized by Department
- Source is departmentally structured data warehouse

Techniques for Creating Departmental Data Mart



- OLAP
- Subset
- Summarized
- Superset
- Indexed
- Arrayed

Data Mart Centric

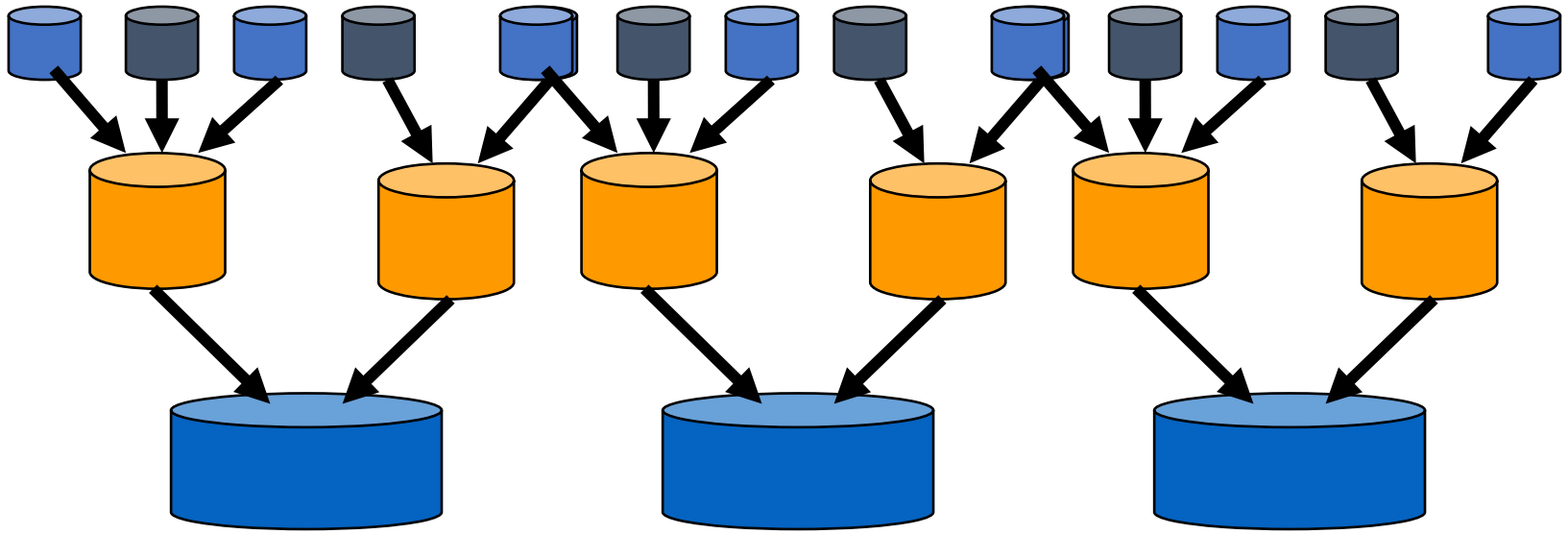


Data Sources

Data Marts

Data Warehouse

Problems with Data Mart Centric Solution



- If you end up creating multiple warehouses, integrating them is a problem

OLAP Is FASMI cont'

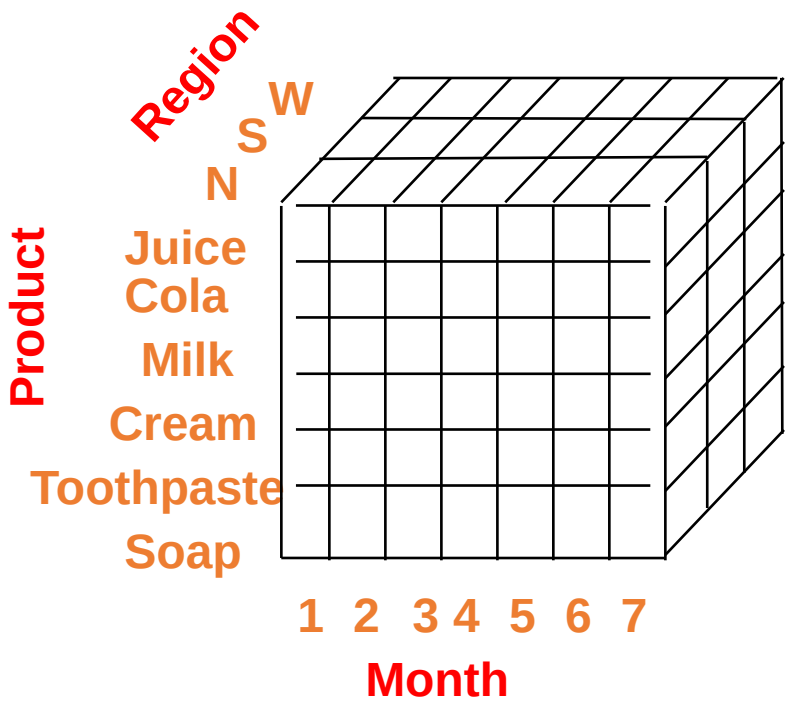
- Fast
- Analysis
- Shared
- Multidimensional
- Information

Fast Analysis of Shared Multidimentional Information is an alternative term of OLAP

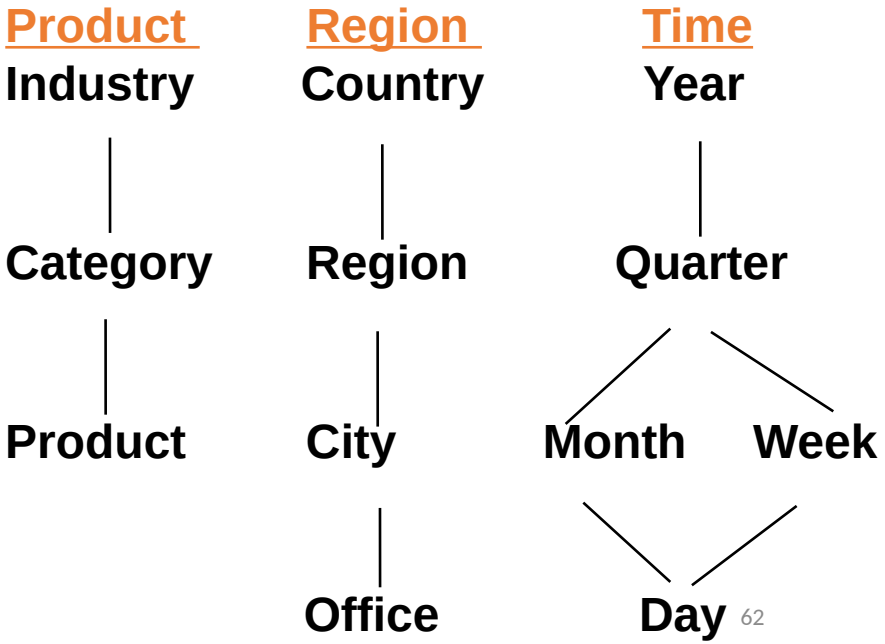
Nigel Pendse, Richard Creath - The OLAP Report

Multi-dimensional Data

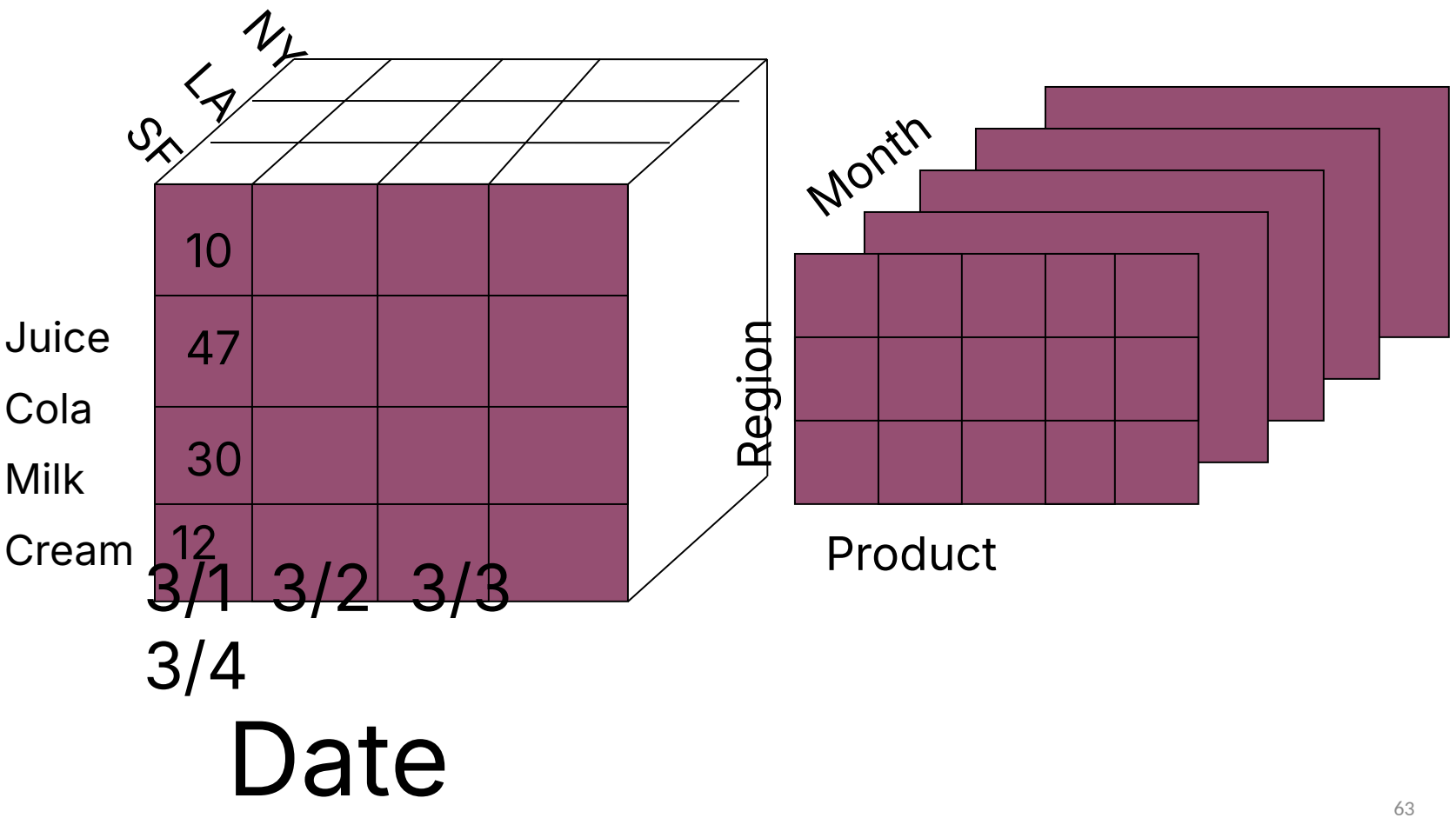
- "Hey…I sold $100M worth of goods"



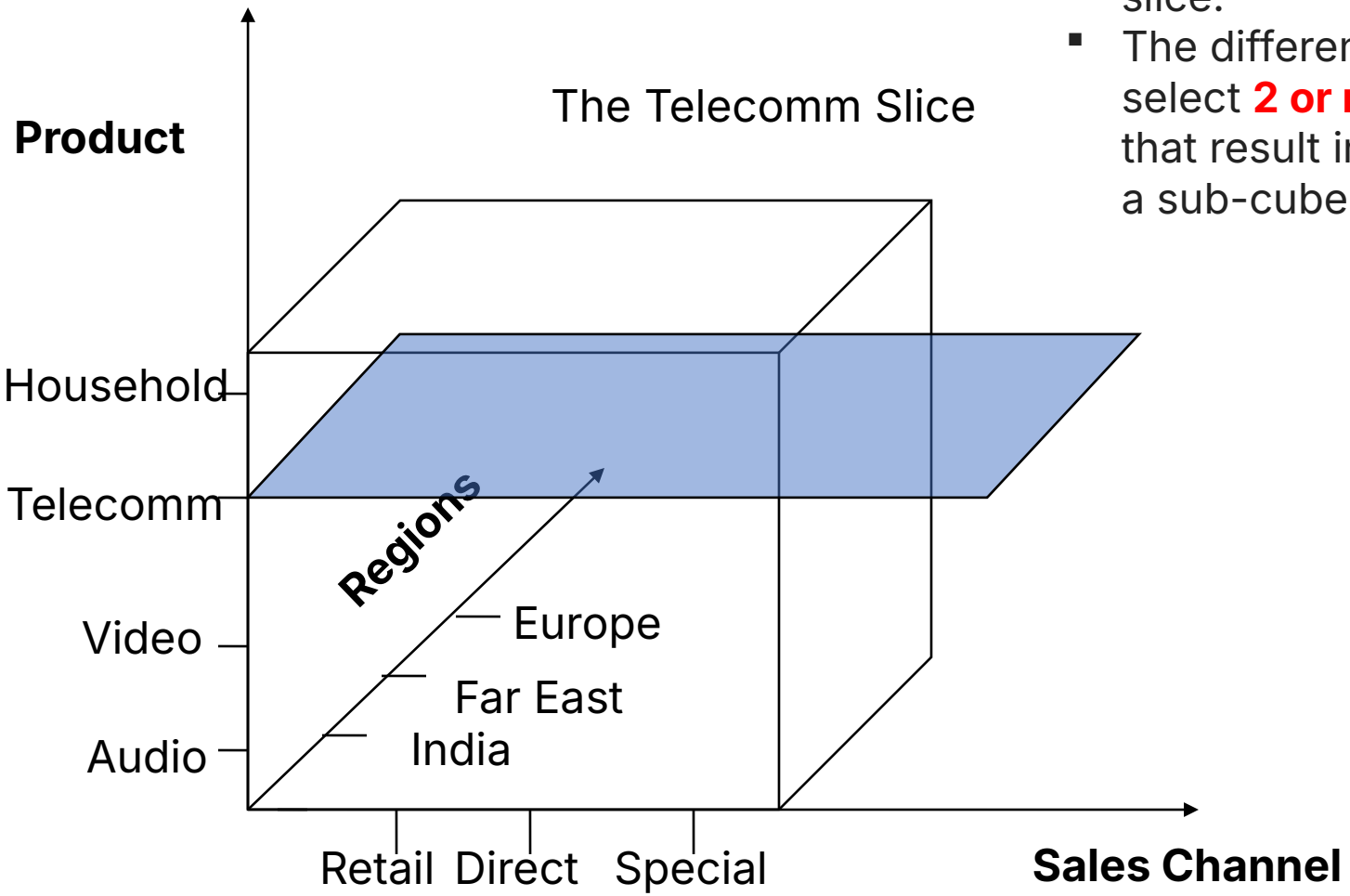
Dimensions: Product, Region, Time
Hierarchical summarization paths



A Visual Operation: Pivot (Rotate)



"Slicing and Dicing"

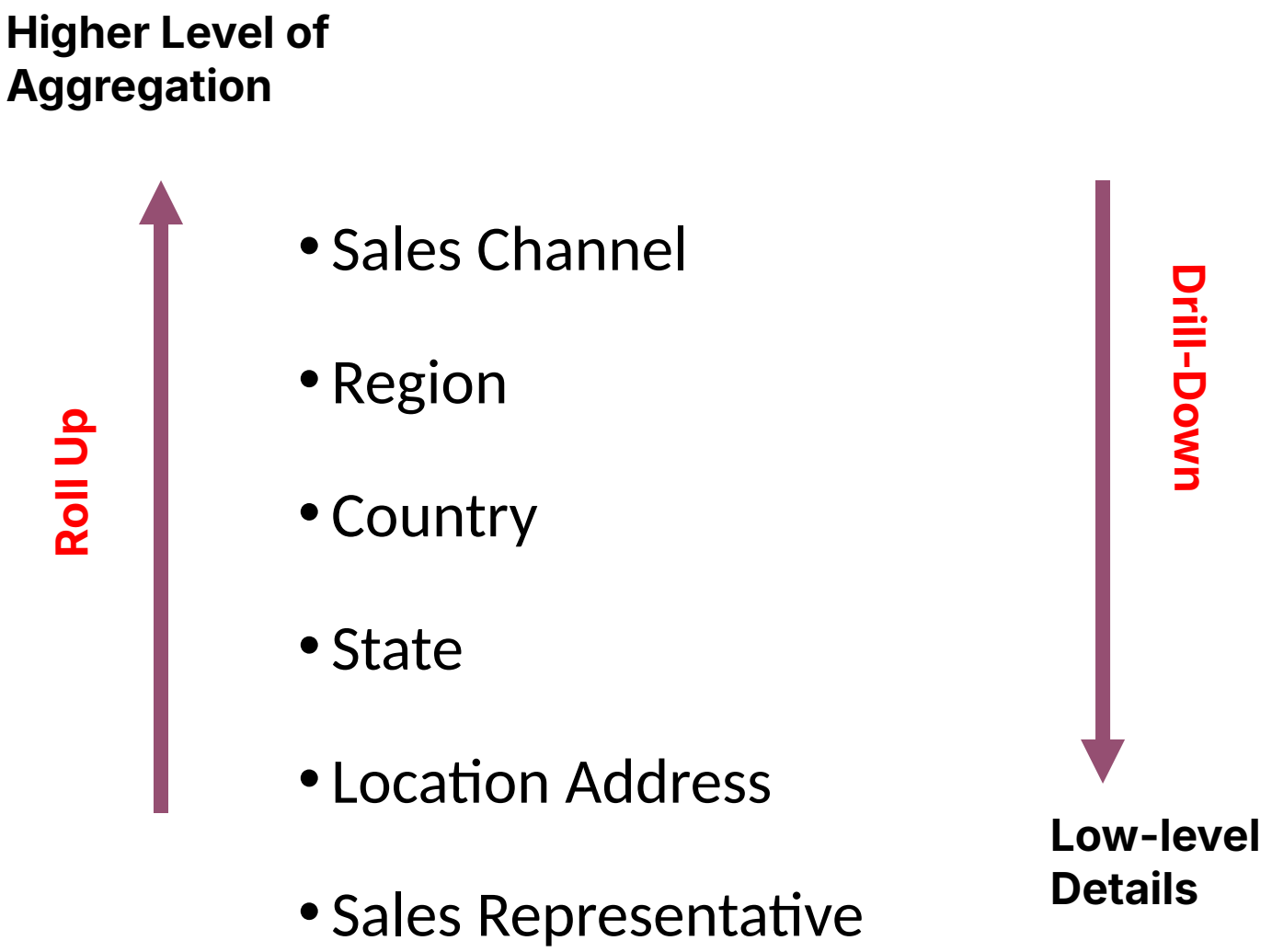


Slicing: Here, one dimension is selected, and a new sub-cube is created.

Dicing:

- This operation is similar to a slice.
- The difference in dice is you select **2 or more dimensions** that result in the creation of a sub-cube.

Roll-up and Drill Down



Data Warehouse Pitfalls

- You are going to spend much time extracting, cleaning, and loading data
- You are going to find problems with systems feeding the data warehouse
- You will find the need to store data not being captured by any existing system
- You will need to validate data not being validated by transaction processing systems

Data Warehouse Pitfalls

- Some transaction processing systems feeding the warehousing system will not contain detail
- Many warehouse end users will be trained and never or seldom apply their training
- Your warehouse users will develop conflicting business rules

Data Warehouse Pitfalls

- 'Overhead' can eat up great amounts of disk space
- The time it takes to load the warehouse will expand to the amount of the time in the available window..
- Assigning security cannot be done with a transaction processing system mindset.
- You are building a HIGH maintenance system.
- You will fail if you concentrate on resource optimization to the neglect of project, data, and customer management issues and an understanding of what adds value to the customer

DW and OLAP Research Issues

- **Data cleaning**
  - ✓focus on data inconsistencies, not schema differences
  - ✓data mining techniques
- **Physical Design**
  - ✓design of summary tables, partitions, indexes
  - ✓tradeoffs in use of different indexes
- **Query processing**
  - ✓selecting appropriate summary tables
  - ✓dynamic optimization with feedback
  - ✓acid test for query optimization: cost estimation, use of transformations, search strategies
  - ✓partitioning query processing between OLAP server and backend server.

DW and OLAP Research Issues .. 2

- **Warehouse Management**
  - ✓ detecting runaway queries
  - ✓ resource management
  - ✓ incremental refresh techniques
  - ✓ computing summary tables during load
  - ✓ failure recovery during load and refresh
  - ✓ process management: scheduling queries, load and refresh
  - ✓ Query processing, caching
  - ✓ use of workflow technology for process management

Data Warehouse (References)

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- W.H. Inmon, J. D. Welch, Katherine L. Glassey, Managing the Data Warehouse, John Wiley and Sons, 1997
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Data Warehouse (References)

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OLAP and DSS

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- Microsoft TechEd Transparencies from Microsoft TechEd 98

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- Michael J.A. Berry and Gordon Linoff, Data Mining Techniques, John Wiley and Sons 1997
- Peter Adriaans and Dolf Zantinge, Data Mining, Addison Wesley Longman Ltd. 1996

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- Donovan Schneider, Data Warehousing Tutorial, Tutorial at International Conference for Management of Data (SIGMOD 1996) and International Conference on Very Large Data Bases 97
- Umeshwar Dayal and Surajit Chaudhuri, Data Warehousing Tutorial at International Conference on Very Large Data Bases 1996
- Anand Deshpande and S. Seshadri, Tutorial on Datawarehousing and Data Mining, CSI-97

Useful URLs

- Data Warehousing
  - https://www.tutorialspoint.com/dwh/index.htm
  - OLAP
  - https://www.tutorialspoint.com/dwh/index.htm

Business Intelligence Overview

Introduction to Business Intelligence

- The Decision Making Process
- Data, Information,knowledge,decision making-Action cycle
- Business Intelligence-overview
- Business Intelligence-Definition
- Business Intelligence Process
- Business Intelligence Components
- Business Intelligence Layers

Equipping the Organization for Effective Decision Making and Decision-Making Process

Introduction

- **Decision making** is a daily **activity for any human being**.
- When it comes to **business organizations**, **decision making** is a **habit and a process** as well.
- **Effective and successful decisions** make **profit to the company** and unsuccessful ones **make losses**.
- Therefore, corporate **decision making process** is the most **critical process in any organization**.

Introduction cont'

- In the **decision-making process**, we choose one course of **action from a few possible alternatives.**
- In the **process of decision making**, we may use **many tools, techniques and perceptions**.
- In addition, we may make our own **private decisions** or may prefer a **collective decision**.
- Usually, **decision making** is hard.
- Majority of corporate decisions involve some level of **dissatisfaction or conflict with another party**.

Introduction cont'

- **Decision making** can be **regarded as the mental processes resulting** in **the selection of a course of action among several alternatives.**
- **Every decision-making process** produces a **final choice**.
- The output can be an **action or an opinion of choice**.

Definition:

In general:

- **Decision making** is the **process of making choices by identifying a decision, gathering information, and assessing alternative resolutions.**
- Using **a step-by-step decision-making process** can help you make more **deliberate, thoughtful decisions** by organizing relevant information and dening alternatives.
- This approach increases the chances that you will choose the most satisfying alternative possible.

Definition cont’

- **Decision :**

Making a choice from two or more alternative.

- **Decision Making:**

The **process of examining your possibilities and options, comparing them, and choosing a course of action**.

Decision Making Process

- **There are 8 steps of Decision Making:**

✓ Identification of problem

✓ Identification of decision Criteria

✓ Allocating weight to criteria

✓ Develop alternatives

✓ Analysis of Alternatives

✓ Selection of alternatives

✓ Implementation of the best alternatives

✓ Evaluation of decision effectiveness

Step1: Identifying The Problem

- A problem is defined as a ***discrepancy between an existing and a desired state of affairs.***
- Some cautions about problem identification include the following:
  - 1.Make sure it's a problem and not just a symptom of a problem.
  - 2.Problem identification is subjective.
  - 3.**Discrepancies** can be found by comparing **current results with some standard.**
  - 4.**Managers aren't likely to characterize a discrepancy** as **a problem** if they perceive that they **don't have the authority, information, or other resources needed to act on it.**



Identifying The Problem



Step2: Identifying Decision Criteria

- **Decision criteria** are factors that are Important (relevant) to Resolving the problem.

A. Costs that will be incurred(investment required)

B. Risk likely to be encountered(chance of failure)

C. Outcomes that are desired (growth of the firm)

Step3: Allocating Weights to the Criteria

- **Decision criteria are not of equal importance**:
  - **Assigning a weight to each item places the items in the correct priority order of their importance in the decision making.**
- Each possible solution must have features or cons and pros.
- A manager should **score the criteria according** to the need of the **situation**.



Step4: developing Alternatives

- As **manager** it is advised you select a **viable alternative that could also resolve the problem.**
- At this point we are **just listing** not **evaluating the alternatives.**



Step5: Analyzing Alternatives

- Appraising each alternative **strengths and weaknesses**
  - An alternative's appraisal is based on **its ability to resolve the issues identified**



Step6: Selecting Alternatives

Choosing the best alternative or the one that scored the highest total in step 5.





Step 7: Implementing the Decision

- Putting the decision into action by conveying into those affected and getting their commitment to it.

Step8: Evaluating the Decision's Effectiveness

- Evaluating the outcome or result of the decision to see if the problem was resolved.
- If the problem still persists, the managers should re-assess the problem and start over again.



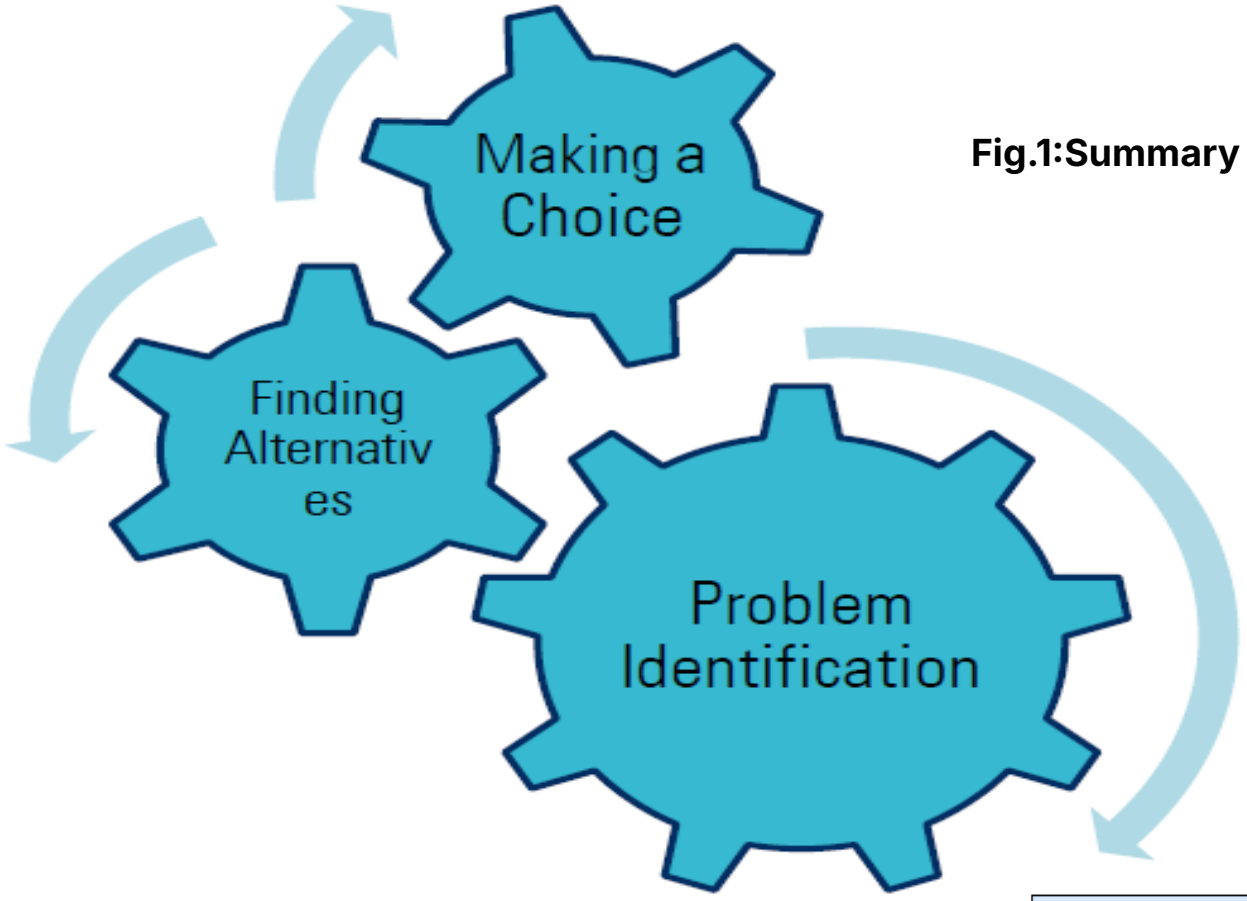


Fig.1:Summary of Decision making steps

Decision making steps include

– Problem identification,

– Finding alternative solutions,

– Making a choice

"Information and knowledge form the backbone of the decision making process"

Who Makes Decisions?

Decision making at different levels:

– Operational

- Related to daily activities with **short-term effect**
- Structured decisions taken by **lower management**

– Tactical

- Semi-structured decisions taken by **middle management**

– Strategic

- Long-term effect
- Unstructured decisions taken by **top management**



Fig.2:Who makes Decisions

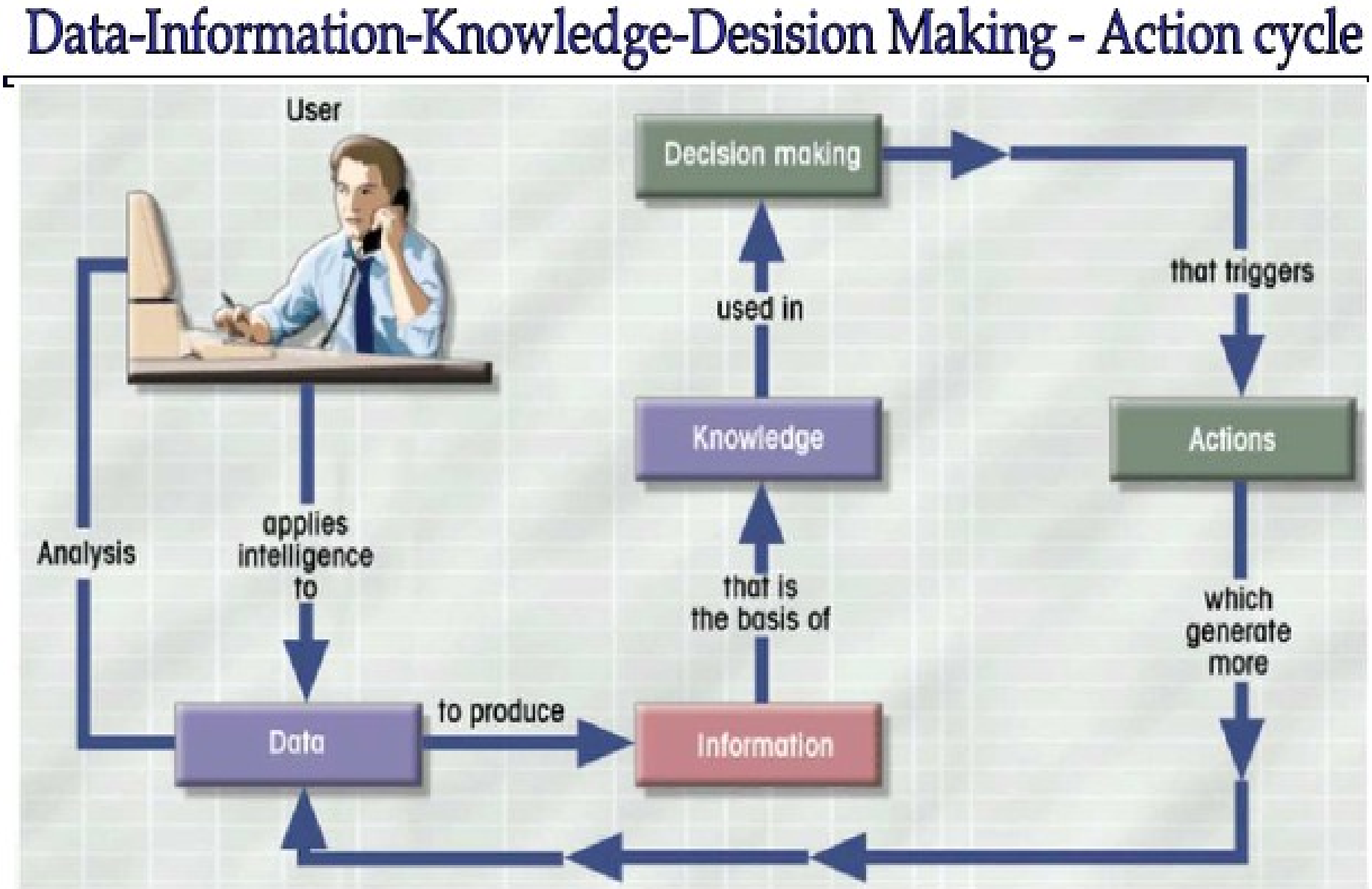
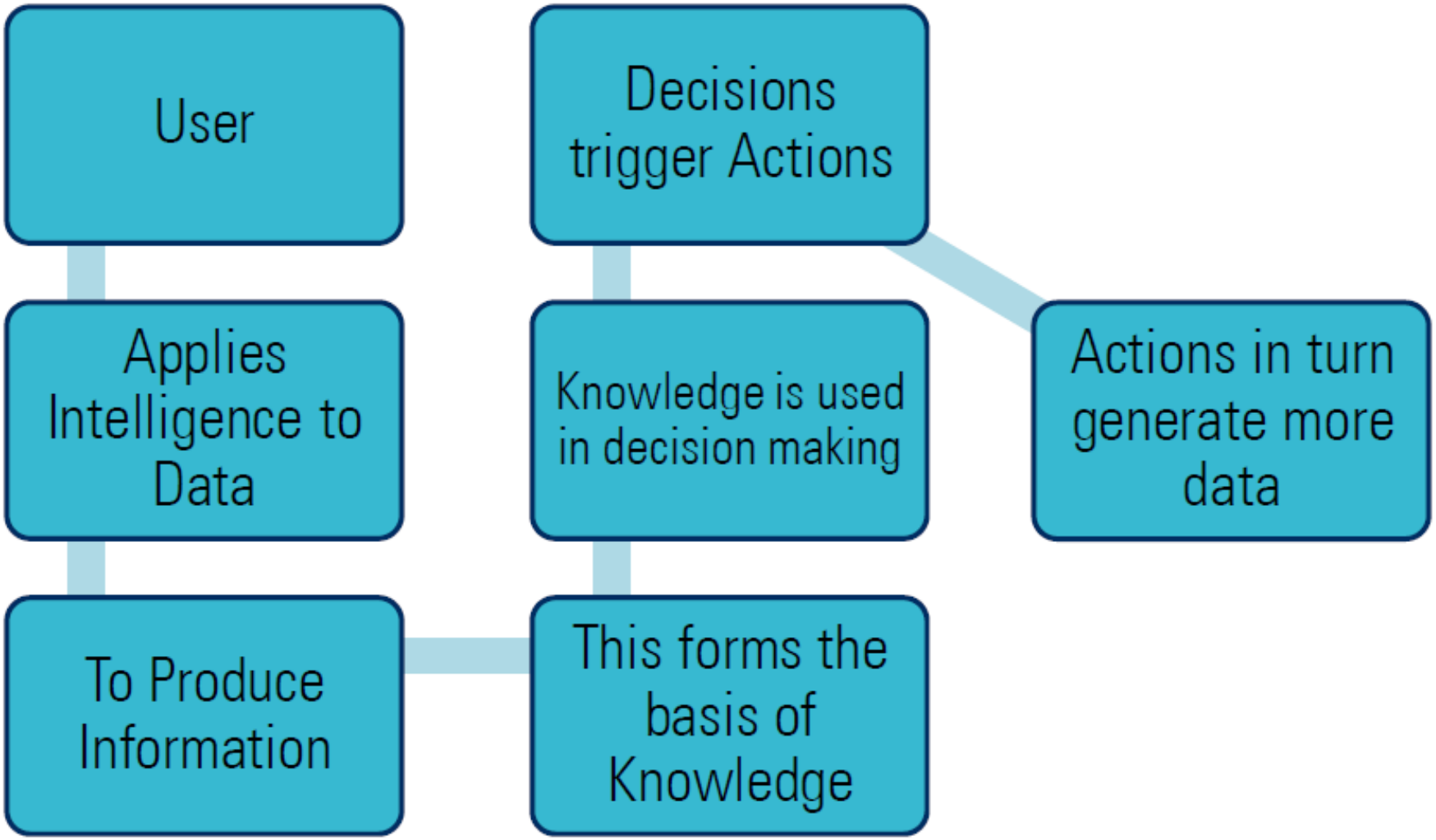
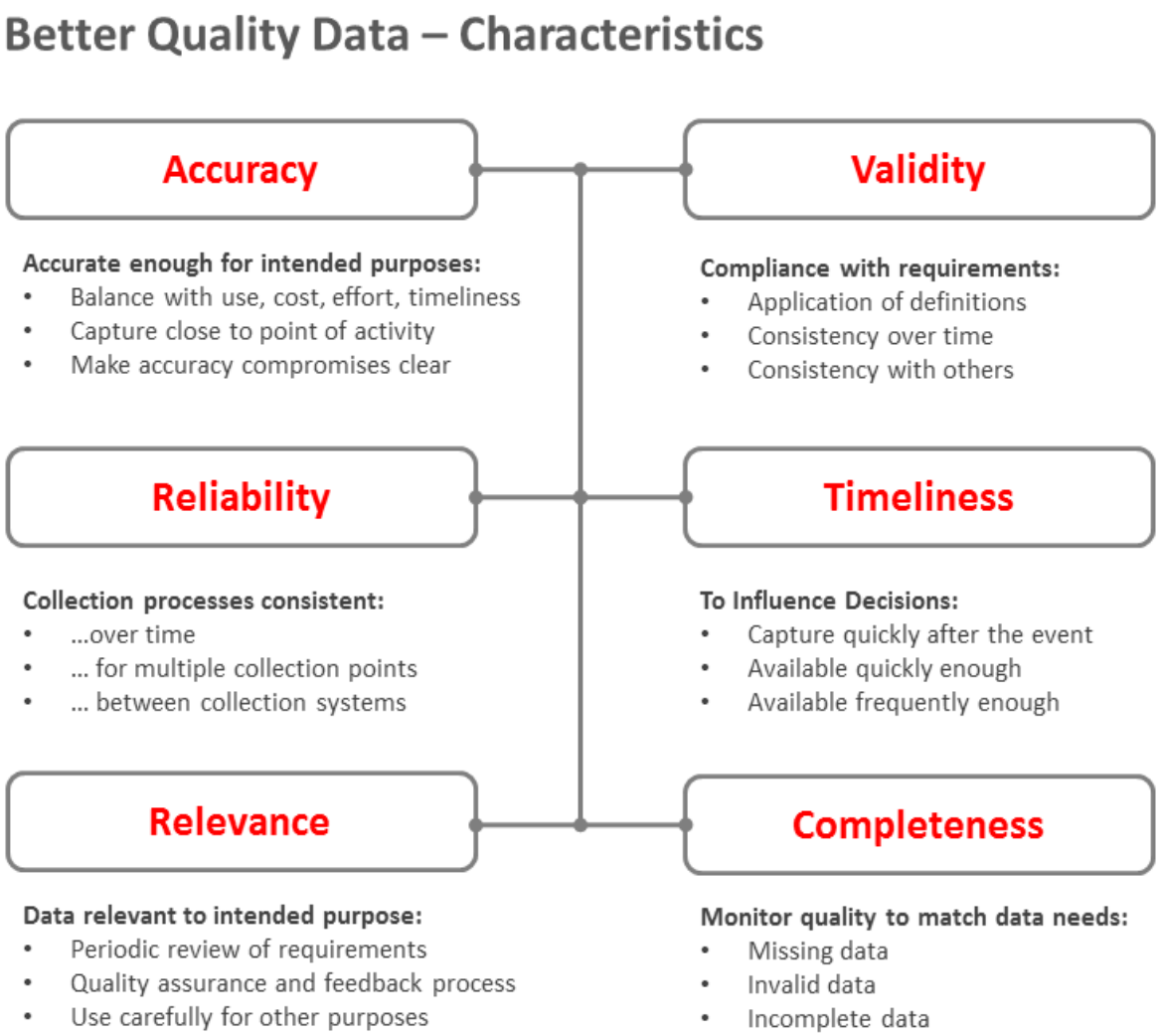


Fig.3:Action cycle of Decision making

Explanation:



Characteristics of Data for Good Decision Making



Conclusion

- When it comes to making decisions, one should always weigh the **positive and negative business consequences** and **should favor the positive outcomes.**
- This avoids the **possible losses to the organization** and **keeps the company running with a sustained growth**.
- But, making the **decisions and accepting its consequences is the only way to stay in control of your corporate life and time**.

Business Intelligence

Overview & introduction

Business Analytics, BI, Big Data, Data Mining - What's the difference?

- **Business Analytics –** Tools to explore past data to gain insight into future business decisions.
- **BI –** Tools and techniques to turn data into meaningful information.
- **Big Data –**data sets that are so large or complex that traditional data processing applications are inadequate.
- **Data Mining -** Tools for discovering patterns in large data sets.

Overview of Business Intelligence

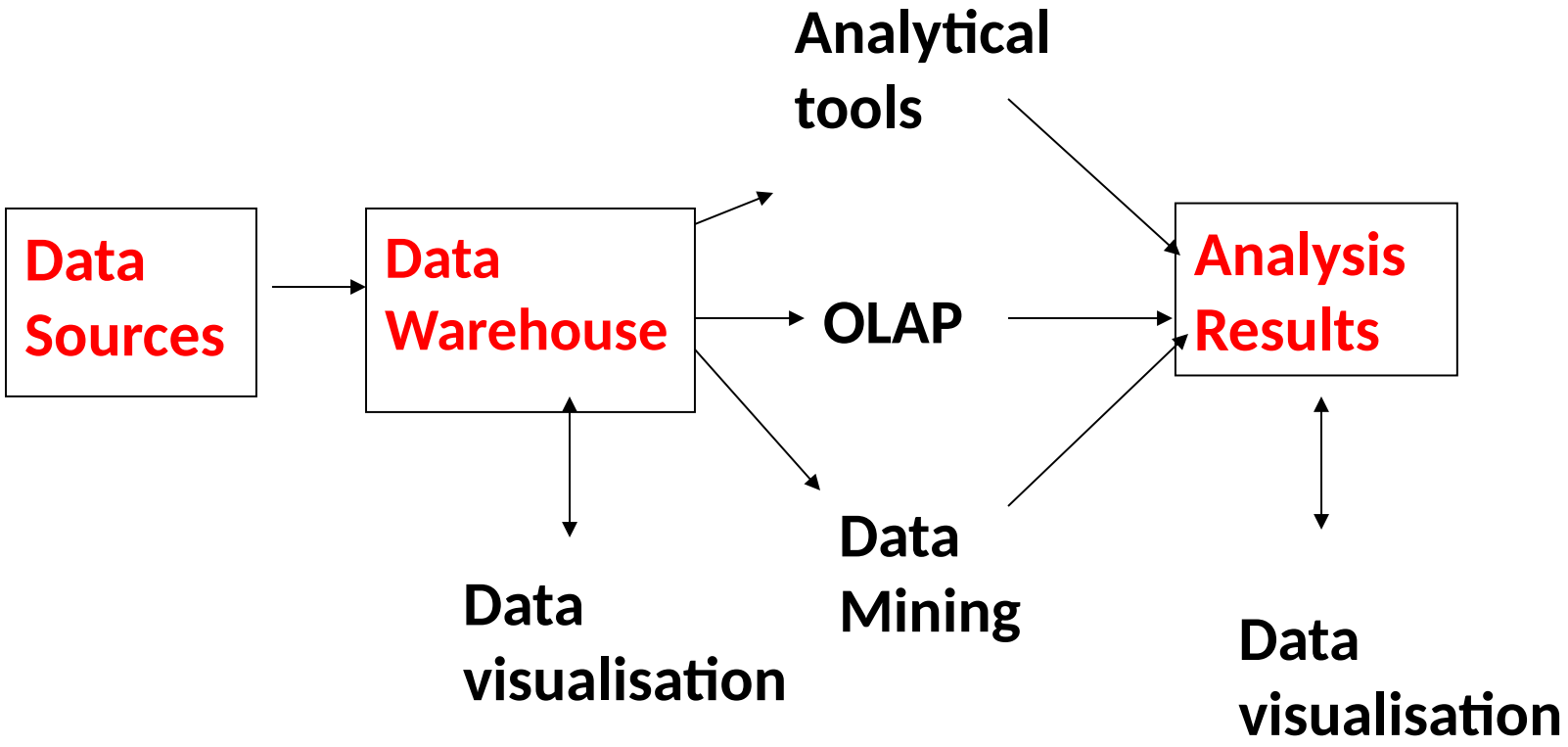


Fig.4:BI Overview

What is Business Intelligence?

- There are many definitions of **Business Intelligence** as below, each definition raises a prominent characteristic of **BI** yet the general rules are all mentioned the **possibility of helping to make efficiency decision** of **Business Intelligence**.

- ✓**Collecting and refining information from many sources** (internal and external)
- ✓**Analyzing and presenting the information in useful ways** (dashboards, visualizations)
- ✓So **that people can make better decisions**
- ✓**That help build and retain competitive advantage.**

- **Business Intelligence (BI)** refers to skills, processes, technologies, applications and practices used to support decision making.
- **Business intelligence - BI** is the set of theories, methodologies, architectures and techniques and tools for the **transformation of raw data** into **meaningful and useful information for business Analysis purposes.**
- **BI** is the applications and technologies transforming Business Data into Action.
- **BI** is the **new technology for understanding the past & predicting the future.**

Summary on BI definition:

- **BI** refers to computer-based techniques used in spotting, digging-out, and analyzing business data, such as sales revenue by products and/or departments or associated costs and incomes.
- **BI technologies** provide historical, current, and predictive views of business operations.
- **Business Intelligence** often aims to support better business decision-making.

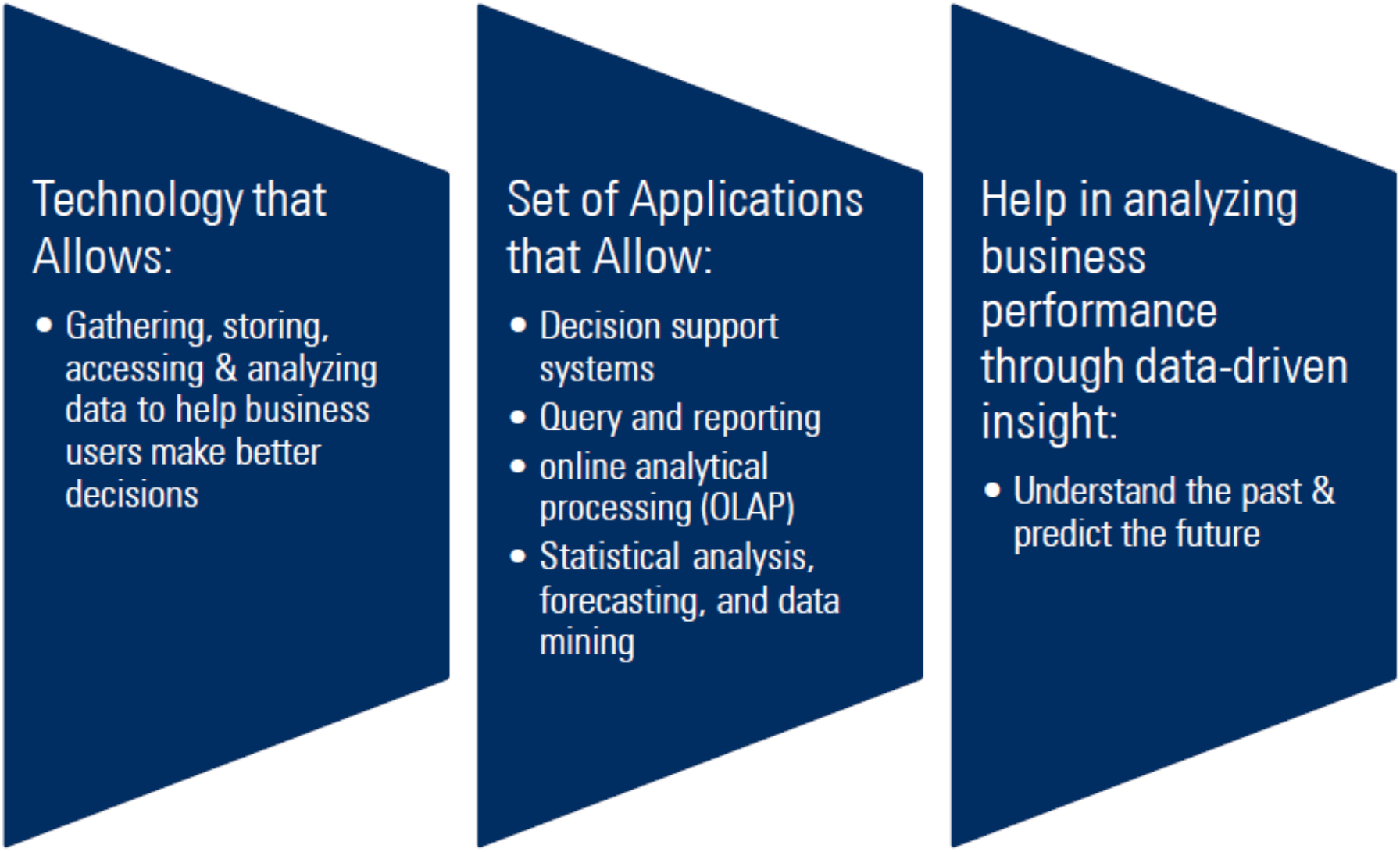
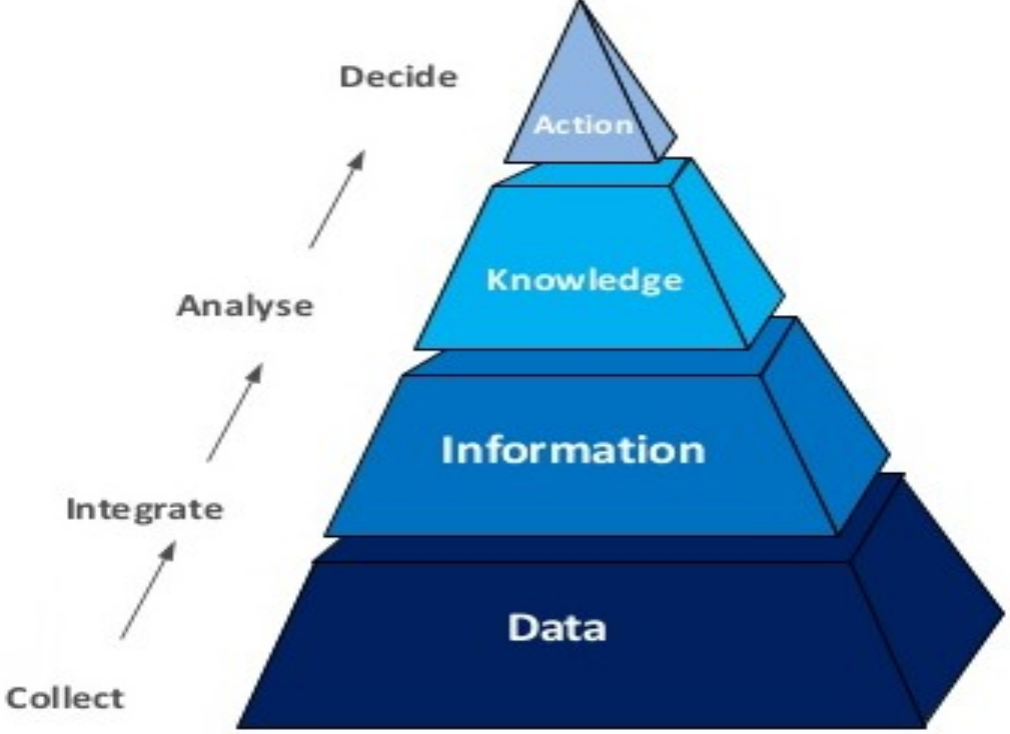
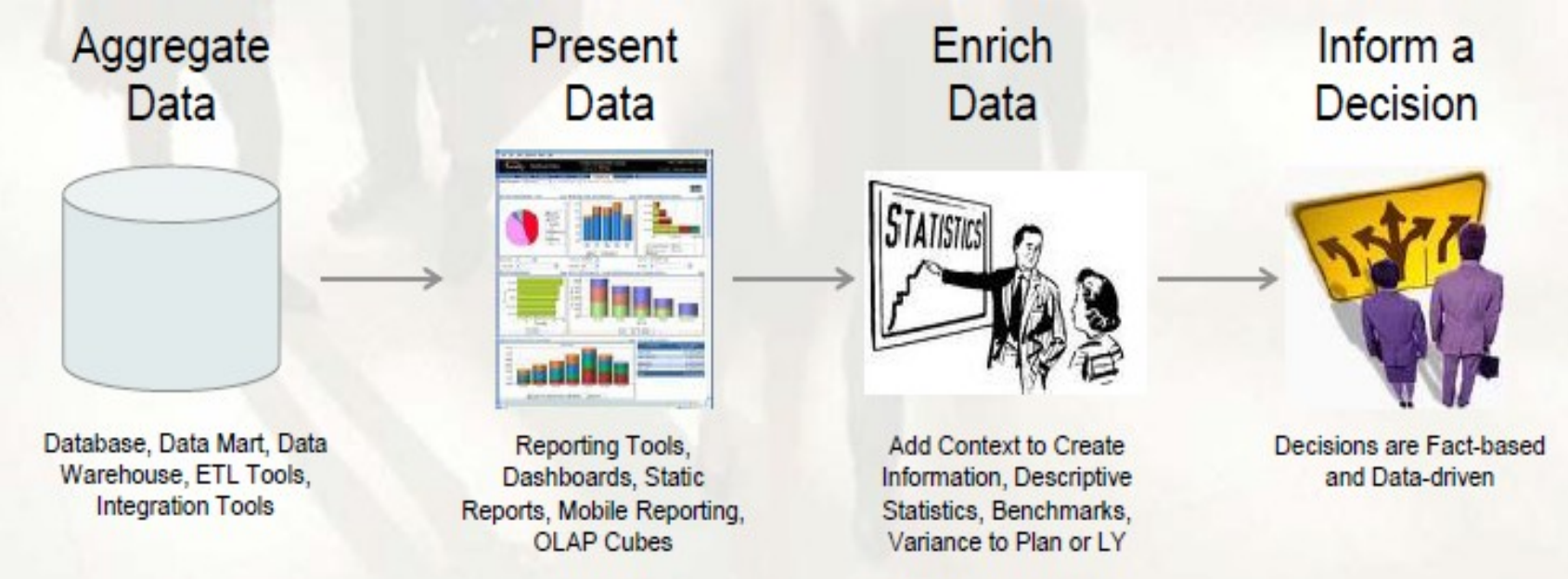


Fig.4:BI Definition



- **BI** is a process and technology that enterprises use to control the huge **number of data, mining data to make the decision- making** more efficient in **their business activity.**
- The target of **BI** is to help enterprises make better decision
- Therefore, **BI** system is also known as Decision Support System **(DSS).**
- **BI** technology provides an overview on activities of enterprises from the **past, present to future predictions**.

- Business Intelligence enables **the business to make intelligent, fact-based decisions**.



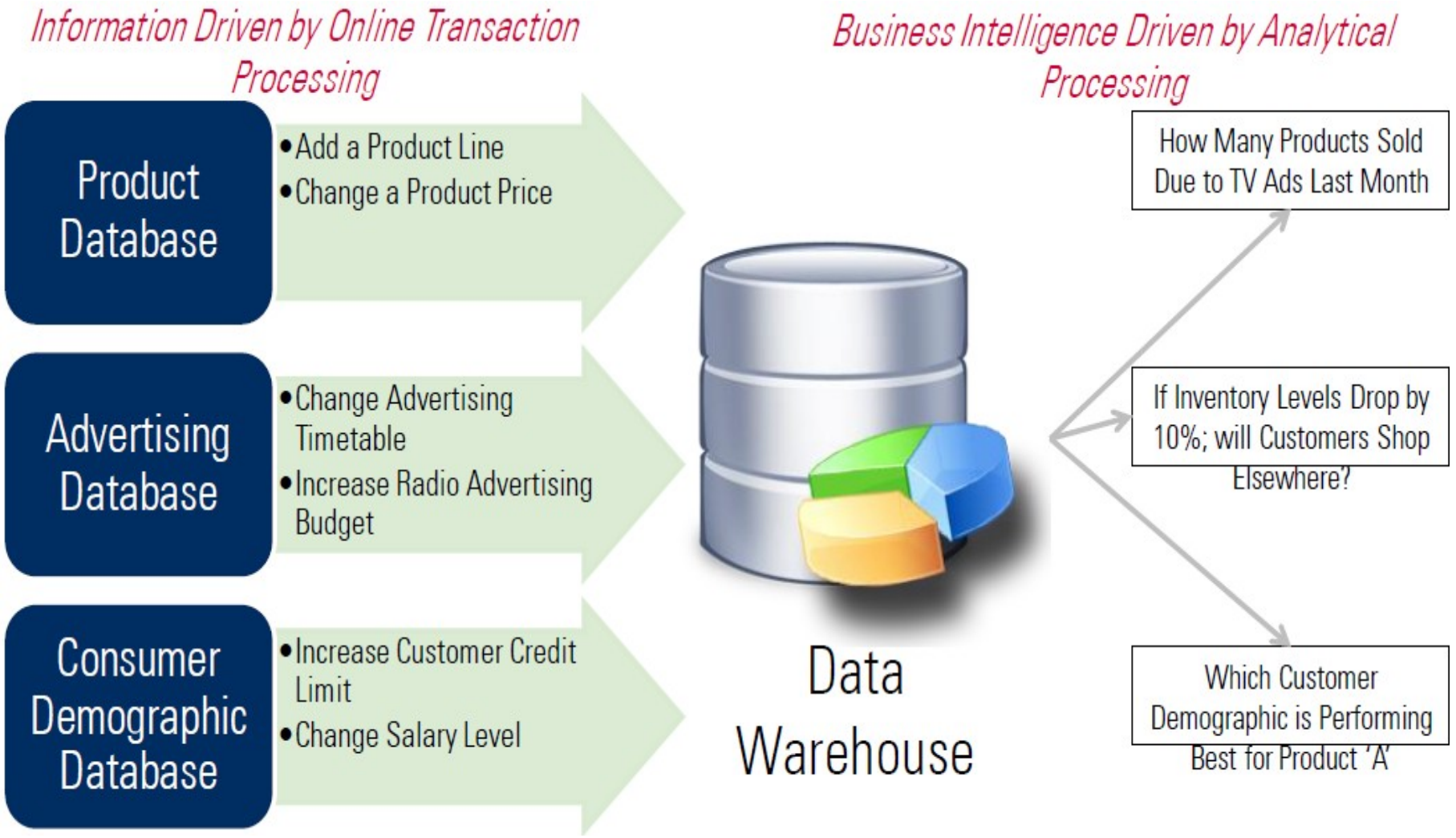
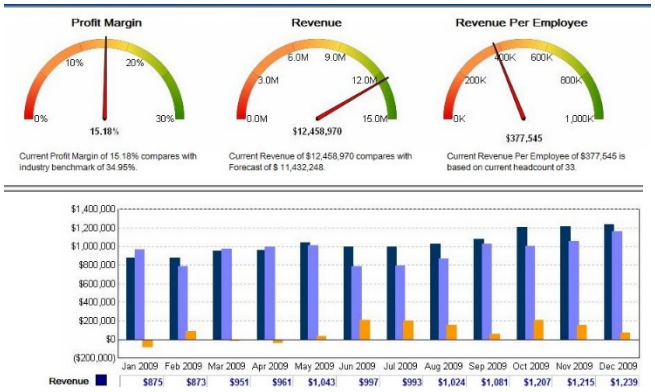
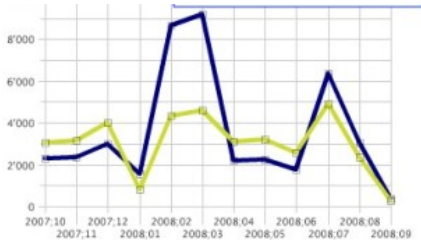
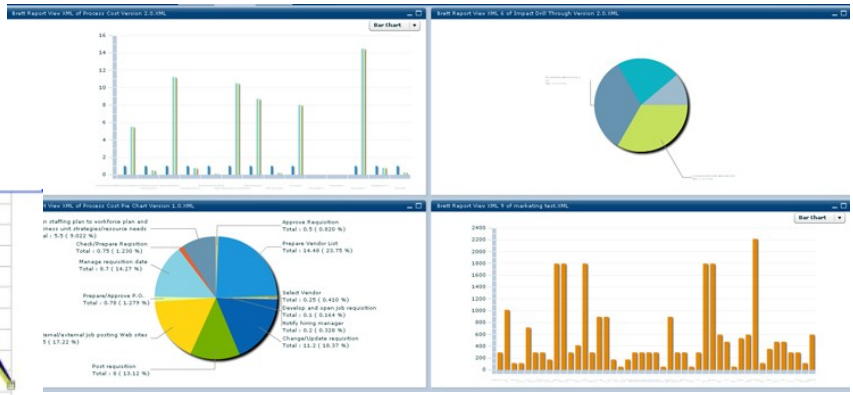


Fig.6:BI Example







Reports

Data repository

Different systems

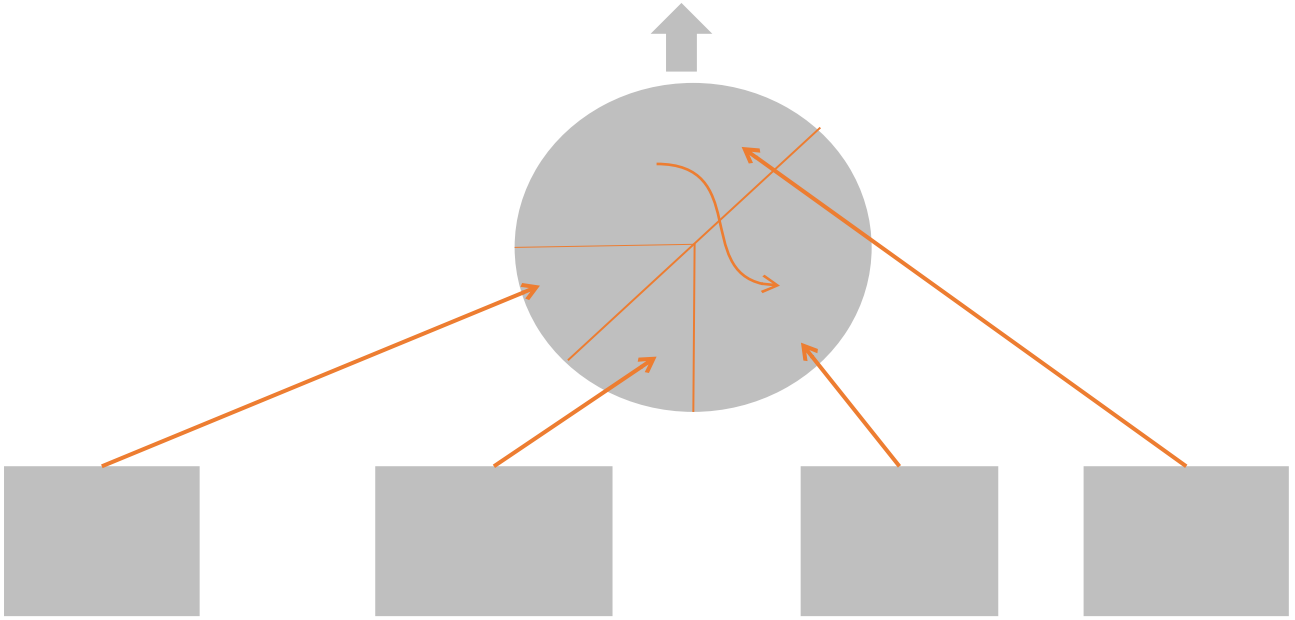
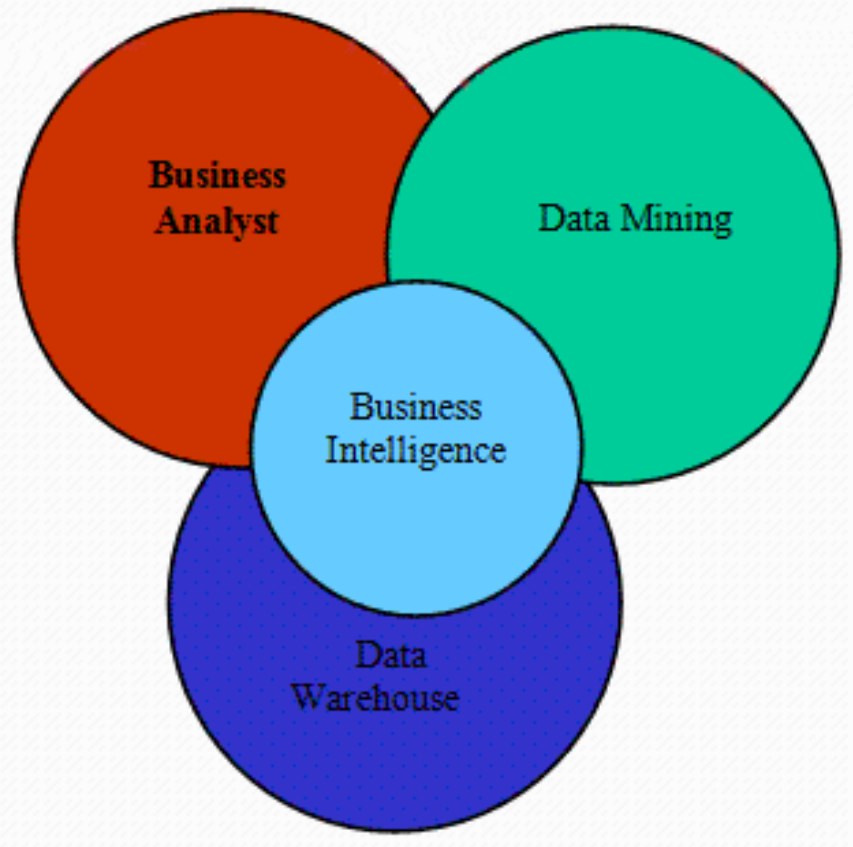
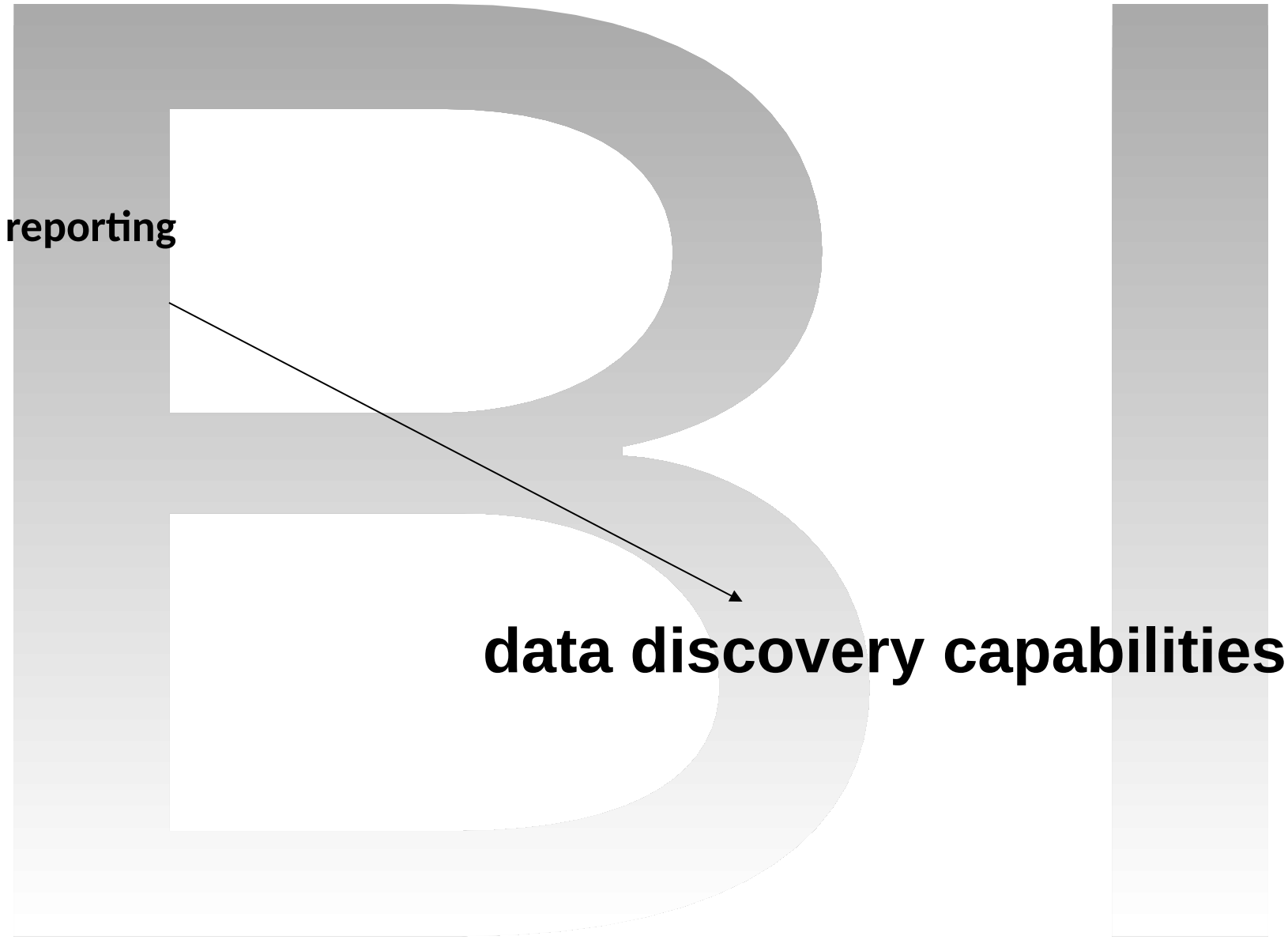


Fig.6:BI Example Cont'

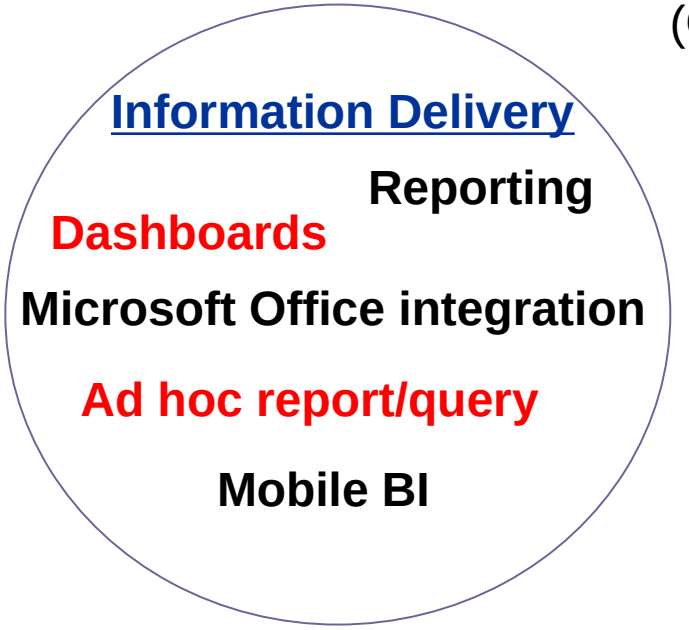
- A simple BI system can be viewed as a combination of three main components as follows:

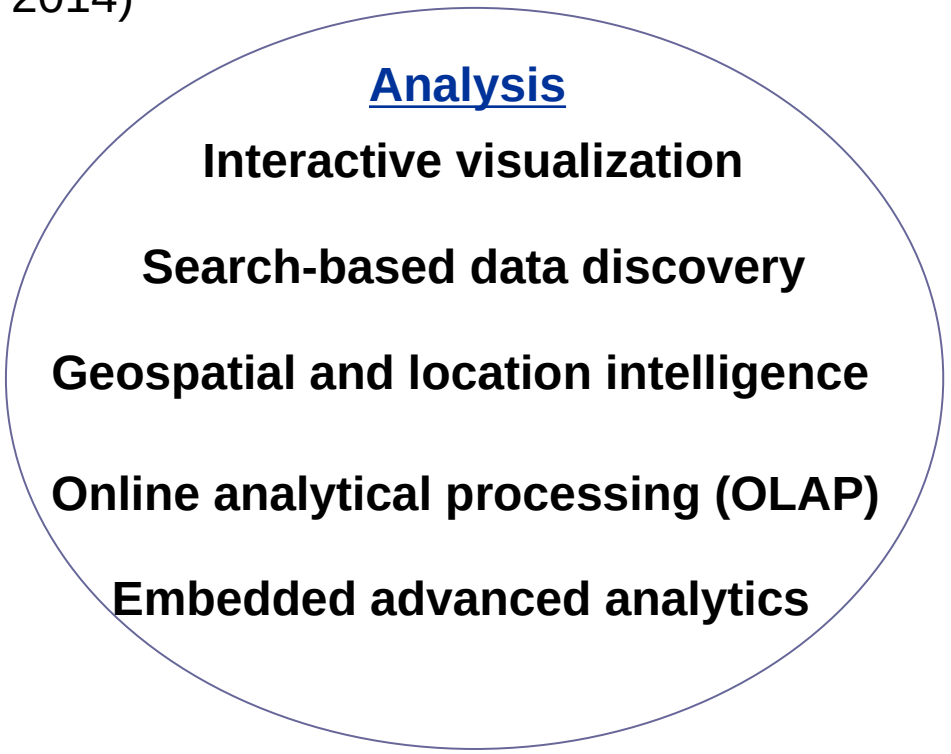


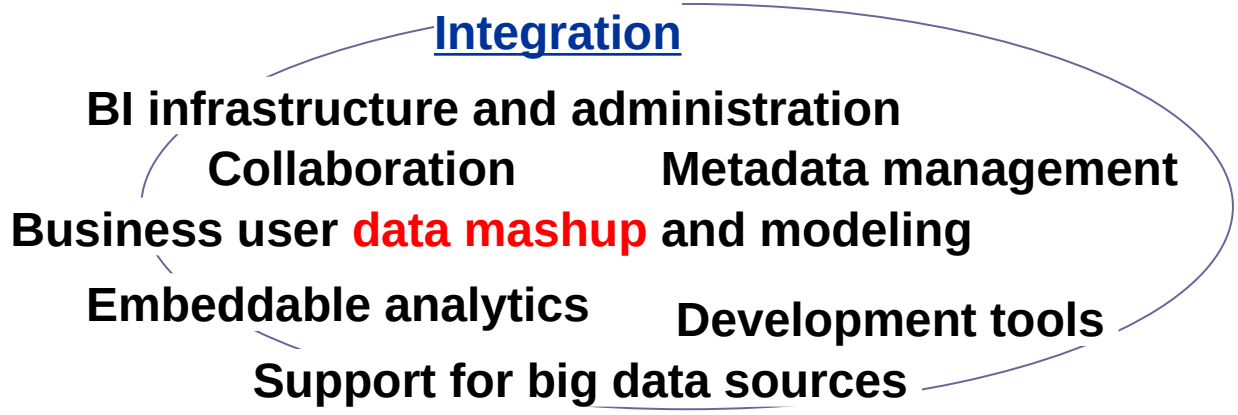
- **Data Warehouse**: contains comprehensive data of the firm.
- **Data Mining**: includes techniques to exploit data and **discover information** like **Classification, Clustering, Association Rule, and Prediction**.
- **Business Analyst:** Leaders of enterprises make **strategic decisions on activities of the firm.**


reporting
data discovery capabilities

BI and analytics
(Gartner, 2014)

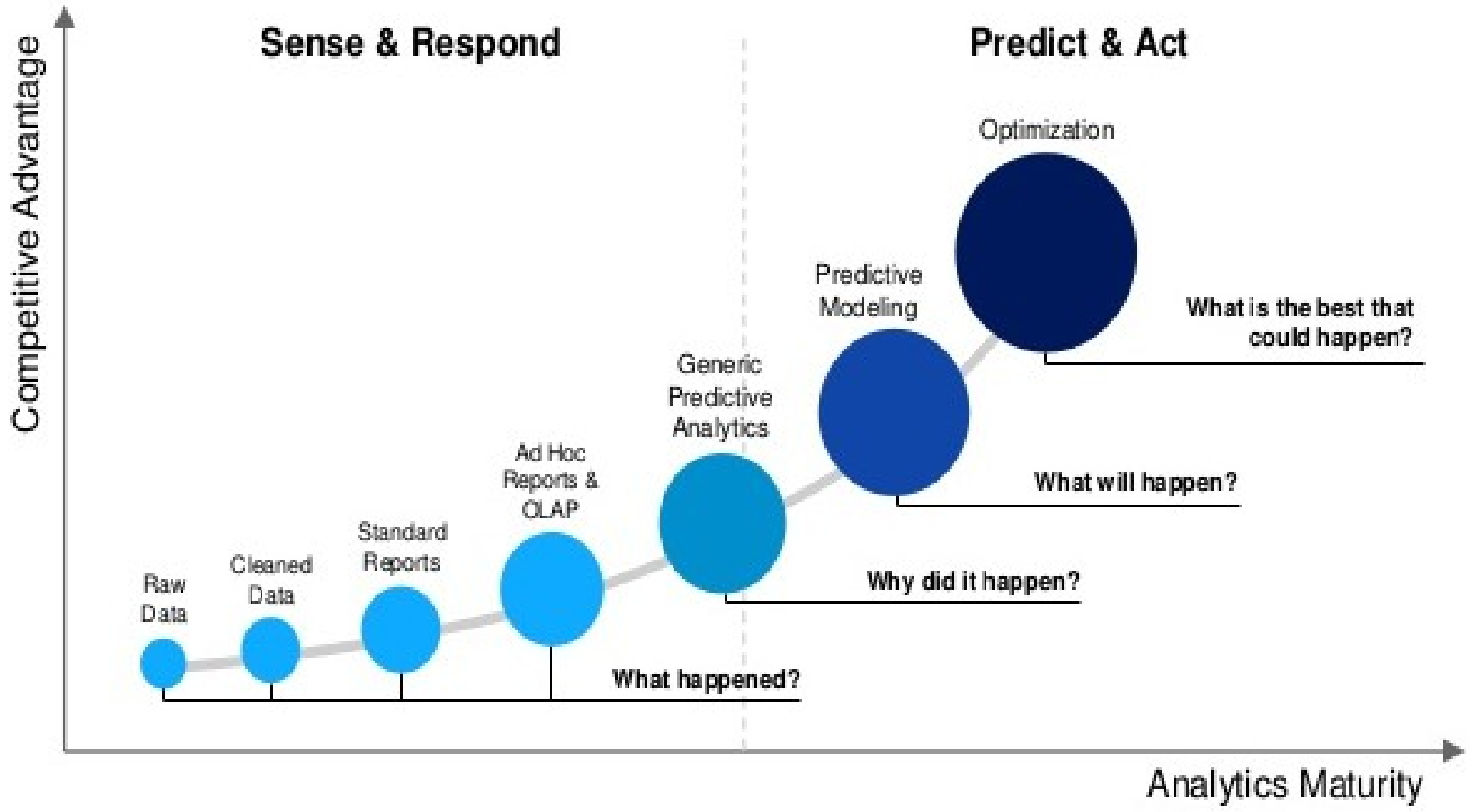






- **Data mashup** is an integration of two or more **data** sets in a single graphical interface.
- **Dashboard** is a prime example of a **data mashup** hard at work – most dashboards pull **data** from multiple, disparate sources

Evolution of BI



Why is BI So Important?



Making Business Decisions is a Balance

In the absence of data, business decisions are often made by the HiPPO.

With Business Intelligence, we can get data to you in a timely manner.



11

Why is BI So Important (Cont')?

- Corporate data is doubling every 2-3 years
- Barriers of entry (costs/technology) are being removed
- Continued pressure on businesses to find efficiencies and new market opportunities, client expectations
- More different data sources than ever before



12

BI Business Areas

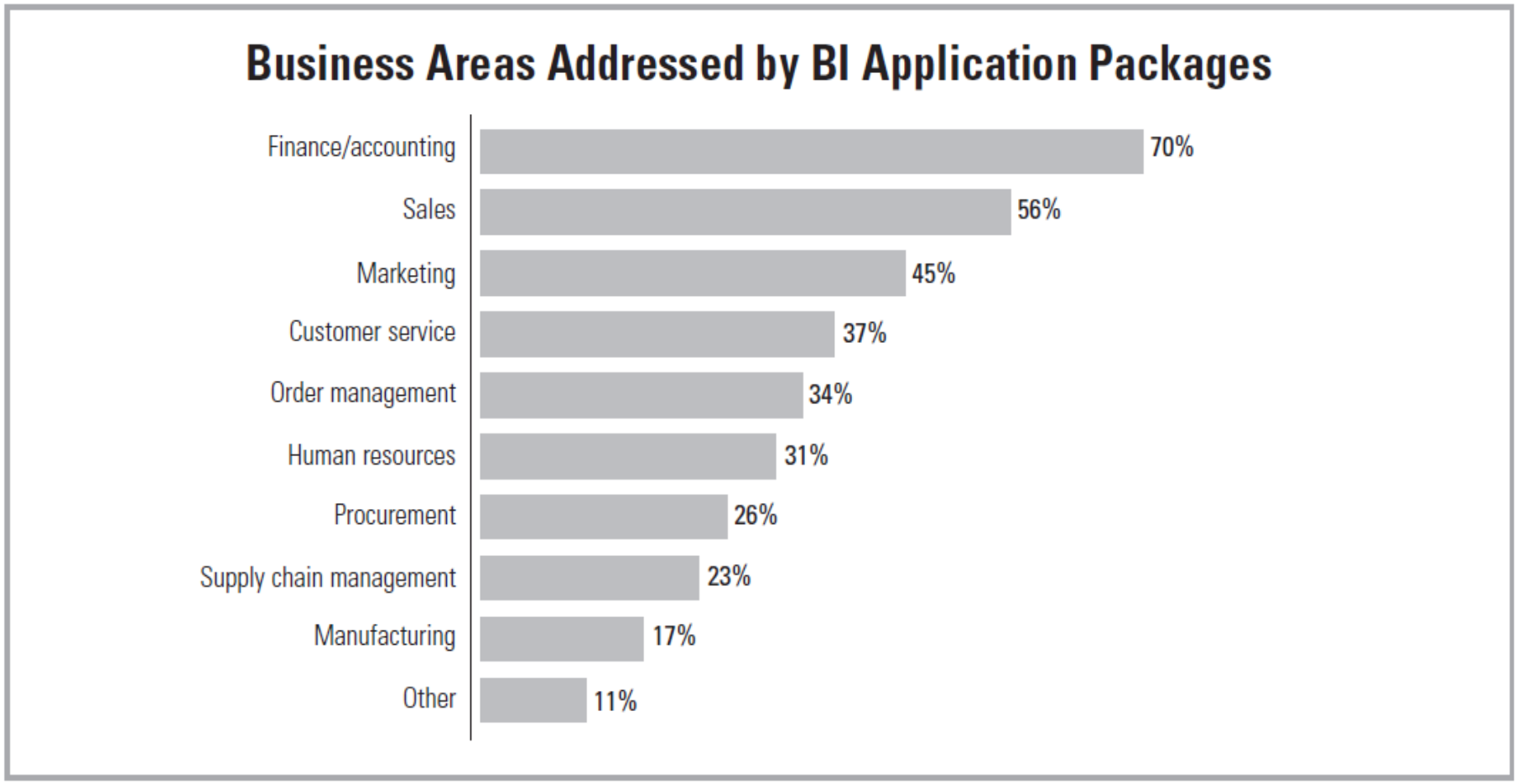


Illustration 1.7: Which business areas do your BI application packages (from BTx application vendors) support? Based on 482 respondents.

The Data Warehousing Institute (TDWI)

BI Business Areas (examples)

Marketing: Helps in analyzing campaign returns, promotional yields and fine tune spending to get better ROI, tracking social media marketing

Sales: Finding best path and best practices, customer acquisition cost and length, process improvement, year-on-year turnover and sales analysis

Inventory: Monitoring and adjusting inventory levels.

Human Resource: Track and manage things like employee turnover, attrition rate, managing recruitment process etc.

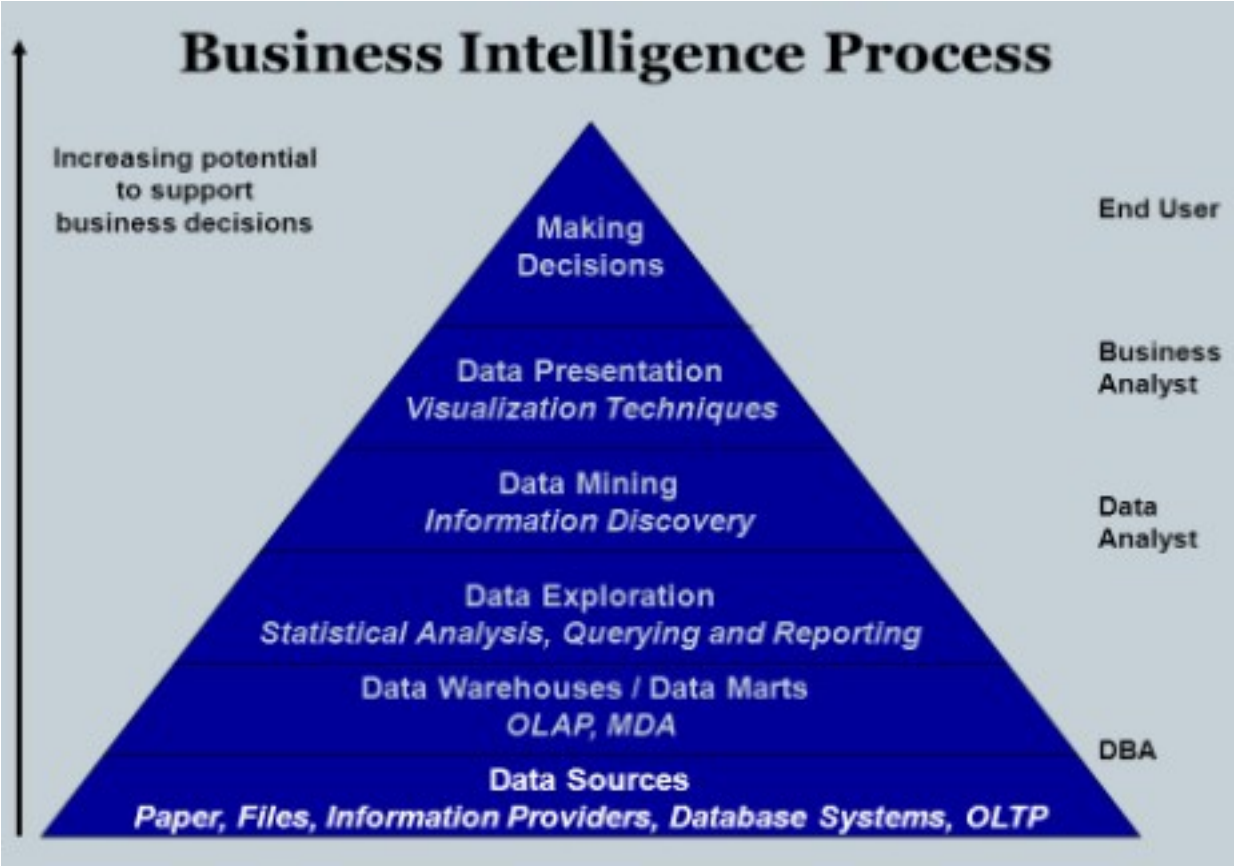
Business Intelligence Process

"Every Level Helps Increase the Potential to Support Business Decisions"



ey aspect of BI is
Warehouse and **Data**
g since data in BI is
esized data.

Data Warehouse is characterized by various sources and formats, and it is **clustering and historical.**



- Besides, **analyzing data in BI** is not simple **(query, Filtering),** but it uses **Data Mining technique to do classification, clustering, or prediction.**
- Thus, BI has a very tight relationship with **Data Warehouse** and **Data mining**.

BI Strategy

- Improving organizations by providing business insights to **all** employees leading to better, faster, more relevant decisions.

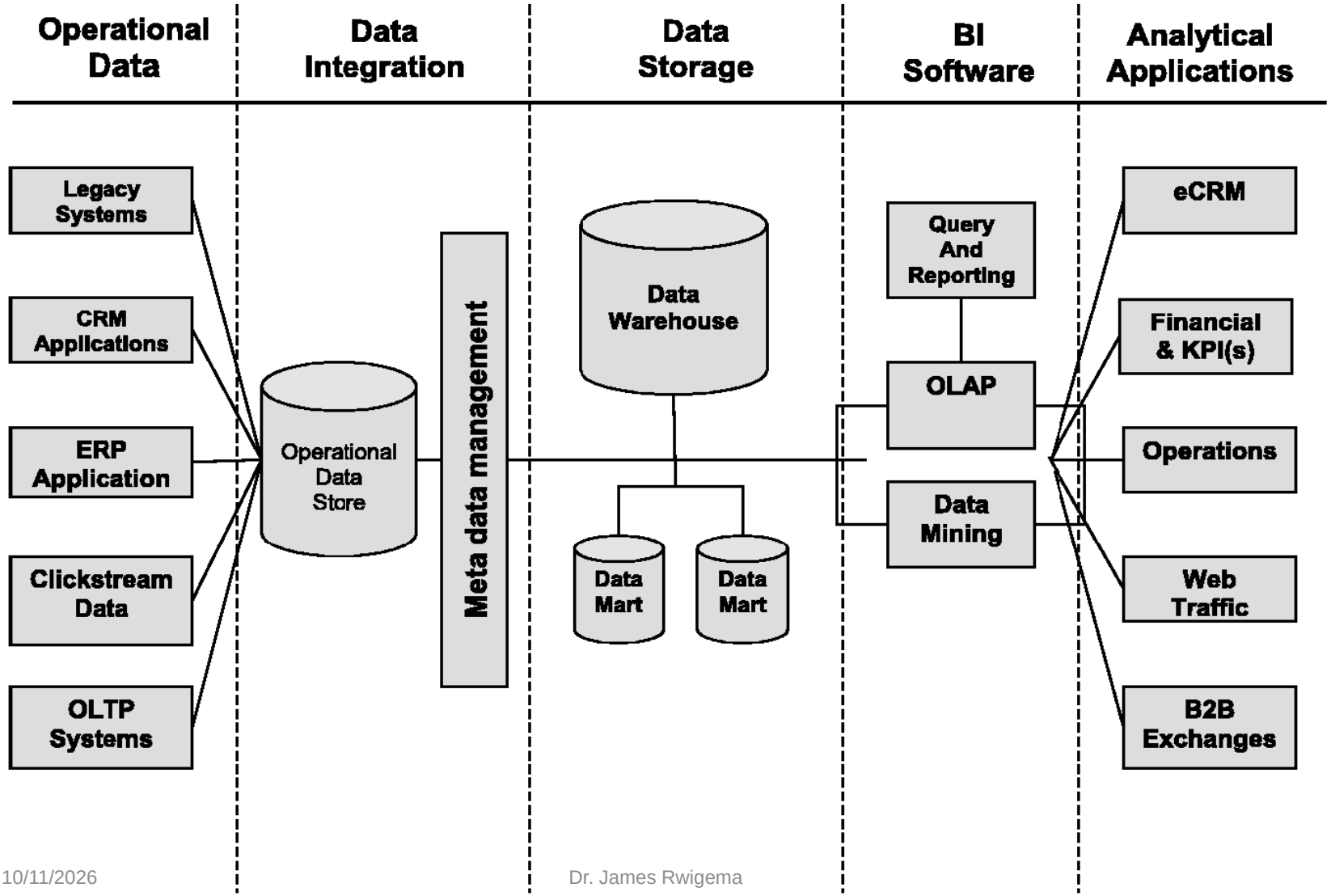


- Allow business users the ability to query and write **reports**
- To simplify reporting across **multiple** transaction systems
- To store **historical data longer** than you can/would in transaction system

Why do companies need BI?



Business Intelligence Components



Data Warehouse

A Data Warehouse Can be Defined as:

A decision support database that is maintained separately from the organization's operational database

A consistent database source that bring together information from multiple sources for decision support queries

Support information processing by providing a solid platform of consolidated, historical data for analysis

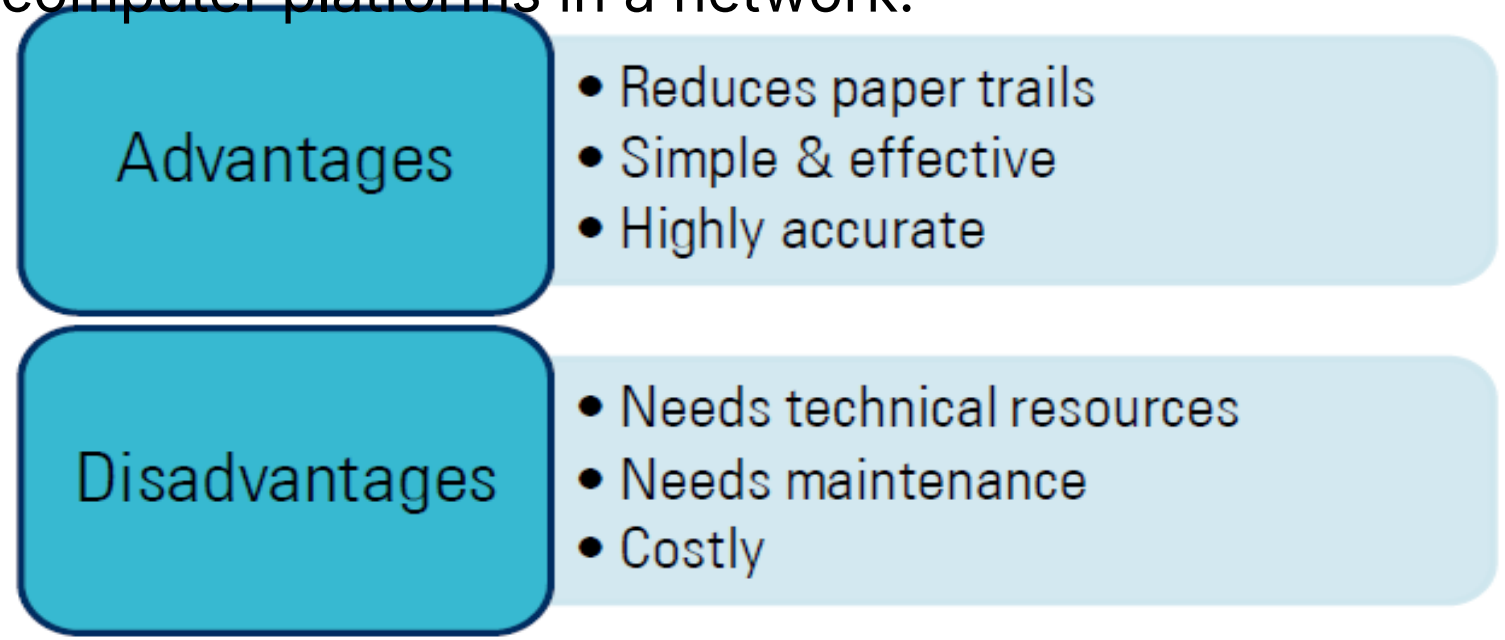
"Data warehousing is the process of constructing and using data warehouses"

- **A data warehouse** is a **subject-oriented**, **integrated**, **time-variant** and **non-volatile** collection of data in support of management decision making process.

OLTP – Online Transaction Processing

- **Online transaction processing**, or OLTP, is a class of information systems that facilitate and manage transaction-oriented applications, typically for **data entry and retrieval transaction processing**.
- Online transaction processing increasingly requires support for transactions that span a network and may include more than one company.

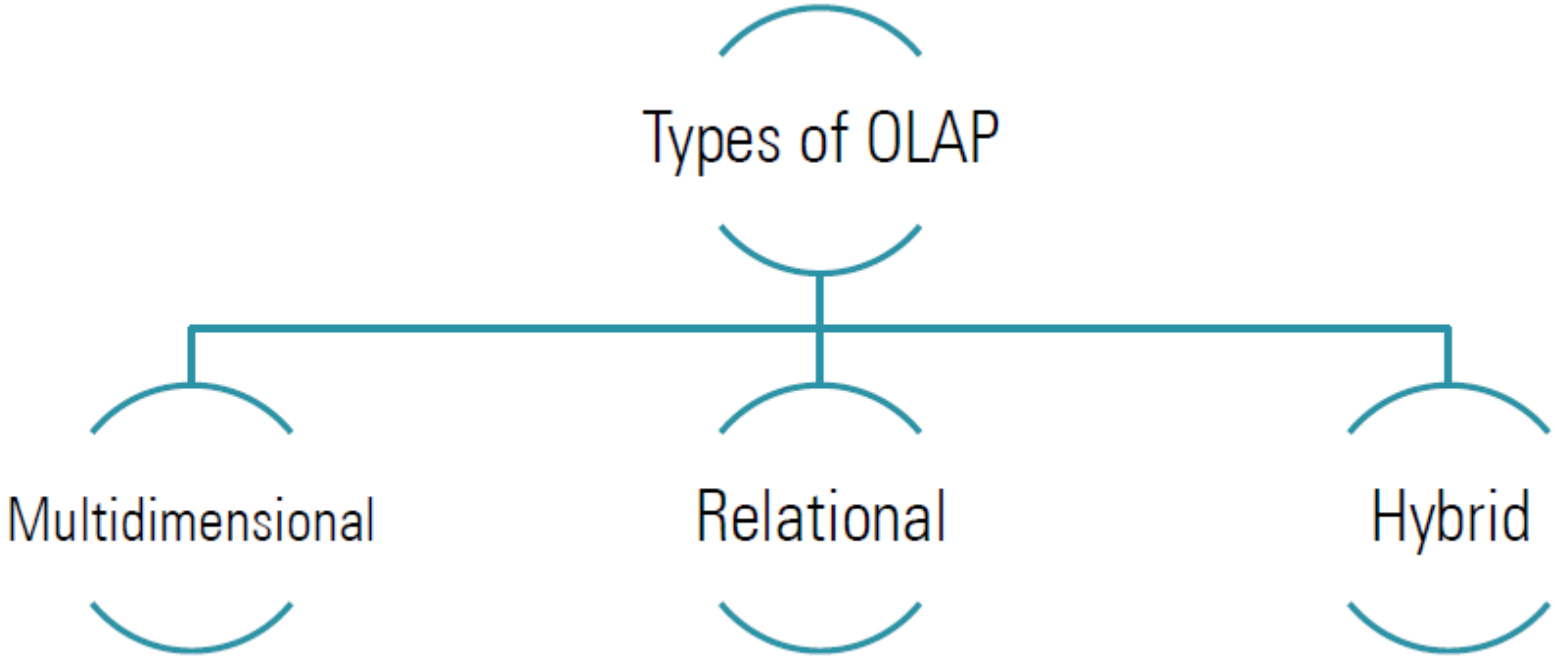
- For this reason, new online transaction processing software **uses client or server processing and brokering software** that allows transactions to run on different computer platforms in a network.



OLAP – Online Analytical Processing

- Online analytical processing, or OLAP is an **approach to answering multi-dimensional analytical (MDA) queries swiftly**.
- OLAP tools enable users **to analyze multidimensional data interactively from multiple perspectives.**
- OLAP consists of three basic analytical operations: **consolidation** (roll-up), **drill-down**, and **slicing and dicing**.

- **Consolidation** involves the **aggregation of data that can be accumulated and computed in one or more dimensions.**

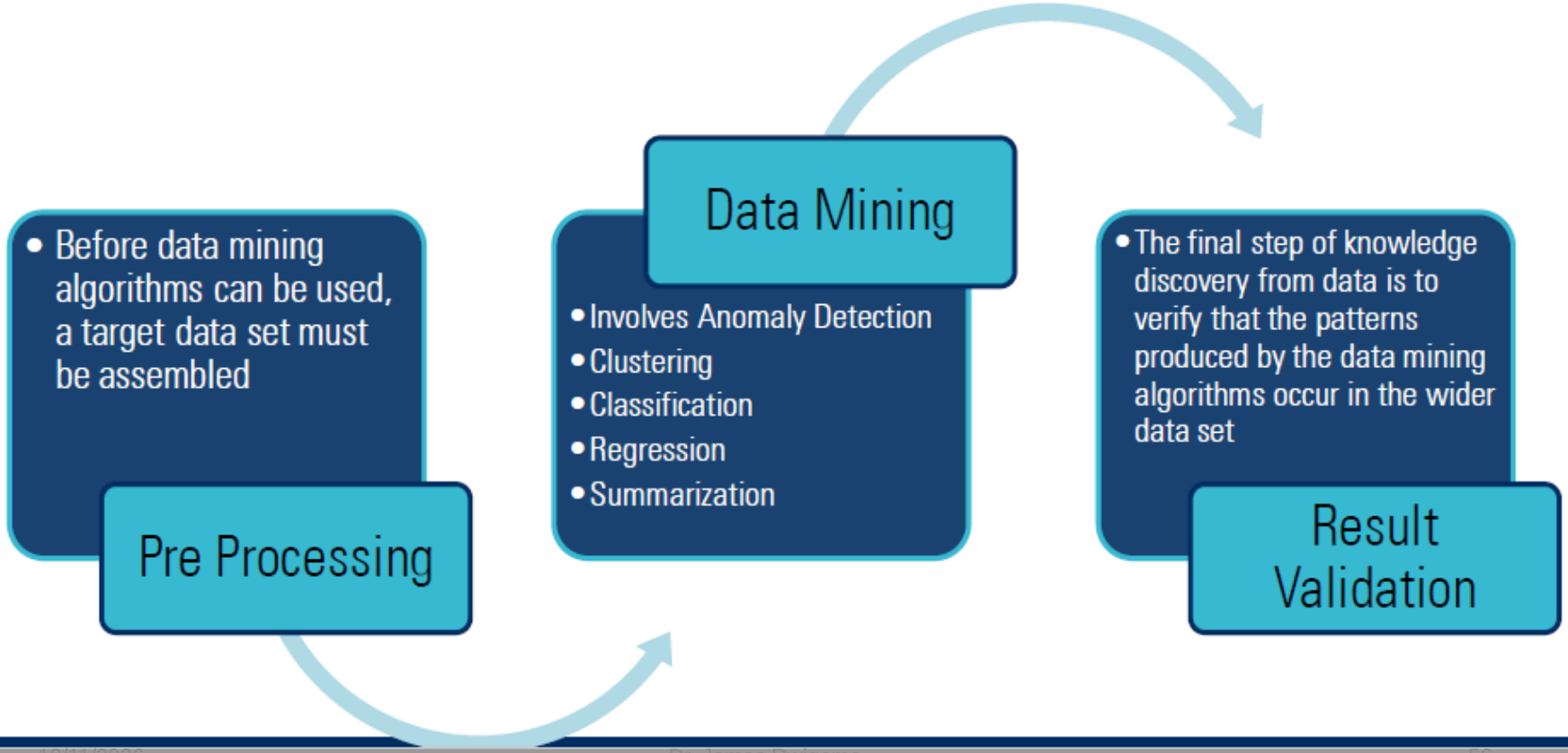


OLAP Vs. OLTP

	OLTP	OLAP
User	Clerk/IT professional	Knowledge Worker
Function	Day to Day Operations	Decision Support
Database Design	Application Oriented	Subject Oriented
Data	Current, Isolated	Historical, Consolidated
View	Detailed, Flat Relational	Summarized, Multidimensional
Usage	Structured, Repetitive	Ad Hoc
Access	Read/Write	Read

Data Mining & it's Process

- **Data Mining** is the computational process of discovering **patterns in large data sets** involving **methods at the intersection of artificial intelligence, machine learning, statistics, and database systems**.



Business Intelligence Layers

Presentation Layer



Reporting



Analysis

Data Warehouse Layer



Dara Warehouse

Source Layer



Finance Dept.



HR Dept.



CRM

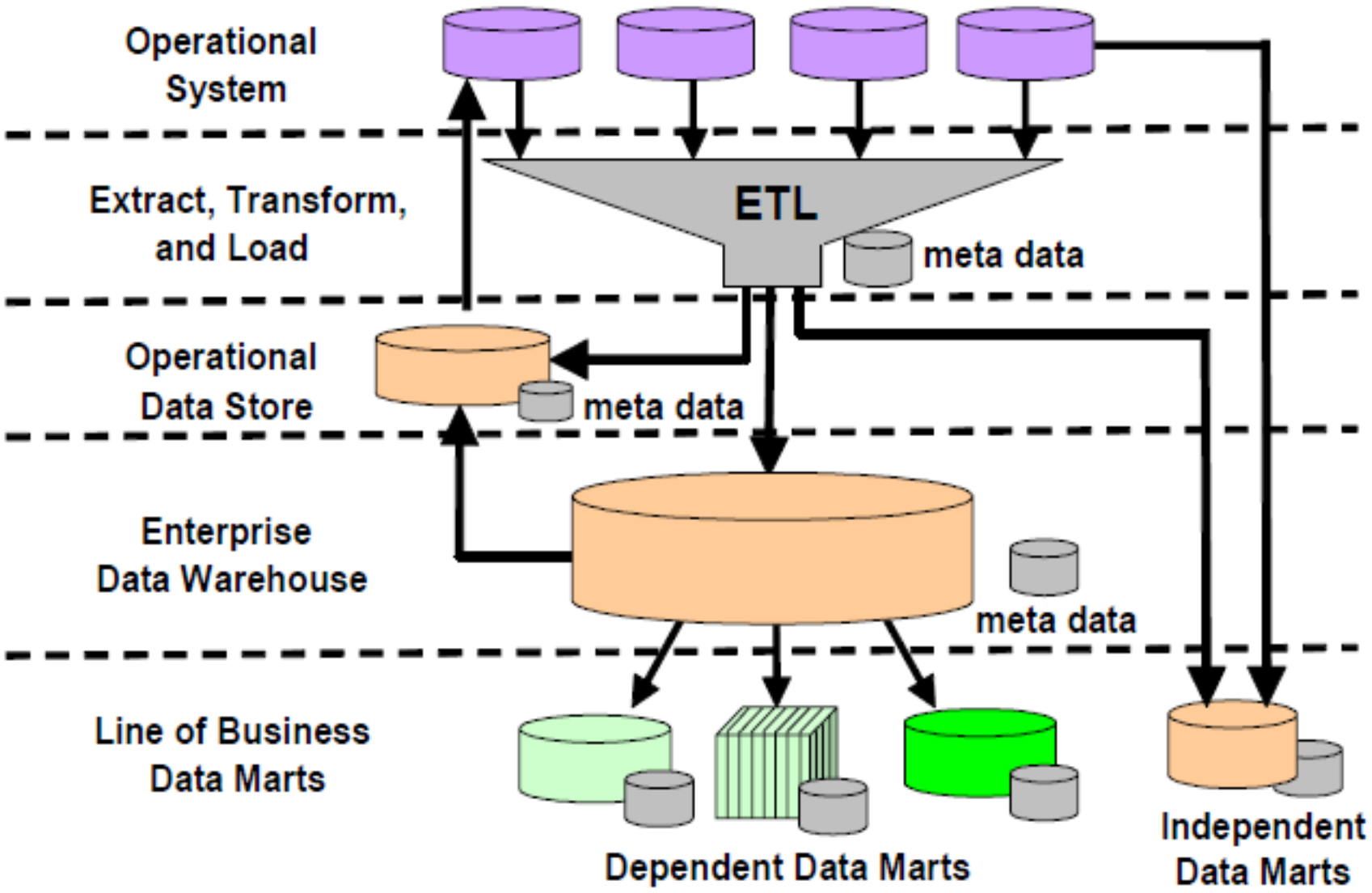


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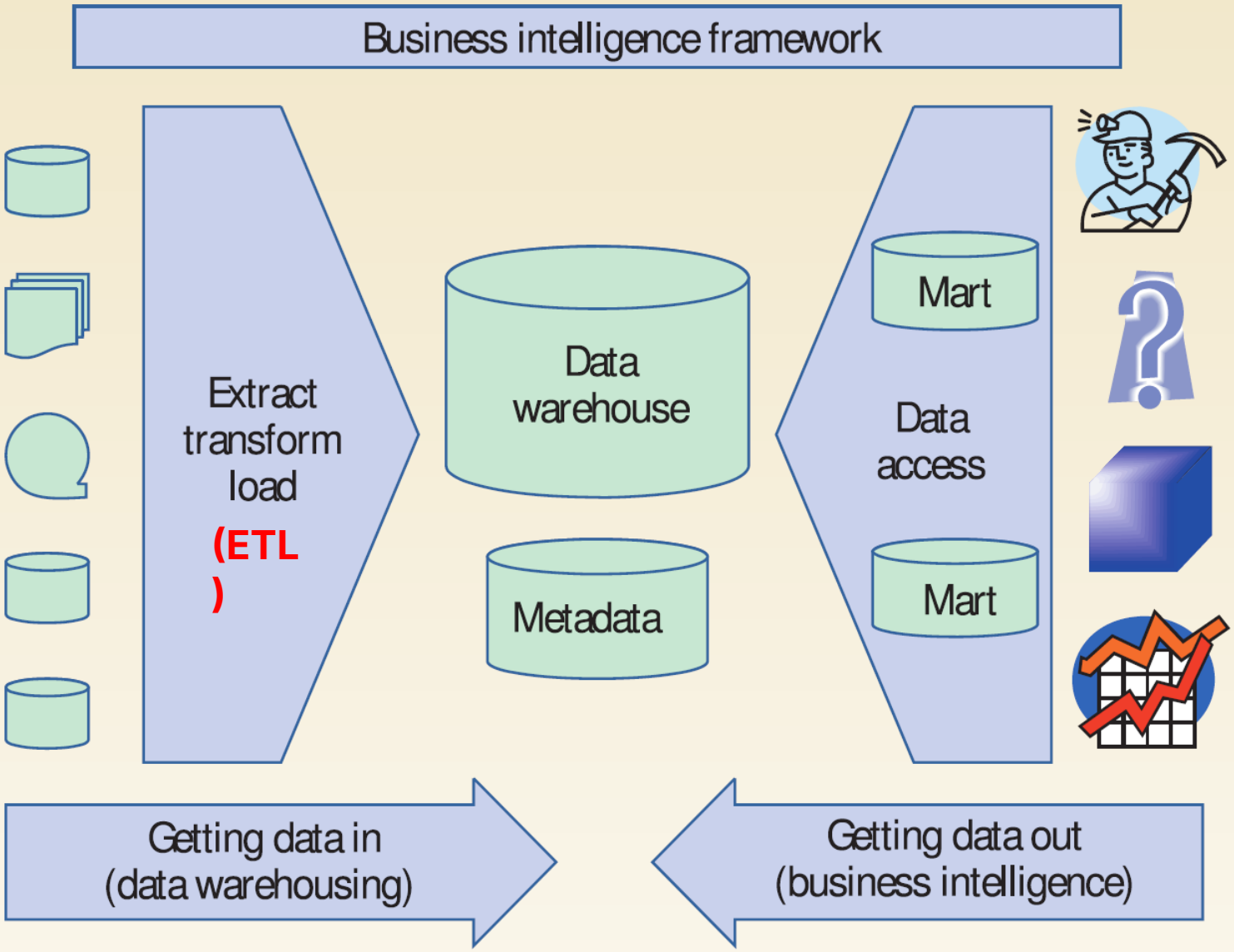


Excel

BI Architecture



BI framework



A Framework for Business Intelligence (BI)

- **Business intelligence (BI)**

➢A conceptual (umbrella) framework for decision support.

➢ It combines **architecture, databases (or data warehouse), analytical tools and applications** .

- **BI Objective**: enable interactive access to data to enable manipulation of data and to give business managers to take a good decision.

A Framework for Business Intelligence (BI) cont′

- The process of BI is based on the *transformation of data to information, then to decisions, and finally* to actions.

A Framework for Business Intelligence (BI) cont'

Benefits of BI

- Time savings
- Single version of truth
- Improved strategies and plans
- Improved tactical decisions
- More efficient processes
- Cost savings
- Faster, more accurate reporting
- Improved decision making
- Improved customer service
- Increased revenue

Many benefits are intangible